

Studies in Algebra and Group Theory
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December 21, 2025

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Set Theory

1.1 Ordered Pairs

We begin with a discussion of ordered pairs. Consider a set of numbers $\{1, 2, 3, 4\}$. For any set there is no particular order that triumphs over another, i.e., $\{1, 2, 3, 4\} = \{1, 3, 2, 4\} = \{3, 4, 2, 1\} = \dots$, an obvious truth. Say we want $\{1, 2, 3, 4\}$ to be in the order of $\{3, 2, 1, 4\}$ for some arbitrary reason; we may then consider the collection,

$$\mathfrak{C} = \{\{3\}, \{3, 2\}, \{3, 2, 1\}, \{3, 2, 1, 4\}\}.$$

It becomes obvious what is happening here: the method by which we get order is dependent on the number of iterations in which a particular element appears in any number of sets in a given collection (in comparison to the other elements in the other sets of the respective collection).

Then consider this process but for a nested collection (a collection within a collection) which contains two sets, namely the arbitrary elements a, b in opposing orders, and notice that it gives us the definition of a 2-tuple (ordered pair),

$$\mathfrak{C}_1 = \{\{a\}, \{a, b\}\} := (a, b) \text{ and } \mathfrak{C}_2 = \{\{b\}, \{b, a\}\} := (b, a).$$

Note:-

This form of generating or representing ordered pairs is called the Kuratowski of ordered pairs.

This, of course, naturally extends to greater n-tuples. There are some obvious consequences of this that will not be gone over here.

The **Cartesian product** is a set of ordered pairs which can be given by taking power sets.

Definition 1.1.1: Cartesian Product

The Cartesian product is the cross product of two sets given by

$$A \times B = \{(a, b) \mid a \in A, b \in B\},$$

where, by the Kuratowski encoding,

$$(a, b) := \{\{a\}, \{a, b\}\} \in \mathcal{P}(\mathcal{P}(A \cup B)).$$

1.2 Relations

The review of ordered pairs leads us to our next discussion on relations. A relation in more colloquial terms is a dynamic or characteristic of a given element shared with another given element such that they are arranged in the form of ordered pairs. Think of equality, then think of equality as equating $(2 + 3, 1 + 4)$, or perhaps something more interesting, the set of all (x, y) such that x is a man, y is an enby (slay), and x is married to y (many such cases). We can then think of relations as a set of ordered pairs $R \subset A \times B$. Consider the following examples:

1. $R := \{(x, y) \in \mathbb{N}^2 \mid y - x > 0\} \iff xRy \equiv y > x$, the relation of greater than;
2. $R := \{(a, B) \in A \times \mathcal{P}(A) \mid a \in B\} \iff aRB \equiv a \in B$, the relation of belonging (being an element of);
3. $R := \{(c, d) \in \{\text{all cats}\} \times \{\text{all dogs}\} \mid c, d \text{ live in the same house}\}.$

Naturally, there are an infinite number of possible relations, however there is a particular class of these which we are interested in for the time being—that which relates two equivalent elements (the definition of what equivalence means is, for now, notably ambiguous).

If a relation R exists s.t. xRx for any $x \in X$, then we say R is reflexive. If $xRy \iff yRx$, it is symmetric, and if $xRy, yRz \implies xRz$, it is transitive. If a given relation R satisfies all three of these properties (reflexivity, symmetry, transitivity), then we say it is an **equivalence relation**.

Definition 1.2.1: Equivalence relations

An equivalence relation, usually denoted $\sim$, is a relation that is reflexive, symmetric, and transitive $\forall (a, b) \in A \times B$.

Question 1

Consider each of the three qualifiers (axioms) for the existence of an equivalence relation. Find an explicit relation that has one property but not the other for each of the three axioms.

Solution

I apologize for the terse abstraction involved in this first example, but let

$$R := \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3) \in \{1, 2, 3\}^2\},$$

notice that $(1, 1), (2, 2), (3, 3)$ is true for R , thus reflexivity holds; however, it is quick to see that $(1, 2)$ exists in R but $(2, 1) \notin R$, thus symmetry cannot hold. For transitivity, notice that we have $(1, 2)$ and $(2, 3)$, so we can set up the first step, but $(1, 3) \notin R$! Thus transitivity fails. This example is a reminder that we need not think of concrete properties that can be applied in front of our eyes (shapes, dogs, even equality!) in order to have or manipulate some sort of mathematical relation. Next, consider a the relation $R := \{(a, b) \mid a \neq b\}$. First, aRa is categorically untrue thus reflexivity means nothing to us. Symmetry, however, tells us that if aRb then bRa , which is true! Transitivity may seem sneaky true at first, but notice that $aRb, bRa \implies aRa$ is not true (we simply replaced c with another instance of a). Finally, transitivity as a singular property is pretty straightforward. Consider $x > y$, notice that $x > x \implies x < x$ is categorically untrue since x can never be greater or less than itself, additionally, $x > y$ never implies $y > x$. For transitivity however, notice that $x > y, y > z \implies x > z$ is true!

Returning to our discussion of relations and partitions we can have this thing called an induced relation. Let $R := X/\mathcal{C}$, we say that if $x, y \in \mathcal{C}$ and $xX/\mathcal{C}y$, then $X/\mathcal{C}$ is the relation induced by the partition $\mathcal{C}$.

Proposition 1.2.1 Relationship between equivalence relations, classes, and set partitions

Explicitly the relationship between the two is given as follows: if R is an equivalence relation on X , then the set of equivalence classes form a partition of X that induces the relation R , and if $\mathcal{C}$ is a partition of X , then the induced relation is an equivalence relation whose set of equivalence classes is exactly $\mathcal{C}$.

1.3 Functions

First I want to remind us of some notation and terminology things (mathematicians are very specific):

Note:-

To say a function is on X into Y , means that the function is from X to Y

Definition 1.3.1: Functions

{We all know what functions are at some level, but here we are going to be extraordinarily explicit (though you likely have already heard it in this form, regardless). A function on X into Y is a relation (R) given by a letter like $f, g, h, \varphi, \dots$, such that:

1. $\text{dom } f = X$;
2. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$ (uniqueness of y for each x); additionally, $f(x) = y$ is conventional notation (instead of $(x, y) \in f$ or xfy);
3. y is called the **value** that f **assumes** at the **argument** x ; i.e., f **maps** **sends**/ **transforms**/ x to/ into y . We often use the terms map, mapping, transformation, correspondence, operator, and function interchangeably.

We denote this correspondence via

$$f : X \rightarrow Y.$$

Finally, we note that the set of all functions from X to Y is a subset of the power set $\mathcal{P}(X \times Y)$ and is denoted Y^X .

Something to note is that a function is not an active entity despite action-invoking words such as **transforms**; it merely *is*. The use of the word function to describe the undefined object that is somehow active in relation to a graph is nevertheless a static set, similar to that of a directory of names that correspond to addresses. Reminder that while the $\text{dom } f = X$, it need not be the case that $\text{ran } f = Y$, instead the range is reserved for the subset of all $y \in Y$ that have a corresponding x such that $f(x) = y$, i.e., is mapped to by some x . We call this property being **onto** and we say f maps X **onto** Y ; more commonly however we say that f is **surjective**—we explicitly write this as $\forall y \in Y, \exists x \in X : f(x) = y$ —and we denote this surjectivity via

$$f : X \twoheadrightarrow Y,$$

and if $A \subset X$, we may consider all $y \in Y$ such that there is an $x \in A$ such that $f(x_A) = y$ (here I use x_A as crude shorthand for $f(x)$ s.t. $x \in A$, where Halmos uses the more primitive $f(A)$), the set of those $y \in Y$ is known as the **image** of f under A . If it is the case that $X \subset Y$, the function f defined by $f(x) = x, \forall x \in X$, we call this the **inclusion map** **embedding**/ **injection**/ or we say f is **one-to-one**; we explicitly write this as $\forall a \neq b, f(a) \neq f(b)$. We denote injectivity via

$$f : A \hookrightarrow Y.$$

A special case of injectivity comes when we inject $X \hookrightarrow X$, relation-wise this is equality in X , language-wise, we call this the **identity map** on X .

Proposition 1.3.1 Restriction and extensions

Consider a function $f : Y \rightarrow Z$ and a set $X \subset Y$. We say that there is a **natural way** of constructing a new function $g : X \rightarrow Z$; let $g(x) = f(x), \forall x \in X$, we then say g is a **restriction** of f to X and f is the **extension** of g to Y . We denote this as $g = f|X$ where we can express the definition as $(f|X)(x) = f(x) \forall x \in X$ and $\text{ran}(f|X) = f(x_X)$.

There are a few other terminologies that follow from all that has been mentioned so far, and I will go through them rather quickly:

Definition 1.3.2: Projection

f (also pr is a **projection** from $X \times Y$ onto X if $f(x, y) = x$; similarly, if $R = X \times Y$, then instead of being a projection of R onto X , it is now the range of the projection f ($\text{ran}(\text{pr})$).

Definition 1.3.3: Bijection

Expounding on our notion of inclusion mappings and surjections from earlier, **bijection**—one-to-one correspondence-ness—is a function that is both injective and surjective.

Bijections are derived from an equivalence relation R in X with $aRb \iff f(a) = f(b)$ for $a, b \in X$; so given a function/ set $g(y) = \{x \in X \mid f(x) = y, \forall y \in Y\}$, the equivalence relation R tells us that $g(y)$ is an equivalence class of the relation R ; that is to say, we define the function as $g : Y \rightarrow X/R$, notice that for g , if $c \neq d$, $c, d \in Y$, due to the nature of equivalence classes (remember, all the classes are disjoint), $g(c) \neq g(d)$; notice this is the exact explicit definition of injectivity, thus this is a surjective-injective, i.e., bijective mapping. Although there is no universally agreed on notation for bijection, we shall denote this mapping as $g : Y \leftrightarrow X/R$.

Definition 1.3.4: Characteristic functions

If $A \subset X$ the characteristic function of A is the function $\chi : X \rightarrow 2$ given by

$$\chi(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X - A \end{cases}$$

, however, we may write χ_A to indicate that it depends on A . The function that assigns to each $A \subset X$, i.e., $\mathcal{P}(X)$, the characteristic function of $A \in 2^X$ is a bijection, i.e., $f : \mathcal{P}(X) \leftrightarrow 2^X$.

Question 2

Concerning bijectivity (point iii.), the examples have all the inclusion maps as one-to-one, however, except in a few trivial special cases, the projections are not. What are these special cases?

Solution

Consider a function's projection as the canonical map $\gamma : X \rightarrow X/R$, the following cases are trivial examples of non-injective projection maps:

1. $X = \emptyset$, vacuously fulfilled (it matters not what R is);
2. $R := \{(x, x) \mid x \in X\}$, so every equivalence class is a singleton.

Question 3

Prove:

1. $Y^\emptyset$ has exactly one element, namely $\emptyset$, regardless of if Y is empty or not.
2. If X is non-empty, then $\emptyset^X$ is empty.

Solution

1. Given that $Y^\emptyset \subset \mathcal{P}(\emptyset \times Y)$, and the definition of Cartesian sets has **and** as a qualifier, i.e., $A \times B = \{(a, b) \mid a \in A \text{ AND } b \in B\}$, we can see that $\emptyset \times Y$ contains in fact no elements which satisfy this, and thus, it is the empty set. So $Y^\emptyset \subset \mathcal{P}(\emptyset) = \{\emptyset\}$, thus the only element that can be in $Y^\emptyset$ is the singleton $\{\emptyset\}$. This is a vacuous satisfaction and is non-intuitive; some intuition, perhaps, may be gained but thinking “the only thing nothing can map to is nothing,” or by taking the cardinality of these two sets with each other given by $|\emptyset| \cdot |Y| = 0|Y| = 0$ (though this is mainly for intuition).
2. $\emptyset^X$ means that something must map to the emptyset, meaning there must be an element within the emptyset, but this cannot be true! Thus, by definition, the set must be empty.

1.4 Families

Forget literally everything you know about notational convention! Why? Idk, that's just how mathematicians decided to approach this topic—the family (awwwwww). Consider a function x (yes, this is strange) s.t. $x : I \rightarrow X$. We call I the **index set**, and thereby each $i \in I$ is an **index**; the range of the function is then called the **indexed set** (why, this is so stupid?!); the function has a special name, being called a **family**, and the value of x at a given i is called a **term** of the family (denoted x_i). Other even more bewildering notation may be used such as $\{x_i\}$ being a family in X or, more horrifically, a family $\{A_i\}$ of subsets of X , being a function $A : I \rightarrow \mathcal{P}(X)$ (now that I think of it, this notation, sickeningly, is becoming somewhat preferable).

Given that $\text{ran}(\{A_i\} = \bigcup_{i \in I} A_i)$, we notice that every arbitrary collection of sets is in fact the range of some arbitrary family: consider I and X equal to $\mathfrak{C}$ s.t. $\{A_i\} : \mathfrak{C} \rightarrow \mathfrak{C}$. I am realizing it is the terminology which is so obtuse, that what I do not like.

We may think of $\bigcup_{i \in I} A_i$ as $\bigcup_{i=1}^n A_i$ (similar to what we have seen for summations and products), where n is the order of i and we assume 1 is the first element in i .

Example 1.4.1

For some index set $i = \{1, 2, 3\}$, and an indexed set A_i we have $\bigcup_{i=1}^3 A_i = A_1 \cup A_2 \cup A_3$. Another example may be given as follows: let $I = \mathbb{N}$, and $A_i = \{-i, 0, i\}$, that is to say for every i we have

$A_1 = \{-1, 0, 1\}, A_2 = \{-2, 0, 2\}, \dots$, and so on. Notice that the infinite union of A_i produces $\mathbb{Z}$ and the infinite intersection produces $\{0\}$.

Example 1.4.2

Let us now consider an uncountably infinite indexing set it a plane $I = [-1, 1], A_i = \{i\} \times [0, 1] \subset \mathbb{R}^2$. Notice this is going to be a subset of ordered pairs, i.e., $\{(i, y) \mid 0 \leq y \leq 1\}$. Geometrically, the union over all $i \in I$ of A_i gives us a rectangle in $\mathbb{R}^2$ defined by $[-1, 1] \times [0, 1]$; similarly the intersection of these sets, notice, are intersections of finite parallel lines, thus there exists no element in common with these sets, thus it is $\emptyset$.

Theorem 1.4.1 Associative law of unions

Let $\{I_j\} : J \rightarrow K \mid K = \bigcup_{j \in I}$ and $\{A_k\} : K \rightarrow X$, then

$$\bigcup_{k \in K} A_k = \bigcup_{j \in J} \left(\bigcup_{i \in I_j} A_i \right).$$

Question 4

Prove the theorem above.

Sorry! I am leaving this one as an exercise for the reader (sorrriyyyyy:)

De Morgan's laws apply as standard for index/ indexed sets and their unions. From such, it is easy to see that if $\{A_i\}, \{B_i\}$ are families of sets, then $\bigcup_i A_i \cap \bigcup_j B_j = \bigcup_{i,j} (A_i \cap B_j)$, a similarly for $\bigcap_{i,j} (A_i \cup B_j)$.

Let us consider a special type of family: let $I = \{a, b\}$, $a \neq b$ and let Z be the set of all families z indexed by I such that $z_a \in X, z_b \in Y$. Now consider $f : Z \leftrightarrow X \times Y$ such that $f(z) = (z_a, z_b)$. This is the **generalized Cartesian product**! There is further detail one can go into about families of sets and sets of all given families of sets, and even sets of those sets (including their Cartesian products), but these are so disgusting as to induce spontaneous retching in even the most tough-stomached mathematicians. So we will avoid it, if possible.

1.5 Inverses and Composites

Consider a function $f : X \rightarrow Y$, a natural mapping that might arise (when thinking on power sets) is some function $F : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$; just as each element of X is mapped to an element Y under f , each subset of X maps to some subset of Y under F , so if $A \subset X$ then $\exists (A \mapsto \text{Im}(f_A)) \forall A \in \mathcal{P}(X)$. Cool? I guess? What is to be done about this? In all honesty, I am not sure this factoid has much use other than being a pedagogical tool (though I am almost certainly wrong and I am sure someone will rush to correct me, wherein I might toil and languish over my incorrectness and stupidity, suffering immensely—but probably not, because I simply do not care), but for our cases we use it to introduce its behavior in the inverse.

Note:-

Regarding notation—as we all too often are— $\text{Im}(f_A)$ is the same thing as $f(A)$ when regarding images of subsets. I prefer the former notation as it feels a bit more natural to me.

Definition 1.5.1: Inverses and inverse images

Given a function $f : X \rightarrow Y$ we say f^{-1} is the inverse of f given by $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ such that if $B \subset Y$, then

$$\text{Im}(f_B^{-1}) = \{x \in X \mid f(x) \in B\},$$

is the inverse image of B under f

This definition results in some immediate necessary and sufficient conditions: first, for $f : X \rightarrow Y$ to exist, the inverse image under f of each non-empty subset of Y must be a non-empty subset of X ; second, for f to be injective is that the inverse image under f of each singleton in the range of f be a singleton in X .

Note:-

We often use f^{-1} for another purpose: for some $f : X \rightarrow Y$, $f^{-1} : \text{ran}(f) \rightarrow X \implies f^{-1}(y) = x \iff f(x) = y$. This will be the most common use case for f^{-1} as the inverse function $g : Y \rightarrow X$

There are some connections between images and inverse images which I will quickly enumerate, but I will not provide proof for (for the proofs see Halmos, pp. 59): Let f be a function such that $f : X \rightarrow A$ for each of the following,

1. $B \subset Y \implies f(f^{-1}(B)) \subset B$;
2. $f : X \rightarrow Y, B \subset Y \implies f(f^{-1}(B)) = B$;
3. $A \subset X \implies A \subset f^{-1}(f(A))$;
4. $f : X \hookrightarrow Y, A \subset X \implies f^{-1}(f(A)) = A$;
5. If $\{B_i\}$ is a family of subsets of Y , then $f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$ and $f^{-1}(\bigcap_{i \in I} B_i) = \bigcap_{i \in I} f^{-1}(B_i)$;
6. $f^{-1}(Y - B) = X - f^{-1}(B)$ for each $B \subset Y$.

We now move onto composites.

Definition 1.5.2: Composition

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ it is clear that we can make a map from $X \rightarrow Z$ since the range of f can be the domain of g . We can represent this new function—i.e., the composite function—as either $h : X \rightarrow Z$ or $g \circ f$, and often simply just gf . We read this right to left.

Notice that function composition is not always commutative, nor need it be. Also notice that this remains true under the special case that $f : X \rightarrow Y$ and $g : Y \rightarrow X$; functional composition, however, is always associative. There will be a strong proof for this later on (in Groups). Notice that if $g : X \rightarrow Y$ for standard f , if it is bijective we have an inverse function and identity map given by h . Functional composition under inverses appears something like this: if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, and $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ and $g^{-1} : \mathcal{P}(Z) \rightarrow \mathcal{P}(Y)$. Then $g \circ f$ has an inverse given by $f^{-1} \circ g^{-1}$. Notice this is not true for $g^{-1} \circ f^{-1}$.

1.5.1 Inverse/ converse and composite relations

The main idea of this comes out of our exploration of inverse functions. Functions are relations after all. Consider the converse relation of R on $X \times Y$ s.t. xRy and $yR^{-1}x$. Similar things can be done for composites by generalizing composition to relations, observe: let S be a relation on $Y \times Z$ and R be a relation on $X \times Y$ such that xRy and ySz . Now let T over $X \times Z$ be given by $S \circ R$ such that xTz exists $\iff \exists y \in Y$ such that xRy and ySz . Congrats! You made a {composite relation}. If it is the case that R and S are functions s.t. $R(x) = y$, $S(y) = z$ and $S(R(x)) = z \iff xTz$ which implies that functional composition is a special case of a greater abstraction called the {relative product}(??).

Group Theory Basics

Groups are the central object of abstract algebra, capturing the essence of symmetry and invertible operations. The definition is deceptively simple—three axioms—yet groups appear everywhere: in geometry (symmetries), number theory (modular arithmetic), topology (fundamental groups), physics (gauge theories), and even chemistry (molecular symmetries). This lecture develops the foundations rigorously.

2.1 The group axioms

Definition 2.1.1: Groups

A **group** is a set G together with a map

$$m : G \times G \rightarrow G$$

called a **law of composition** (also **binary operation**). Writing ab for $m(a, b)$, the law of composition must satisfy three axioms:

1. **Associativity:** $\forall a, b, c \in G, (ab)c = a(bc)$.
2. **Identity:** $\exists e \in G$ such that $ea = ae = a$ for all $a \in G$.
3. **Inverses:** $\forall a \in G, \exists b \in G$ such that $ab = ba = e$.

We write (G, m) or simply G for the group, and call $|G|$ the **order** of G .

The axioms have immediate but important consequences that we now prove carefully.

Proposition 2.1.1 Uniqueness of identity

The identity element e is unique.

Proof: Uniqueness of identity

Suppose e and e' are both identities. Then $e = ee'$ (since e' is an identity) and $ee' = e'$ (since e is an identity). Thus $e = e'$. $\square$

Proposition 2.1.2 Uniqueness of inverses

For each $a \in G$, the inverse is unique. We denote it a^{-1} (or $-a$ in additive notation).

Proof: Uniqueness of inverses

Suppose b and c are both inverses of a , so $ab = ba = e$ and $ac = ca = e$. Then:

$$b = be = b(ac) = (ba)c = ec = c.$$

$\square$

Proposition 2.1.3 Cancellation laws

In a group G :

1. **Left cancellation:** $ab = ac \implies b = c$.
2. **Right cancellation:** $ba = ca \implies b = c$.

Proof: Proof of cancellation

For left cancellation: if $ab = ac$, multiply both sides on the left by a^{-1} : $a^{-1}(ab) = a^{-1}(ac)$, so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$, i.e., $b = c$. Right cancellation is similar. $\square$

Proposition 2.1.4 Inverse of products

For all $a, b \in G$: $(ab)^{-1} = b^{-1}a^{-1}$. Note the reversal of order!

Proof: Proof of inverse formula

We verify $(ab)(b^{-1}a^{-1}) = e$: $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e$. Similarly for the other side. $\square$

Proposition 2.1.5 Inverse of inverse

For all $a \in G$: $(a^{-1})^{-1} = a$.

Proof: Proof of double inverse

By definition, a^{-1} satisfies $a \cdot a^{-1} = e$. This says a is an inverse of a^{-1} , so by uniqueness, $(a^{-1})^{-1} = a$. $\square$

Note:-

It is customary to use additive notation $(+, 0, -a)$ for abelian groups and multiplicative notation $(\cdot, 1$ or $e, a^{-1})$ in general. This does not mean that multiplication implies non-commutativity!

2.1.1 Group variants and related structures

Groups sit in a hierarchy of algebraic structures with fewer or more axioms.

Definition 2.1.2: Abelian groups

A group G is **abelian** (or **commutative**) if $ab = ba$ for all $a, b \in G$.

Definition 2.1.3: Monoid

A **monoid** is a set with an associative binary operation and an identity element, but inverses are not required. Example: $(\mathbb{N}, +)$ with identity 0, or $(\mathbb{Z}, \times)$ with identity 1.

Definition 2.1.4: Semigroup

A **semigroup** is a set with an associative binary operation—no identity or inverses required. Example: $(\mathbb{Z}_{>0}, +)$.

Note:-

There's a pattern here: Group = Monoid + Inverses = Semigroup + Identity + Inverses. Category theory provides a unifying framework: a group is a category with one object where every morphism is invertible; a monoid is a category with one object (morphisms need not be invertible).

2.2 Fundamental examples

2.2.1 Integers and modular arithmetic

Example 2.2.1 (The integers)

The set $(\mathbb{Z}, +)$ is an abelian group with identity 0 and inverse $-n$ for each n . This is the prototypical infinite cyclic group.

Example 2.2.2 (Integers mod n)

$\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, \dots, n-1\}$ with addition mod n :

$$a + b := \begin{cases} a + b & \text{if } a + b < n \\ a + b - n & \text{if } a + b \geq n. \end{cases}$$

Identity is 0, inverse of a is $n - a \pmod{n}$. This is an abelian group of order n .

Proposition 2.2.1 Units mod n

The set $(\mathbb{Z}/n\mathbb{Z})^\times = \{a \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}$ forms a group under multiplication mod n . Its order is $\varphi(n)$, Euler's totient function.

Proof: Proof that units form a group

Closure: If $\gcd(a, n) = \gcd(b, n) = 1$, then $\gcd(ab, n) = 1$ (if a prime p divides both ab and n , it divides a or b , contradicting coprimality).

Identity: 1 is coprime to n .

Inverses: If $\gcd(a, n) = 1$, then by Bézout's identity, $\exists r, s$ with $ra + sn = 1$. Thus $ra \equiv 1 \pmod{n}$, so r is a multiplicative inverse. $\square$

2.2.2 Multiplicative groups of fields

Example 2.2.3 (Multiplicative groups)

$\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, and $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ are abelian groups under multiplication, with identity 1 and inverse $1/x$.

Example 2.2.4 (The circle group)

The unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is a group under multiplication. It is isomorphic to $\mathbb{R}/\mathbb{Z}$ via $t \mapsto e^{2\pi it}$. This is a compact abelian group and plays a fundamental role in Fourier analysis and representation theory.

Example 2.2.5 (Roots of unity)

For each $n \geq 1$, the n th roots of unity $\mu_n = \{z \in \mathbb{C} : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, 1, \dots, n-1\}$ form a cyclic group of order n under multiplication. We have $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$.

2.2.3 Symmetric and permutation groups

Definition 2.2.1: Permutation group

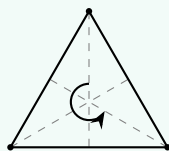
A **permutation** of a set X is a bijection $\sigma : X \rightarrow X$. The set of all permutations of X forms a group $\text{Sym}(X)$ under composition, with identity id_X and inverse σ^{-1} .

Definition 2.2.2: Symmetric group

The **symmetric group** on n letters is $S_n = \text{Sym}(\{1, 2, \dots, n\})$. It has order $n!$ and is non-abelian for $n \geq 3$.

Example 2.2.6 (Symmetries of a triangle)

S_3 can be visualized as symmetries of an equilateral triangle: 3 rotations (including identity) and 3 reflections.



Labeling vertices 1, 2, 3, rotations give permutations $e, (123), (132)$ and reflections give $(12), (13), (23)$. Thus $|S_3| = 6$.

2.2.4 Matrix groups

Definition 2.2.3: General linear group

The **general linear group** $GL_n(\mathbb{F})$ is the group of invertible $n \times n$ matrices over a field $\mathbb{F}$, under matrix multiplication. A matrix is invertible iff its determinant is nonzero.

Definition 2.2.4: Special linear group

The **special linear group** $SL_n(\mathbb{F}) = \{A \in GL_n(\mathbb{F}) : \det(A) = 1\}$ is a subgroup of $GL_n(\mathbb{F})$.

Note:-

Matrix groups are our first examples of Lie groups—groups that are also smooth manifolds. They are central to representation theory: a representation of G is a homomorphism $\rho : G \rightarrow GL_n(\mathbb{F})$, making G act on $\mathbb{F}^n$ by linear transformations. This perspective unifies group theory with linear algebra.

Example 2.2.7 (Non-abelian example)

In $GL_2(\mathbb{R})$, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then:

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

So $GL_2(\mathbb{R})$ is non-abelian.

2.3 Constructions: Products and sums

How do we build new groups from existing ones?

2.3.1 Direct products

Definition 2.3.1: Direct product

Given groups G and H , the **direct product** $G \times H$ is the Cartesian product with componentwise operation:

$$(a, b) \cdot (a', b') = (aa', bb').$$

Identity is (e_G, e_H) , inverse of (a, b) is (a^{-1}, b^{-1}) .

Proposition 2.3.1 Direct product is a group

The direct product $G \times H$ satisfies all group axioms. Moreover, $|G \times H| = |G| \cdot |H|$ (for finite groups).

Proof: Verification of group axioms for $G \times H$

Associativity: $((a, b)(a', b'))(a'', b'') = (aa', bb')(a'', b'') = (aa'a'', bb'b'')$. Similarly, $(a, b)((a', b')(a'', b'')) = (a, b)(a'a'', b'b'') = (aa'a'', bb'b'')$. Equal by associativity in G and H .

Identity: $(e_G, e_H)(a, b) = (e_G a, e_H b) = (a, b)$, and similarly on the right.

Inverses: $(a, b)(a^{-1}, b^{-1}) = (aa^{-1}, bb^{-1}) = (e_G, e_H)$. □

Example 2.3.1

We have $\mathbb{Z}/2 \times \mathbb{Z}/3 \cong \mathbb{Z}/6$ (since $\gcd(2, 3) = 1$), but $\mathbb{Z}/2 \times \mathbb{Z}/2 \not\cong \mathbb{Z}/4$ (the former has no element of order 4).

This generalizes to arbitrary (even infinite) collections of groups.

Definition 2.3.2: Arbitrary direct products

For a family of groups $\{G_i\}_{i \in I}$, the **direct product** is:

$$\prod_{i \in I} G_i = \{(a_i)_{i \in I} : a_i \in G_i\}$$

with componentwise operations.

Definition 2.3.3: Direct sum

The **direct sum** (for abelian groups) is the subgroup:

$$\bigoplus_{i \in I} G_i = \{(a_i)_{i \in I} \in \prod_{i \in I} G_i : a_i = e_{G_i} \text{ for all but finitely many } i\}.$$

Note:-

For finite index sets, $\prod = \bigoplus$. For infinite sets, they differ crucially.

Example 2.3.2 (Power series vs polynomials)

Taking $G_i = (\mathbb{R}, +)$ for all $i \in \mathbb{N}$:

$$\prod_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[[x]] \quad (\text{formal power series under addition})$$

$$\bigoplus_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[x] \quad (\text{polynomials under addition})$$

A power series $\sum_{i=0}^{\infty} a_i x^i$ can have infinitely many nonzero terms; a polynomial cannot.

Question 1

1. Prove that $\mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/mn$ if and only if $\gcd(m, n) = 1$.
2. Find all groups of order 4 up to isomorphism.
3. Prove that $G \times H \cong H \times G$ for any groups G, H .

Solution

(1) ($\Leftarrow$) Assume $\gcd(m, n) = 1$. The element $(1, 1) \in \mathbb{Z}/m \times \mathbb{Z}/n$ has order $\text{lcm}(m, n) = mn$ (since the order of (a, b) is $\text{lcm}(\text{ord}(a), \text{ord}(b))$). Thus $\mathbb{Z}/m \times \mathbb{Z}/n$ contains an element of order $mn = |G|$, so it's cyclic, hence $\cong \mathbb{Z}/mn$.

($\Rightarrow$) If $d = \gcd(m, n) > 1$, then for any $(a, b) \in \mathbb{Z}/m \times \mathbb{Z}/n$, we have $\frac{mn}{d} \cdot (a, b) = (\frac{mn}{d}a, \frac{mn}{d}b) = (0, 0)$ since $m \mid \frac{mn}{d}$ and $n \mid \frac{mn}{d}$. So every element has order dividing $\frac{mn}{d} < mn$, meaning no element has order mn . But $\mathbb{Z}/mn$ has an element of order mn , so they're not isomorphic.

(2) Let G have order 4. Every element has order dividing 4, so orders are 1, 2, or 4.

Case 1: G has an element a of order 4. Then $G = \{e, a, a^2, a^3\} = \langle a \rangle \cong \mathbb{Z}/4$.

Case 2: Every non-identity element has order 2. Then $G = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$. What is ab ? It's not e (else $a = b^{-1} = b$), not a (else $b = e$), not b (else $a = e$). So $ab = c$. Similarly $ba = c$ (check: $(ab)(ba) = a(bb)a = a^2 = e$, so $ba = (ab)^{-1} = ab$ since ab has order 2). So G is abelian, and $G \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).

(3) Define $\varphi : G \times H \rightarrow H \times G$ by $\varphi(g, h) = (h, g)$. This is clearly a bijection. It's a homomorphism: $\varphi((g_1, h_1)(g_2, h_2)) = \varphi(g_1 g_2, h_1 h_2) = (h_1 h_2, g_1 g_2) = (h_1, g_1)(h_2, g_2) = \varphi(g_1, h_1)\varphi(g_2, h_2)$.

2.4 Subgroups

Definition 2.4.1: Subgroup

A **subgroup** of a group G is a nonempty subset $H \subseteq G$ such that:

1. **Closure:** $a, b \in H \implies ab \in H$;
2. **Inverses:** $a \in H \implies a^{-1} \in H$.

We write $H \leq G$ (or $H < G$ if $H \neq G$, a **proper subgroup**).

Proposition 2.4.1 Subgroup criterion

A nonempty subset $H \subseteq G$ is a subgroup iff $a, b \in H \implies ab^{-1} \in H$.

Proof: One-step subgroup test

($\Rightarrow$) If $H \leq G$ and $a, b \in H$, then $b^{-1} \in H$ (inverses) and $ab^{-1} \in H$ (closure).

($\Leftarrow$) Assume the condition. Since $H \neq \emptyset$, pick $a \in H$. Then $e = aa^{-1} \in H$. For any $b \in H$, $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we have $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$. $\square$

Example 2.4.1 (Subgroups of $\mathbb{Z}$)

Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\}$ for some $n \geq 0$. These form a chain: $\mathbb{Z} = 1\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset 8\mathbb{Z} \supset \dots$ (not all subgroups are comparable, though: $2\mathbb{Z}$ and $3\mathbb{Z}$ are incomparable).

Theorem 2.4.1 Subgroups of $\mathbb{Z}$

Let H be a subgroup of $(\mathbb{Z}, +)$. Then either $H = \{0\}$ or $H = n\mathbb{Z}$ where n is the smallest positive element of H .

Proof: Classification of subgroups of $\mathbb{Z}$

If $H = \{0\}$, we're done. Otherwise, H contains some nonzero element a ; then $-a \in H$ too, so H contains a positive element. Let n be the smallest positive element of H .

We show $H = n\mathbb{Z}$. Clearly $n\mathbb{Z} \subseteq H$ since H is closed under addition: $n \in H \implies 2n = n + n \in H \implies 3n \in H \implies \dots$, and $-n \in H \implies -2n \in H \implies \dots$

For $H \subseteq n\mathbb{Z}$: take any $m \in H$. By division algorithm, $m = qn + r$ with $0 \leq r < n$. Then $r = m - qn \in H$ (since $m, qn \in H$). By minimality of n , we must have $r = 0$, so $m = qn \in n\mathbb{Z}$. $\square$

Proposition 2.4.2 Greatest common divisor

Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then:

1. $\mathbb{Z}a + \mathbb{Z}b = \{ra + sb : r, s \in \mathbb{Z}\} = d\mathbb{Z}$;
2. d divides both a and b ;
3. If n divides both a and b , then n divides d ;
4. $\exists r, s \in \mathbb{Z}$ such that $d = ra + sb$ (Bézout's identity).

Proof: Proof of GCD properties

The set $S = \mathbb{Z}a + \mathbb{Z}b$ is a subgroup of $\mathbb{Z}$ (verify: closed under addition and negation). By the theorem, $S = d\mathbb{Z}$ for some $d \geq 0$. Since $a = 1 \cdot a + 0 \cdot b \in S$ and $b \in S$, and $S = d\mathbb{Z}$, we have $d \mid a$ and $d \mid b$. Since $d \in S$, we have $d = ra + sb$ for some r, s . If $n \mid a$ and $n \mid b$, then $n \mid (ra + sb) = d$. These properties characterize $d = \gcd(a, b)$. $\square$

Corollary 2.4.1 Coprimality criterion

a and b are coprime (i.e., $\gcd(a, b) = 1$) iff $\exists r, s \in \mathbb{Z}$ with $ra + sb = 1$.

Corollary 2.4.2 Euclid's lemma

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: Proof of Euclid's lemma

Suppose $p \nmid a$. Since p is prime, $\gcd(p, a) = 1$. By Bézout, $\exists r, s$ with $rp + sa = 1$. Multiply by b : $rpb + sab = b$. Since $p \mid rpb$ and $p \mid ab$, we have $p \mid b$. $\square$

2.4.1 Generated subgroups and cyclic groups

Definition 2.4.2: Subgroup generated by a set

For $S \subseteq G$, the **subgroup generated by S** , denoted $\langle S \rangle$, is the smallest subgroup of G containing S . Explicitly:

$$\langle S \rangle = \{s_1^{\epsilon_1} s_2^{\epsilon_2} \cdots s_k^{\epsilon_k} : k \geq 0, s_i \in S, \epsilon_i \in \{1, -1\}\}.$$

Definition 2.4.3: Cyclic groups

A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$. The element a is called a **generator**.

Definition 2.4.4: Order of an element

The **order** of $a \in G$, denoted $\text{ord}(a)$ or $|a|$, is the smallest positive integer n such that $a^n = e$, or ∞ if no such n exists.

Proposition 2.4.3 Structure of cyclic groups

Let $G = \langle a \rangle$.

1. If $\text{ord}(a) = \infty$, then $G \cong \mathbb{Z}$ via $n \mapsto a^n$.
2. If $\text{ord}(a) = n < \infty$, then $G = \{e, a, a^2, \dots, a^{n-1}\}$ and $G \cong \mathbb{Z}/n\mathbb{Z}$.

Proof: Proof of cyclic group structure

Define $\varphi : \mathbb{Z} \rightarrow G$ by $\varphi(k) = a^k$. This is a homomorphism: $\varphi(k+l) = a^{k+l} = a^k a^l = \varphi(k)\varphi(l)$. It's surjective since $G = \langle a \rangle$.

Case 1: $\text{ord}(a) = \infty$. If $\varphi(k) = \varphi(l)$ with $k > l$, then $a^{k-l} = e$, contradicting infinite order. So φ is injective, hence an isomorphism $\mathbb{Z} \xrightarrow{\sim} G$.

Case 2: $\text{ord}(a) = n$. Then $\ker(\varphi) = \{k \in \mathbb{Z} : a^k = e\}$. Since $a^n = e$, we have $n\mathbb{Z} \subseteq \ker(\varphi)$. Conversely, if $a^k = e$, write $k = qn + r$ with $0 \leq r < n$. Then $a^r = a^{k-qn} = a^k(a^n)^{-q} = e$. By minimality of n , $r = 0$, so $k \in n\mathbb{Z}$. Thus $\ker(\varphi) = n\mathbb{Z}$, and $G \cong \mathbb{Z}/n\mathbb{Z}$ by the First Isomorphism Theorem. $\square$

Proposition 2.4.4 Subgroups of cyclic groups

Every subgroup of a cyclic group is cyclic. If $G = \langle a \rangle$ has order n , the subgroups of G are exactly $\langle a^d \rangle$ for divisors $d \mid n$, and $|\langle a^d \rangle| = n/d$.

Proof: Subgroups of cyclic groups

Let $H \leq \langle a \rangle$. If $H = \{e\}$, it's cyclic. Otherwise, let d be the smallest positive integer with $a^d \in H$. We claim $H = \langle a^d \rangle$.

Clearly $\langle a^d \rangle \subseteq H$. For the reverse: take $a^k \in H$. Write $k = qd + r$ with $0 \leq r < d$. Then $a^r = a^k(a^d)^{-q} \in H$. By minimality of d , $r = 0$, so $a^k = (a^d)^q \in \langle a^d \rangle$.

If $|G| = n$, then a^d has order $n/\gcd(d, n)$. The subgroups correspond to divisors via $d \leftrightarrow \langle a^{n/d} \rangle$ of order d . $\square$

Question 2

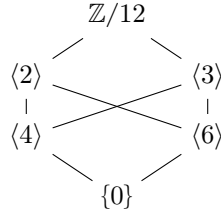
1. Find all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and draw the subgroup lattice.
2. Prove that $\mathbb{Z}/p\mathbb{Z}$ has no proper nontrivial subgroups for p prime.
3. How many generators does $\mathbb{Z}/n\mathbb{Z}$ have?

4. Prove: if G is a group where every element satisfies $x^2 = e$, then G is abelian.

Solution

(1) Divisors of 12: 1, 2, 3, 4, 6, 12. Subgroups: $\langle 0 \rangle = \{0\}$, $\langle 6 \rangle = \{0, 6\}$, $\langle 4 \rangle = \{0, 4, 8\}$, $\langle 3 \rangle = \{0, 3, 6, 9\}$, $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$, $\langle 1 \rangle = \mathbb{Z}/12$.

Lattice (larger subgroups higher):



(2) By Lagrange's theorem, the order of any subgroup divides p . So subgroups have order 1 or p , meaning $\{0\}$ or $\mathbb{Z}/p$.

(3) a generates $\mathbb{Z}/n$ iff $\text{ord}(a) = n$ iff $\gcd(a, n) = 1$. The count is $\varphi(n)$, Euler's totient.

(4) For any $a, b \in G$: $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$ (using $x^{-1} = x$ since $x^2 = e$). So G is abelian.

2.5 Homomorphisms

Definition 2.5.1: Homomorphism

A homomorphism $\varphi : G \rightarrow H$ between groups is a map satisfying

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

Proposition 2.5.1 Basic properties of homomorphisms

Let $\varphi : G \rightarrow H$ be a homomorphism.

1. $\varphi(e_G) = e_H$.
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all $a \in G$.
3. $\varphi(a^n) = \varphi(a)^n$ for all $n \in \mathbb{Z}$.
4. If $K \leq G$, then $\varphi(K) \leq H$.
5. If $L \leq H$, then $\varphi^{-1}(L) \leq G$.

Proof: Proofs of homomorphism properties

(1) $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)\varphi(e_G)$. Multiply both sides by $\varphi(e_G)^{-1}$: $e_H = \varphi(e_G)$.

(2) $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$. So $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

(3) For $n > 0$: induction. For $n = 0$: part (1). For $n < 0$: $\varphi(a^n) = \varphi((a^{-1})^{-n}) = \varphi(a^{-1})^{-n} = (\varphi(a)^{-1})^{-n} = \varphi(a)^n$.

(4) For $\varphi(a), \varphi(b) \in \varphi(K)$ (with $a, b \in K$): $\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(K)$ since $ab^{-1} \in K$.

(5) For $a, b \in \varphi^{-1}(L)$: $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L$ (since $L \leq H$), so $ab^{-1} \in \varphi^{-1}(L)$. $\square$

Note:-

The homomorphism condition can be expressed as a commutative diagram: the two paths $G \times G \xrightarrow{m_G} G \xrightarrow{\varphi} H$ and $G \times G \xrightarrow{\varphi \times \varphi} H \times H \xrightarrow{m_H} H$ give the same result.

$$\begin{array}{ccc} G \times G & \xrightarrow{\varphi \times \varphi} & H \times H \\ m_G \downarrow & & \downarrow m_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

This diagrammatic perspective is fundamental to category theory.

Definition 2.5.2: Types of homomorphisms

Let $\varphi : G \rightarrow H$ be a homomorphism.

1. φ is a **monomorphism** (or **embedding**) if it's injective.
2. φ is an **epimorphism** if it's surjective.
3. φ is an **isomorphism** if it's bijective. We write $G \cong H$.
4. An isomorphism $G \rightarrow G$ is called an **automorphism**. The set of all automorphisms of G , denoted $\text{Aut}(G)$, forms a group under composition.

Example 2.5.1 (Important homomorphisms)

1. **Reduction mod n** : $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $\pi(a) = a \bmod n$. Surjective, kernel $n\mathbb{Z}$.
2. **Quotient mod n** : If $n \mid m$, then $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $a \mapsto a \bmod n$. Example: last-two-digits to last-digit, $\mathbb{Z}/100 \rightarrow \mathbb{Z}/10$.
3. **Determinant**: $\det : \text{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}^*$. Homomorphism since $\det(AB) = \det(A)\det(B)$. Kernel is $\text{SL}_n(\mathbb{R})$.
4. **Exponential**: $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \times)$, $x \mapsto e^x$. Isomorphism with inverse $\ln$.
5. **Complex exponential**: $\exp : (\mathbb{R}, +) \rightarrow (S^1, \times)$, $t \mapsto e^{2\pi it}$. Surjective, kernel $\mathbb{Z}$. Induces $\mathbb{R}/\mathbb{Z} \cong S^1$.

2.5.1 Isomorphisms and invariants**Definition 2.5.3: Isomorphic groups**

Groups G and H are **isomorphic**, written $G \cong H$, if there exists an isomorphism $\varphi : G \rightarrow H$. Isomorphic groups are “the same” for all group-theoretic purposes.

What properties are preserved by isomorphisms? These are called invariants.

Proposition 2.5.2 Isomorphism invariants

If $G \cong H$, then:

1. $|G| = |H|$ (same order);
2. G is abelian iff H is abelian;
3. G is cyclic iff H is cyclic;

4. For each n , G and H have the same number of elements of order n ;
5. For each n , G and H have the same number of subgroups of order n .

Proof: Why these are invariants

Let $\varphi : G \xrightarrow{\sim} H$ be an isomorphism.

(1) Bijections preserve cardinality.

(2) If G is abelian and $h_1, h_2 \in H$, write $h_i = \varphi(g_i)$. Then $h_1 h_2 = \varphi(g_1) \varphi(g_2) = \varphi(g_1 g_2) = \varphi(g_2 g_1) = \varphi(g_2) \varphi(g_1) = h_2 h_1$.

(3) If $G = \langle a \rangle$, then $H = \varphi(G) = \varphi(\langle a \rangle) = \langle \varphi(a) \rangle$.

(4) $\text{ord}(\varphi(a)) = \text{ord}(a)$ since $\varphi(a)^n = \varphi(a^n) = e_H \iff a^n = e_G$.

(5) φ induces a bijection between subgroups of G and subgroups of H . □

Example 2.5.2 (Non-isomorphic groups of same order)

$\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ both have order 4, but:

- $\mathbb{Z}/4$ has an element of order 4 (namely, 1);
- $\mathbb{Z}/2 \times \mathbb{Z}/2$ has no element of order 4 (all non-identity elements have order 2).

So they're not isomorphic. These are the only groups of order 4 up to isomorphism.

Example 2.5.3 (All groups of order 2 are isomorphic)

If $|G| = 2$, write $G = \{e, x\}$. Then $x \neq e$, so $x^2 \neq x$ (else $x = e$). Thus $x^2 = e$, and $G = \langle x \rangle \cong \mathbb{Z}/2$.

Question 3

1. Prove that $\text{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
2. Find $\text{Aut}(\mathbb{Z})$.
3. Prove that if $\varphi : G \rightarrow H$ is an isomorphism, then $\varphi^{-1} : H \rightarrow G$ is also an isomorphism.
4. Prove that “is isomorphic to” is an equivalence relation on groups.

Solution

(1) An automorphism φ of $\mathbb{Z}/n$ is determined by $\varphi(1)$, since $\varphi(k) = k\varphi(1)$. We need $\varphi(1)$ to generate $\mathbb{Z}/n$, i.e., $\gcd(\varphi(1), n) = 1$, i.e., $\varphi(1) \in (\mathbb{Z}/n)^\times$. Conversely, any $a \in (\mathbb{Z}/n)^\times$ defines an automorphism $\varphi_a : k \mapsto ka$. The map $a \mapsto \varphi_a$ is an isomorphism $(\mathbb{Z}/n)^\times \rightarrow \text{Aut}(\mathbb{Z}/n)$ (check it's a homomorphism and bijection).

(2) An automorphism of $\mathbb{Z}$ is determined by $\varphi(1)$. We need $\langle \varphi(1) \rangle = \mathbb{Z}$, so $\varphi(1) = \pm 1$. Thus $\text{Aut}(\mathbb{Z}) = \{\text{id}, -\text{id}\} \cong \mathbb{Z}/2$.

(3) φ^{-1} is a bijection. For $h_1, h_2 \in H$, let $g_i = \varphi^{-1}(h_i)$. Then $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1) \varphi(g_2)) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1) \varphi^{-1}(h_2)$.

(4) Reflexive: $\text{id}_G : G \rightarrow G$ is an isomorphism. Symmetric: if $\varphi : G \rightarrow H$ is an isomorphism, so is $\varphi^{-1} : H \rightarrow G$ by part (3). Transitive: if $\varphi : G \rightarrow H$ and $\psi : H \rightarrow K$ are isomorphisms, so is $\psi \circ \varphi : G \rightarrow K$.

2.6 The order of an element and its properties

We've introduced order; now we develop its theory more fully.

Proposition 2.6.1 Powers and order

Let $a \in G$ have finite order n . Then:

1. $a^k = e \iff n \mid k$.
2. $a^i = a^j \iff n \mid (i - j)$.
3. The elements $e, a, a^2, \dots, a^{n-1}$ are all distinct.

Proof: Properties of order

(1) ($\Leftarrow$) If $k = mn$, then $a^k = (a^n)^m = e^m = e$. ($\Rightarrow$) Write $k = qn + r$ with $0 \leq r < n$. Then $e = a^k = a^{qn+r} = (a^n)^q a^r = a^r$. By minimality of n , $r = 0$, so $n \mid k$.

(2) $a^i = a^j \iff a^{i-j} = e \iff n \mid (i - j)$ by (1).

(3) If $a^i = a^j$ for $0 \leq i < j < n$, then $a^{j-i} = e$ with $0 < j - i < n$, contradicting minimality of n . $\square$

Proposition 2.6.2 Order of powers

Let $\text{ord}(a) = n$. Then $\text{ord}(a^k) = \frac{n}{\gcd(k, n)}$.

Proof: Order of powers

Let $d = \gcd(k, n)$. We show $\text{ord}(a^k) = n/d$.

First, $(a^k)^{n/d} = a^{kn/d} = (a^n)^{k/d} = e$, so $\text{ord}(a^k) \mid n/d$.

Conversely, let $m = \text{ord}(a^k)$. Then $a^{km} = e$, so $n \mid km$. Write $k = d \cdot k'$ and $n = d \cdot n'$ where $\gcd(k', n') = 1$. Then $dn' \mid dk'm$, so $n' \mid k'm$. Since $\gcd(k', n') = 1$, we have $n' \mid m$, i.e., $n/d \mid m$. Thus $m = n/d$. $\square$

Corollary 2.6.1 Generators of cyclic groups

In $\langle a \rangle$ of order n , the element a^k is a generator iff $\gcd(k, n) = 1$.

Example 2.6.1

In $\mathbb{Z}/12$, the generators are 1, 5, 7, 11 (the four elements coprime to 12). The element 4 has order $12/\gcd(4, 12) = 12/4 = 3$.

Question 4

1. In a group G , if $a^{12} = e$ and $a^8 = e$, what are the possible orders of a ?
2. Prove: if $\text{ord}(a) = n$ and $\text{ord}(b) = m$ with $\gcd(n, m) = 1$ and $ab = ba$, then $\text{ord}(ab) = nm$.
3. Find the order of $(2, 3)$ in $\mathbb{Z}/6 \times \mathbb{Z}/8$.

Solution

(1) $\text{ord}(a)$ divides both 12 and 8, so $\text{ord}(a) \mid \gcd(12, 8) = 4$. Possible orders: 1, 2, 4.

(2) Let $k = \text{ord}(ab)$. Then $e = (ab)^{nm} = a^{nm}b^{nm} = (a^n)^m(b^m)^n = e$ (using commutativity), so $k \mid nm$.

Now $e = (ab)^{kn} = a^{kn}b^{kn} = b^{kn}$ (since $a^n = e$), so $m \mid kn$. Since $\gcd(m, n) = 1$, $m \mid k$. Similarly, $n \mid k$. Since $\gcd(m, n) = 1$, $mn \mid k$. Thus $k = nm$.

(3) $\text{ord}(2, 3) = \text{lcm}(\text{ord}(2), \text{ord}(3)) = \text{lcm}(6/\gcd(2, 6), 8/\gcd(3, 8)) = \text{lcm}(3, 8) = 24$.

Group Relations

We say that two sets have the same **cardinality** if $\exists f : X \hookrightarrow Y$, and we write $|X| \leq |Y|$. This allows us to inter-relate cardinality and function relations via a special theorem.

Theorem 3.0.1 Schröder-Berstein Theorem

If $\exists f : X \hookrightarrow Y$ and $g : Y \hookrightarrow X$ then $|X| = |Y|$.

The Schröder-Berstein theorem provides some immediate results for allowing explicit bijections.

Proposition 3.0.1 Cardinality of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ to any power

Using Schröder-Berstein we can create explicit bijections $f : \mathbb{Z} \leftrightarrow \mathbb{N}$ given by $n \mapsto$

$$\begin{cases} 2n & \text{if } n \geq 0; \\ -(2n+1) & \text{if } n < 0. \end{cases}$$

Proposition 3.0.2 $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}| = \aleph_0$

We create the explicit bijection via $f : \mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$, $(k, l) \mapsto 2^k(2l+1) - 1$. It follows then that $\aleph_0^a, \aleph_0^b, \aleph_0^c, \forall a, b, c \in \mathbb{N}$ permits a similar bijection.

Proof: Proof of Prop. 3.1.2

Notice that any natural number can be written as $2^k(m)$ where m is odd, i.e., $m = 2n+1$. A bijection is automatically created here with $2^k(2l+1)$, and the additional -1 at the end of the statement is there to shift the entire bijection down one. So instead of $(0, 0) \mapsto 2^0(2(0)+1) = 1$ we get $(0, 0) \mapsto 2^0(2(0)+1) - 1 = 0$. $\square$

Proposition 3.0.3 Countable vs uncountable infinities

$|\mathbb{R}| > |\mathbb{N}|$.

Proof: Proof by Cantor

Given by Cantor's diagonal argument, no map $f : \mathbb{N} \rightarrow \mathbb{R}$ be surjective, because:

$$\begin{cases} f(0) &= a_{00}a_{01}a_{02}a_{03}\dots \\ f(1) &= a_{10}a_{11}a_{12}a_{13}\dots \\ f(2) &= a_{20}a_{21}a_{22}a_{23}\dots \end{cases}$$

then let $y = b_0b_1b_2b_3\dots$ where $b_j \neq a_{jj}$ for each j . Now look at the j^{th} digit, $y \neq f(j), \forall j \in \mathbb{N}$, so f can't be surjective. $\square$

Proposition 3.0.4 Cardinality of mappings

$|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$

Proof: Proof by Cantor's diagonal used for mappings

We can use the fact that $|(0,1)| = |\mathbb{R}|$ in addition to Cantor's diagonal argument as a decimal expansion— $t = 0.t_1t_2t_3\dots$ —with t_i as an integer between 0 and 9. So now to show $\exists f : (0,1) \leftrightarrow (0,1)^{\mathbb{N}}$, we create some $s_j \in (0,1)^{\mathbb{N}}$ and construct the bijection via:

$$\begin{cases} s_1 &= 0.t_1t_3t_5\dots \\ s_2 &= 0.t_2t_6t_{10}\dots \\ s_3 &= 0.t_4t_{12}t_{20}\dots \end{cases}$$

We say that s_1 is made up of all t_i that are odd; s_2 is made up of all i which are odd such that $i \cdot 2 \nmid i \in s_2$; s_3 is made up of all i which are odd such that $i \cdot 2^2 \nmid i \in s_3$, $i \cdot 2^3 \in s_3$ and so on (the general form being $i \cdot 2^n \mid n = j - 1$). Therefore there exists a bijection between $(0,1) \leftrightarrow (0,1)^{\mathbb{N}}$. $\square$

Proposition 3.0.5 There does not exist a largest cardinality

Let $\mathcal{P}(S)$ be the standard powerset. It follows then that $\mathcal{P}(S) = \{0,1\}^S$ and $|\mathcal{P}(S)| > |S|$ for any S .

Proof: Proof of non-existence of largest cardinality

Recall that $\{0,1\}^S = \{\text{all } f \text{ such that } f : S \rightarrow \{0,1\}\}$. There must exist a bijection since $|\mathcal{P}(S)| = 2^{|S|}$ and $|\{0,1\}^S| = 2^{|S|}$. Now to show $|\mathcal{P}(S)| > |S|$ we use proof by contradiction: suppose $\exists \alpha : S \leftrightarrow \mathcal{P}(S)$ and let $A \subset S$ defined by $A := \{s \in S \mid s \neq \alpha(s)\}$, now for $A = \alpha(t)$ for some $t \in A$ we notice that $t \notin \alpha(t) = A$ means that $t \in A$, but to say $t \in \alpha(t) \implies t \notin A$, thus we have a contradiction. $\square$

Another option for this proof, though less formal, can be found by showing that $\mathcal{P}(S) = \{0,1\}^S$ and that subsets of S have f such that $f \mapsto f^{-1}$; let $A \subset S$ then

$$A \mapsto \left(\chi_A : x \mapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} \right)$$

Note that if S is finite then $|S| = n$ and $\mathcal{P}(S) = 2^n$. This naturally leads us into our next theorem regarding the infinitude of the cardinality of S in relation to $\mathcal{P}(S)$.

3.1 Quotient groups

Consider $\mathbb{Z}/n$. Under the equivalence relation congruence mod n we can take the sum of the various subsets that mod n partitions $\mathbb{Z}$ into as the law of composition given by $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = a + b + n\mathbb{Z}$. Notice that we can let T be the set of equivalence classes under mod n , and thus we have a homomorphic map from $\mathbb{Z} \rightarrow T$. Finally, $T = \mathbb{Z}/n$.

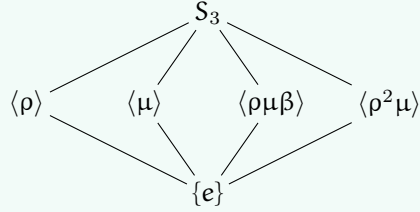
Why stop at integers? What if we want to make congruence mod H where H is a subgroup $H \subset G$? Under what cases, given that we can make mod H equivalence classes (we will define this in a bit) is the quotient map $G \rightarrow T$ a homomorphism? This question is the question of quotient groups, and its answer—normality—is of note and importance to much of group theory (i.e. it is foundational material).

Example 3.1.1 (Quotient subgroups of S_3)

Note that S_3 is isomorphic to $\mathbb{Z}/6$, thus we may think of the subgroups as such:

$$S_3 \cong \mathbb{Z}/6 \cong \mathbb{Z}/3 \times \mathbb{Z}/2$$

and since $\mathbb{Z}/6$'s elements (03) form $\mathbb{Z}/2$ with $\{e, \mu\} \mapsto (03) \mapsto \{\pm 1\}$, we have similar happening for (024) .

**3.1.1 Congruence mod H and cosets**

For any two elements $a, b \in G$ we define $a \sim b \iff ab^{-1} \in H$, this is to say that $a \in bH$. We thus have $a \sim a$ since $e \in H$, $a \sim b \implies ab^{-1} \in H$, and $ba^{-1} = (ab^{-1})^{-1} \in H$, thus $b \sim a$. Finally, $a \sim b, b \sim c$ meaning that ab^{-1} and bc^{-1} are both in H , so $ac^{-1} = (ab^{-1})(bc^{-1}) \in H$ since H is closed under the group operation, thus $a \sim c$ and transitivity holds. We have shown congruence mod H is indeed an equivalence relation.

Definition 3.1.1: Cosets

Given a subgroup $H \subset G$ and an element $a \in G$, the **left coset** of H containing a is the set $aH = \{ah \mid h \in H\}$. Similarly, the **right coset** is $Ha = \{ha \mid h \in H\}$. The equivalence classes under congruence mod H are precisely the left cosets of H .

Proposition 3.1.1 Cosets partition G

The left cosets of H form a partition of G . Every element $g \in G$ belongs to exactly one coset, namely gH , and two cosets aH, bH are either identical or disjoint.

Proof: Proof of coset partition

This follows immediately from the fact that cosets are equivalence classes: $g \in gH$ since $g = ge$ and $e \in H$. For disjointness, suppose $aH \cap bH \neq \emptyset$, so there exists $c \in aH \cap bH$. Then $c = ah_1 = bh_2$ for some $h_1, h_2 \in H$, which gives $a = bh_2h_1^{-1} \in bH$. For any $ah \in aH$, we have $ah = bh_2h_1^{-1}h \in bH$, so $aH \subseteq bH$. By symmetry $bH \subseteq aH$, thus $aH = bH$. $\square$

Here is the key observation that makes quotient theory work: all cosets have the same size!

Proposition 3.1.2 Cosets are equinumerous

For any $a \in G$, the map $\lambda_a : H \rightarrow aH$ given by $h \mapsto ah$ is a bijection. In particular, $|aH| = |H|$ for all $a \in G$.

Proof: Proof of coset bijection

Surjectivity is immediate from the definition of aH . For injectivity, suppose $ah_1 = ah_2$. Left-multiplying by a^{-1} gives $h_1 = h_2$. (This is just left-translation, which is always a bijection in a group—a fact we will revisit when discussing group actions.) $\square$

Theorem 3.1.1 Lagrange's Theorem

Let G be a finite group and $H \subset G$ a subgroup. Then $|H|$ divides $|G|$, and the quotient $|G|/|H| = [G : H]$ equals the number of distinct left cosets of H in G .

Proof: Proof of Lagrange's Theorem

The left cosets of H partition G into disjoint subsets, each of cardinality $|H|$. If there are k distinct cosets, then $|G| = k \cdot |H|$, so $|H| \mid |G|$ and $k = [G : H]$. $\square$

Definition 3.1.2: Index

The **index** of a subgroup H in G , denoted $[G : H]$, is the number of distinct left cosets of H in G . For finite groups, $[G : H] = |G|/|H|$.

Note:-

Lagrange's theorem has immediate corollaries: the order of any element divides $|G|$ (since $\langle g \rangle$ is a subgroup), and groups of prime order are cyclic (the only subgroups have order 1 or p). The converse of Lagrange is false in general: A_4 has order 12 but no subgroup of order 6. However, for abelian groups and more generally for solvable groups, partial converses exist—a theme we'll explore later.

A question naturally arises then: when is G/H a group? If we try to define $(aH)(bH) = abH$, we need this to be well-defined, meaning if $aH = a'H$ and $bH = b'H$, then $abH = a'b'H$. This does not always work!

Example 3.1.2

Consider $G = S_3$ and $H = \{e, (12)\}$. We have three left cosets: $H = \{e, (12)\}$, $(13)H = \{(13), (132)\}$, and $(23)H = \{(23), (123)\}$. Take representatives $a = (13)$ and $a' = (132)$ from the same coset; both satisfy $aH = a'H = (13)H$. Now take $b = (23)$. We compute:

$$\begin{aligned} ab &= (13)(23) = (132), & \text{so } abH &= (132)H = (13)H \\ a'b &= (132)(23) = (13), & \text{so } a'bH &= (13)H \end{aligned}$$

Okay, this worked! But now try $b = (13)$ and $b' = (132)$ (same coset), with $a = (23)$:

$$\begin{aligned} ab &= (23)(13) = (123), & \text{so } abH &= (123)H = (23)H \\ ab' &= (23)(132) = (12), & \text{so } ab'H &= (12)H = H \end{aligned}$$

Disaster! We got $(23)H \neq H$, so coset multiplication is not well-defined for this H .

What went wrong? The issue is that $H = \{e, (12)\}$ is not “symmetric” enough—left cosets and right cosets differ. For instance, $(13)H = \{(13), (132)\}$ but $H(13) = \{(13), (123)\}$.

3.1.2 Normal subgroups and quotient groups

Definition 3.1.3: Normal subgroup

A subgroup $N \subset G$ is **normal** (written $N \trianglelefteq G$) if any of the following equivalent conditions hold:

1. $gNg^{-1} = N$ for all $g \in G$ (conjugation preserves N);
2. $gN = Ng$ for all $g \in G$ (left cosets equal right cosets);
3. $gNg^{-1} \subseteq N$ for all $g \in G$ (one inclusion suffices);
4. N is the kernel of some homomorphism $\varphi : G \rightarrow H$.

Proof: Equivalence of normality conditions

(1) $\Rightarrow$ (2): If $gNg^{-1} = N$, then $gN = Ng$ by right-multiplying by g .

(2) $\Rightarrow$ (3): If $gN = Ng$, then for $n \in N$, we have $gn \in gN = Ng$, so $gn = n'g$ for some $n' \in N$, giving $gng^{-1} = n' \in N$.

(3) $\Rightarrow$ (1): We have $gNg^{-1} \subseteq N$ by assumption. For the reverse, take $n \in N$. We want $n \in gNg^{-1}$, i.e., $g^{-1}ng \in N$. But applying (3) with g^{-1} : $g^{-1}N(g^{-1})^{-1} = g^{-1}Ng \subseteq N$, so $g^{-1}ng \in N$.

(4) $\Rightarrow$ (1): If $N = \ker(\varphi)$, then for $n \in N$ and $g \in G$: $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H$, so $gng^{-1} \in \ker(\varphi) = N$.

(1) $\Rightarrow$ (4): This is the content of the next theorem—every normal subgroup is a kernel! $\square$

Theorem 3.1.2 Quotient groups exist for normal subgroups

If $N \trianglelefteq G$, then G/N is a group under the operation $(aN)(bN) = abN$, called the **quotient group**. The map $\pi : G \rightarrow G/N$ given by $\pi(g) = gN$ is a surjective homomorphism (the **canonical projection**) with $\ker(\pi) = N$.

Proof: Proof of quotient group structure

Well-definedness: Suppose $aN = a'N$ and $bN = b'N$, so $a' = an_1$ and $b' = bn_2$ for some $n_1, n_2 \in N$. Then

$$a'b' = an_1bn_2 = a(n_1b)n_2.$$

Since N is normal, $n_1b = bn_3$ for some $n_3 \in N$ (because $bN = Nb$ implies $n_1 \in Nb$, so $n_1 = bn_3$... wait, let me be more careful). We have $n_1b \in Nb = bN$, so $n_1b = bn_3$ for some $n_3 \in N$. Thus $a'b' = abn_3n_2 \in abN$, so $a'b'N = abN$.

Group axioms: Associativity: $(aN \cdot bN) \cdot cN = abN \cdot cN = (ab)cN = a(bc)N = aN \cdot bcN = aN \cdot (bN \cdot cN)$. Identity: $eN = N$ satisfies $N \cdot aN = eaN = aN$. Inverses: $(aN)^{-1} = a^{-1}N$ since $aN \cdot a^{-1}N = aa^{-1}N = N$.

The projection: $\pi(ab) = abN = aN \cdot bN = \pi(a)\pi(b)$, so π is a homomorphism. Surjectivity is clear. Finally, $\ker(\pi) = \{g \in G : gN = N\} = \{g \in G : g \in N\} = N$. $\square$

Question 1

Let G be a group and $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ the center of G .

1. Prove that $Z(G) \trianglelefteq G$.
2. Prove that if $G/Z(G)$ is cyclic, then G is abelian.
3. Find $Z(D_4)$ and describe $D_4/Z(D_4)$.

Solution

(1) We verify normality via condition (1). For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = zgg^{-1} = z \in Z(G)$ (using that z commutes with everything). Thus $gZ(G)g^{-1} \subseteq Z(G)$ for all g , so $Z(G) \trianglelefteq G$.

(2) Suppose $G/Z(G) = \langle aZ(G) \rangle$ is cyclic. Then every element of G has the form a^kz for some $k \in \mathbb{Z}$ and $z \in Z(G)$. Take any two elements $g = a^i z_1$ and $h = a^j z_2$. Then:

$$gh = a^i z_1 a^j z_2 = a^i a^j z_1 z_2 = a^{i+j} z_1 z_2$$

(since z_1 commutes with a^j and z_2), and similarly $hg = a^{j+i} z_2 z_1 = a^{i+j} z_1 z_2 = gh$ (since z_1, z_2 commute with everything and with each other). Thus G is abelian.

(3) Recall $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. For $z \in Z(D_4)$, we need $zr = rz$ and $zs = sz$.

For rotations: $r^k \in Z(D_4)$ requires $sr^k = r^k s$. We have $sr^k = r^{-k}s$ (by induction using $srs = r^{-1}$), so $r^{-k}s = r^k s$, giving $r^{2k} = e$, so $k \in \{0, 2\}$. Thus e, r^2 are central among rotations.

For reflections: $sr^j \in Z(D_4)$ requires $r(sr^j) = (sr^j)r$, i.e., $rsr^j = sr^{j+1}$. But $rsr^j = r \cdot r^{-j}s = r^{1-j}s$,

so we need $r^{1-j}s = sr^{j+1}$, giving $r^{1-j} = r^{-(j+1)} = r^{-j-1}$, so $r^2 = e$, contradiction since r has order 4. No reflections are central. Therefore $Z(D_4) = \{e, r^2\} \cong \mathbb{Z}/2$. The quotient $D_4/Z(D_4)$ has order $8/2 = 4$. The cosets are $\{e, r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\}$. One can verify this is $\cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group), not $\mathbb{Z}/4$, since every non-identity element has order 2 in the quotient.

As a final remark before moving onto our next topic, I wish to introduce the concept of simple groups. Simply put, no pun intended, a simple group is a group that cannot be decomposed into further normal subgroups other than the trivial subgroup and itself. Formally:

Definition 3.1.4: Simple groups

We say a group is simple if it has no normal subgroups other than itself and the trivial subgroup.

3.2 Group actions and symmetry

The notion of a group “acting” on a set is perhaps the single most important concept connecting abstract algebra to the rest of mathematics. When we study symmetries of geometric objects, solutions to polynomial equations, or even the structure of groups themselves, we are studying group actions. This section develops the theory systematically—pay close attention, as these ideas will reappear throughout when we discuss representation theory (where groups act on vector spaces via linear maps).

3.2.1 Group actions

Definition 3.2.1: Group action

A (left) group action of a group G on a set X is a map $G \times X \rightarrow X$, written $(g, x) \mapsto g \cdot x$, satisfying:

1. $e \cdot x = x$ for all $x \in X$ (identity acts trivially);
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$ (compatibility).

We say G acts on X , or that X is a G -set.

Note:-

There is an equivalent, perhaps more illuminating, formulation: a group action is a homomorphism $\rho : G \rightarrow \text{Sym}(X)$, where $\text{Sym}(X)$ is the group of all bijections $X \rightarrow X$. Given an action, define $\rho(g)(x) = g \cdot x$; the axioms ensure each $\rho(g)$ is a bijection (with inverse $\rho(g^{-1})$) and that ρ is a homomorphism. Conversely, any such homomorphism defines an action. This perspective is crucial: group actions are the same as representations of G into symmetric groups.

Example 3.2.1

Some fundamental examples of group actions:

1. **Left multiplication:** G acts on itself by $g \cdot h = gh$. This action is always faithful (only e acts trivially) and transitive (any element can be sent to any other).
2. **Conjugation:** G acts on itself by $g \cdot h = ghg^{-1}$. The kernel is $Z(G)$, the center.
3. **Conjugation on subgroups:** G acts on the set of its subgroups by $g \cdot H = gHg^{-1}$. Normal subgroups are precisely the fixed points!
4. **Coset action:** If $H \subset G$, then G acts on G/H (left cosets) by $g \cdot (aH) = (ga)H$.

5. **Linear actions:** $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by matrix multiplication. This is the prototype for representation theory.

Definition 3.2.2: Orbits and stabilizers

Let G act on X . For $x \in X$:

1. The **orbit** of x is $G \cdot x = \{g \cdot x : g \in G\} \subseteq X$.
2. The **stabilizer** (or **isotropy group**) of x is $G_x = \text{Stab}_G(x) = \{g \in G : g \cdot x = x\} \subseteq G$.

Proposition 3.2.1 Stabilizers are subgroups

For any $x \in X$, the stabilizer G_x is a subgroup of G .

Proof: Proof that stabilizers are subgroups

Identity: $e \cdot x = x$, so $e \in G_x$. Closure: if $g, h \in G_x$, then $(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$, so $gh \in G_x$. Inverses: if $g \in G_x$, then $x = e \cdot x = (g^{-1}g) \cdot x = g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x$, so $g^{-1} \in G_x$. $\square$

Proposition 3.2.2 Orbits partition X

The orbits of a group action form a partition of X . Define $x \sim y$ iff $y = g \cdot x$ for some $g \in G$; this is an equivalence relation whose equivalence classes are exactly the orbits.

Proof: Proof that orbits partition

Reflexivity: $x = e \cdot x$, so $x \sim x$. Symmetry: if $y = g \cdot x$, then $x = g^{-1} \cdot y$, so $y \sim x$ implies $x \sim y$. Transitivity: if $y = g \cdot x$ and $z = h \cdot y$, then $z = h \cdot (g \cdot x) = (hg) \cdot x$, so $x \sim z$. $\square$

The following theorem is one of the most useful results in all of group theory. It connects the size of an orbit to the index of a stabilizer.

Theorem 3.2.1 Orbit-Stabilizer Theorem

Let G act on X and let $x \in X$. There is a bijection between the orbit $G \cdot x$ and the set of left cosets G/G_x , given by $g \cdot x \mapsto gG_x$. In particular, if G is finite,

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|}.$$

Proof: Proof of Orbit-Stabilizer

Define $\varphi : G/G_x \rightarrow G \cdot x$ by $\varphi(gG_x) = g \cdot x$.

Well-defined: If $gG_x = hG_x$, then $h = gk$ for some $k \in G_x$, so $h \cdot x = (gk) \cdot x = g \cdot (k \cdot x) = g \cdot x$.

Injective: If $g \cdot x = h \cdot x$, then $h^{-1}g \cdot x = x$, so $h^{-1}g \in G_x$, giving $gG_x = hG_x$.

Surjective: Every element of $G \cdot x$ has the form $g \cdot x = \varphi(gG_x)$. $\square$

Example 3.2.2

Consider S_4 acting on $\{1, 2, 3, 4\}$ in the natural way. Take $x = 1$. The orbit is $S_4 \cdot 1 = \{1, 2, 3, 4\}$ (any element can be sent to any other). The stabilizer is $(S_4)_1 = \{\sigma \in S_4 : \sigma(1) = 1\}$, which consists of all permutations of $\{2, 3, 4\}$, so $(S_4)_1 \cong S_3$ and $|(S_4)_1| = 6$. Check: $|S_4 \cdot 1| = 4 = 24/6 = |S_4|/|(S_4)_1|$. $\checkmark$

3.2.2 Cayley's theorem and applications

Theorem 3.2.2 Cayley's Theorem

Every group G is isomorphic to a subgroup of $\text{Sym}(G)$. If $|G| = n$, then G embeds into S_n .

Proof: Proof of Cayley's Theorem

Consider the left multiplication action of G on itself: $g \cdot h = gh$. This gives a homomorphism $\rho : G \rightarrow \text{Sym}(G)$ where $\rho(g)$ is the bijection $h \mapsto gh$.

Injectivity: If $\rho(g) = \rho(g')$, then for all $h \in G$, $gh = g'h$. Taking $h = e$: $g = g'$. Thus $\ker(\rho) = \{e\}$, so ρ is injective.

By the First Isomorphism Theorem, $G \cong \text{Im}(\rho) \subset \text{Sym}(G)$. If $|G| = n$, then $\text{Sym}(G) \cong S_n$. $\square$

Note:-

Cayley's theorem is simultaneously profound and useless. Profound: every abstract group is “really” a permutation group; the group axioms capture exactly the structure of bijection composition. Useless: the embedding is almost always far larger than necessary. S_n has order $n!$, so embedding a group of order n into S_n wastes enormous space. Better embeddings come from finding smaller faithful actions.

Proposition 3.2.3 Improved Cayley embedding

Let $H \subset G$ with $[G : H] = n$. The action of G on G/H gives a homomorphism $\rho : G \rightarrow S_n$ with $\ker(\rho) = \bigcap_{g \in G} gHg^{-1}$, the largest normal subgroup of G contained in H .

Proof: Proof of improved embedding

The action $g \cdot (aH) = (ga)H$ is well-defined (check!). The kernel consists of all g fixing every coset: $gaH = aH$ for all a , i.e., $a^{-1}ga \in H$ for all a , i.e., $g \in aHa^{-1}$ for all a . Thus $\ker(\rho) = \bigcap_{a \in G} aHa^{-1}$. $\square$

Example 3.2.3

Let $G = S_4$ and $H = S_3$ (permutations fixing 4). Then $[G : H] = 4$, and the action on G/H identifies with the natural action on $\{1, 2, 3, 4\}$. The kernel is trivial (since the natural action is faithful), recovering the identity $S_4 \hookrightarrow S_4$. But: if we take $G = A_5$ and $H = A_4$ (stabilizer of a point), then $[G : H] = 5$ and we get $A_5 \hookrightarrow S_5$. Since $|A_5| = 60 < 120 = |S_5|$, this is a non-surjective embedding, showing A_5 as a proper subgroup of S_5 .

Question 2

Let G be a group of order $2p$ where p is an odd prime.

1. Prove that G has a unique subgroup of order p , which is therefore normal.
2. Conclude that $G \cong \mathbb{Z}/2p$ or $G \cong D_p$ (the dihedral group of order $2p$).

Solution

(1) By Cauchy's theorem (or direct counting), G has an element a of order p , so $H = \langle a \rangle$ is a subgroup of order p . Since $[G : H] = 2$, we have $H \trianglelefteq G$ (index 2 subgroups are always normal: $G = H \sqcup gH = H \sqcup Hg$ for any $g \notin H$, so $gH = Hg$).

For uniqueness: suppose K is another subgroup of order p . Since $|H| = |K| = p$ is prime, $H \cap K$ is a subgroup of both with order dividing p , so $|H \cap K| \in \{1, p\}$. If $|H \cap K| = p$, then $H = K$. Otherwise, $H \cap K = \{e\}$, and since elements of H and K (excluding e) have order p , we have $|H \cup K| = 1 + (p-1) + (p-1) = 2p-1$ elements of orders 1 or p . But then G has only one element not in $H \cup K$, call it b . We must have $b^2 \in H \cup K$ (since $b \notin H \cup K$), but b has order dividing $2p$... this gets complicated.

Easier: if $H \neq K$, then HK is a subgroup (since $H \trianglelefteq G$) of order $|H||K|/|H \cap K| = p^2 > 2p = |G|$, contradiction.

(2) Let $H = \langle a \rangle \cong \mathbb{Z}/p$ be the unique subgroup of order p . Pick $b \in G \setminus H$; then b has order 2 (if $|b| = p$, then $b \in H$ by uniqueness; if $|b| = 2p$, then $G = \langle b \rangle$ is cyclic and $G \cong \mathbb{Z}/2p$).

Assume $|b| = 2$. Since $H \trianglelefteq G$, conjugation by b gives an automorphism of $H \cong \mathbb{Z}/p$. We have $bab^{-1} = a^k$ for some k . Since $b^2 = e$: $a = b^2ab^{-2} = b(bab^{-1})b^{-1} = ba^kb^{-1} = (bab^{-1})^k = a^{k^2}$, so $k^2 \equiv 1 \pmod{p}$, giving $k \equiv \pm 1 \pmod{p}$.

If $k \equiv 1$: then $bab^{-1} = a$, so $ab = ba$, and $G = \langle a, b \rangle$ is abelian of order $2p$ with $|a| = p, |b| = 2$, so $G \cong \mathbb{Z}/p \times \mathbb{Z}/2 \cong \mathbb{Z}/2p$.

If $k \equiv -1$: then $bab^{-1} = a^{-1}$, i.e., $ba = a^{-1}b$. This is exactly the dihedral relation! We have $G = \langle a, b : a^p = b^2 = e, bab^{-1} = a^{-1} \rangle \cong D_p$.

3.3 The isomorphism theorems

The isomorphism theorems are the structural backbone of group theory. They describe how quotients, subgroups, and homomorphisms interact. These results will reappear essentially unchanged when we study rings, modules, and vector spaces—they are instances of a more general phenomenon that category theory makes precise.

3.3.1 Kernels, images, and the First Isomorphism Theorem

Definition 3.3.1: Kernel

The kernel of a group homomorphism $\varphi : G \rightarrow H$ is

$$\ker(\varphi) = \{g \in G : \varphi(g) = e_H\} = \varphi^{-1}(\{e_H\}).$$

Definition 3.3.2: Image

The image of a group homomorphism $\varphi : G \rightarrow H$ is

$$\text{Im}(\varphi) = \varphi(G) = \{h \in H : \exists g \in G, \varphi(g) = h\}.$$

Proposition 3.3.1 Basic properties of kernels and images

Let $\varphi : G \rightarrow H$ be a homomorphism.

1. $\ker(\varphi)$ is a normal subgroup of G .
2. $\text{Im}(\varphi)$ is a subgroup of H (not necessarily normal).
3. φ is injective $\iff \ker(\varphi) = \{e_G\}$.
4. $\varphi(g_1) = \varphi(g_2) \iff g_1^{-1}g_2 \in \ker(\varphi) \iff g_1 \ker(\varphi) = g_2 \ker(\varphi)$.

Proof: Proofs of kernel/image properties

(1) First, $\ker(\varphi)$ is a subgroup: $\varphi(e) = e$, so $e \in \ker(\varphi)$; if $\varphi(a) = \varphi(b) = e$, then $\varphi(ab) = ee = e$; if $\varphi(a) = e$, then $\varphi(a^{-1}) = \varphi(a)^{-1} = e^{-1} = e$. For normality: if $k \in \ker(\varphi)$, then $\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g) \cdot e \cdot \varphi(g)^{-1} = e$.

(2) Subgroup: $e = \varphi(e) \in \text{Im}(\varphi)$; if $\varphi(a), \varphi(b) \in \text{Im}(\varphi)$, then $\varphi(a)\varphi(b) = \varphi(ab) \in \text{Im}(\varphi)$; $\varphi(a)^{-1} = \varphi(a^{-1}) \in \text{Im}(\varphi)$.

(3) If $\ker(\varphi) = \{e\}$ and $\varphi(g_1) = \varphi(g_2)$, then $\varphi(g_1^{-1}g_2) = e$, so $g_1^{-1}g_2 = e$, giving $g_1 = g_2$. Conversely, if φ is injective and $k \in \ker(\varphi)$, then $\varphi(k) = e = \varphi(e)$, so $k = e$.

(4) $\varphi(g_1) = \varphi(g_2) \iff \varphi(g_1)^{-1}\varphi(g_2) = e \iff \varphi(g_1^{-1}g_2) = e \iff g_1^{-1}g_2 \in \ker(\varphi)$. The last equivalence with cosets is the definition of coset equality. $\square$

Theorem 3.3.1 First Isomorphism Theorem

Let $\varphi : G \rightarrow H$ be a homomorphism. Then:

1. $\ker(\varphi) \trianglelefteq G$;
2. The map $\bar{\varphi} : G/\ker(\varphi) \rightarrow H$ defined by $\bar{\varphi}(g\ker(\varphi)) = \varphi(g)$ is a well-defined injective homomorphism;
3. $\text{Im}(\bar{\varphi}) = \text{Im}(\varphi)$, so $G/\ker(\varphi) \cong \text{Im}(\varphi)$.

In slogan form: the image of a homomorphism is isomorphic to the domain modulo the kernel.

Proof: Proof of First Isomorphism Theorem

Part (1) was proved above. For (2): $\bar{\varphi}$ is well-defined because $g_1\ker(\varphi) = g_2\ker(\varphi)$ implies $\varphi(g_1) = \varphi(g_2)$ by property (4). It's a homomorphism: $\bar{\varphi}(g_1K \cdot g_2K) = \bar{\varphi}(g_1g_2K) = \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = \bar{\varphi}(g_1K)\bar{\varphi}(g_2K)$ (writing $K = \ker(\varphi)$). It's injective: $\bar{\varphi}(gK) = e \implies \varphi(g) = e \implies g \in K \implies gK = K$.

For (3): $\text{Im}(\bar{\varphi}) = \{\bar{\varphi}(gK) : g \in G\} = \{\varphi(g) : g \in G\} = \text{Im}(\varphi)$. Since $\bar{\varphi}$ is an injective homomorphism onto $\text{Im}(\varphi)$, it's an isomorphism $G/\ker(\varphi) \xrightarrow{\sim} \text{Im}(\varphi)$. $\square$

Note:-

The First Isomorphism Theorem is sometimes called the “fundamental theorem of homomorphisms.” It says that every homomorphism $\varphi : G \rightarrow H$ factors as $G \twoheadrightarrow G/\ker(\varphi) \xrightarrow{\sim} \text{Im}(\varphi) \hookrightarrow H$: a surjection, followed by an isomorphism, followed by an inclusion. This factorization is unique and natural—in the categorical sense, which you'll see later.

3.3.2 The Second and Third Isomorphism Theorems

Theorem 3.3.2 Second Isomorphism Theorem (Diamond Theorem)

Let G be a group, $H \subset G$ a subgroup, and $N \trianglelefteq G$ a normal subgroup. Then:

1. $HN = \{hn : h \in H, n \in N\}$ is a subgroup of G ;
2. $H \cap N \trianglelefteq H$;
3. $H/(H \cap N) \cong HN/N$.

Proof: Proof of Second Isomorphism Theorem

(1) HN is a subgroup: $e = e \cdot e \in HN$. For closure: $(h_1n_1)(h_2n_2) = h_1(n_1h_2)n_2$. Since N is normal, $n_1h_2 = h_2n_3$ for some $n_3 \in N$, so $(h_1n_1)(h_2n_2) = h_1h_2n_3n_2 \in HN$. For inverses: $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$ since $hn^{-1}h^{-1} \in N$ by normality.

(2) $H \cap N$ is a subgroup of H (intersection of subgroups). For normality in H : if $x \in H \cap N$ and $h \in H$, then $h x h^{-1} \in H$ (since H is a subgroup) and $h x h^{-1} \in N$ (since $N \trianglelefteq G$), so $h x h^{-1} \in H \cap N$.

(3) Define $\varphi : H \rightarrow HN/N$ by $\varphi(h) = hN$. This is a homomorphism: $\varphi(h_1h_2) = h_1h_2N = (h_1N)(h_2N) = \varphi(h_1)\varphi(h_2)$.

Surjectivity: Any element of HN/N has the form $hnN = hN$ (since $n \in N$), which equals $\varphi(h)$.

Kernel: $\ker(\varphi) = \{h \in H : hN = N\} = \{h \in H : h \in N\} = H \cap N$.

By the First Isomorphism Theorem, $H/(H \cap N) = H/\ker(\varphi) \cong \text{Im}(\varphi) = HN/N$. $\square$

Note:-

The name “Diamond Theorem” comes from the lattice diagram: draw G at top, HN below it, H and N below HN on left and right, and $H \cap N$ at the bottom. The isomorphism $H/(H \cap N) \cong HN/N$ says the “left edge” quotient equals the “right edge” quotient. This is a manifestation of a general principle called the modular law in lattice theory.

Theorem 3.3.3 Third Isomorphism Theorem

Let G be a group and $N \subset M$ be normal subgroups of G (so $N \trianglelefteq G$ and $M \trianglelefteq G$ with $N \subset M$). Then:

1. $M/N \trianglelefteq G/N$;
2. $(G/N)/(M/N) \cong G/M$.

Proof: Proof of Third Isomorphism Theorem

(1) Elements of G/N are cosets gN ; elements of M/N are cosets mN for $m \in M$. We verify $M/N \trianglelefteq G/N$: for $gN \in G/N$ and $mN \in M/N$, we have $(gN)(mN)(gN)^{-1} = gm g^{-1}N$. Since $M \trianglelefteq G$, $gm g^{-1} \in M$, so $gm g^{-1}N \in M/N$.

(2) Define $\pi : G/N \rightarrow G/M$ by $\pi(gN) = gM$.

Well-defined: If $g_1N = g_2N$, then $g_1^{-1}g_2 \in N \subset M$, so $g_1M = g_2M$.

Homomorphism: $\pi(g_1N \cdot g_2N) = \pi(g_1g_2N) = g_1g_2M = (g_1M)(g_2M) = \pi(g_1N)\pi(g_2N)$.

Surjective: For any $gM \in G/M$, we have $\pi(gN) = gM$.

Kernel: $\ker(\pi) = \{gN : gM = M\} = \{gN : g \in M\} = M/N$.

By the First Isomorphism Theorem, $(G/N)/\ker(\pi) = (G/N)/(M/N) \cong \text{Im}(\pi) = G/M$. □

Note:-

The Third Isomorphism Theorem says “quotients of quotients are quotients”: $(G/N)/(M/N) \cong G/M$. The N ’s “cancel.” This is exactly what you’d hope for, and it’s a sign that quotient groups are a natural and well-behaved construction.

3.3.3 The Correspondence Theorem

The final structural theorem describes the relationship between subgroups of a quotient and subgroups of the original group.

Theorem 3.3.4 Correspondence Theorem (Fourth Isomorphism Theorem)

Let $N \trianglelefteq G$ and $\pi : G \rightarrow G/N$ be the canonical projection. There is a bijection

$$\{\text{subgroups of } G \text{ containing } N\} \longleftrightarrow \{\text{subgroups of } G/N\}$$

given by $H \mapsto H/N = \pi(H)$ with inverse $\bar{H} \mapsto \pi^{-1}(\bar{H})$. Moreover:

1. $H_1 \subset H_2 \iff H_1/N \subset H_2/N$, and $[H_2 : H_1] = [H_2/N : H_1/N]$;
2. $H \trianglelefteq G \iff H/N \trianglelefteq G/N$.

Proof: Proof of Correspondence Theorem

Given $H \supset N$, we have $H/N = \{hN : h \in H\}$, which is a subgroup of G/N . Given a subgroup $\bar{H} \subset G/N$, define $H = \pi^{-1}(\bar{H}) = \{g \in G : gN \in \bar{H}\}$. This contains N (since $eN = N \in \bar{H}$) and is a subgroup.

These operations are inverses: starting with $H \supset N$, we get $\pi^{-1}(H/N) = \{g : gN \in H/N\} = \{g : gN = hN \text{ for some } h \in H\} = \{g : g \in hN \text{ for some } h \in H\} = HN = H$ (since $N \subset H$).

Starting with $\bar{H} \subset G/N$: $\pi(\pi^{-1}(\bar{H})) = \{gN : gN \in \bar{H}\} = \bar{H}$.

(1) $H_1 \subset H_2 \implies H_1/N \subset H_2/N$ is clear. Conversely, if $H_1/N \subset H_2/N$ and $h \in H_1$, then $hN \in H_1/N \subset H_2/N$, so $hN = h'N$ for some $h' \in H_2$, giving $h \in h'N \subset H_2$. The index statement follows since the correspondence preserves cosets.

(2) If $H \trianglelefteq G$, then for $gN \in G/N$ and $hN \in H/N$: $(gN)(hN)(gN)^{-1} = ghg^{-1}N \in H/N$ since $ghg^{-1} \in H$. Conversely, if $H/N \trianglelefteq G/N$, then for $g \in G, h \in H$: $ghg^{-1}N = (gN)(hN)(gN)^{-1} \in H/N$, so $ghg^{-1} \in H$. $\square$

Question 3

Let $G = \mathbb{Z}$ and $N = 12\mathbb{Z}$.

1. List all subgroups of $G/N \cong \mathbb{Z}/12\mathbb{Z}$.
2. For each, find the corresponding subgroup of $\mathbb{Z}$ containing $12\mathbb{Z}$.
3. Which of these are normal in $\mathbb{Z}$? (Trick question!)
4. Verify the Third Isomorphism Theorem: check that $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.

Solution

(1) $\mathbb{Z}/12$ is cyclic of order 12, so its subgroups correspond to divisors of 12: they are $\langle \bar{0} \rangle, \langle \bar{6} \rangle, \langle \bar{4} \rangle, \langle \bar{3} \rangle, \langle \bar{2} \rangle, \langle \bar{1} \rangle$ of orders 1, 2, 3, 4, 6, 12 respectively.

(2) The corresponding subgroups of $\mathbb{Z}$ are those containing $12\mathbb{Z}$, i.e., of the form $d\mathbb{Z}$ where $d \mid 12$:

$$\begin{aligned} \langle \bar{0} \rangle &\leftrightarrow 12\mathbb{Z}, & \langle \bar{6} \rangle &\leftrightarrow 6\mathbb{Z}, & \langle \bar{4} \rangle &\leftrightarrow 4\mathbb{Z}, \\ \langle \bar{3} \rangle &\leftrightarrow 3\mathbb{Z}, & \langle \bar{2} \rangle &\leftrightarrow 2\mathbb{Z}, & \langle \bar{1} \rangle &\leftrightarrow \mathbb{Z}. \end{aligned}$$

(3) All subgroups of $\mathbb{Z}$ are normal because $\mathbb{Z}$ is abelian!

(4) We have $4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\} \cong \mathbb{Z}/3$. The quotient $(\mathbb{Z}/12)/(4\mathbb{Z}/12\mathbb{Z})$ has order $12/3 = 4$. Explicitly, the cosets are $\bar{0} + 4\mathbb{Z}/12\mathbb{Z} = \{\bar{0}, \bar{4}, \bar{8}\}$, $\bar{1} + \dots = \{\bar{1}, \bar{5}, \bar{9}\}$, $\bar{2} + \dots = \{\bar{2}, \bar{6}, \bar{10}\}$, $\bar{3} + \dots = \{\bar{3}, \bar{7}, \bar{11}\}$. This is cyclic of order 4, generated by $\bar{1} + 4\mathbb{Z}/12\mathbb{Z}$, hence $\cong \mathbb{Z}/4$.

The Third Isomorphism Theorem says $(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$, which checks out!

3.4 The symmetric group

The symmetric group S_n is arguably the most important finite group. By Cayley's theorem, every finite group embeds in some S_n . But S_n is also interesting in its own right: its structure illuminates concepts like conjugacy classes, normal subgroups, and simplicity. Moreover, symmetric groups connect to linear algebra through the sign homomorphism and determinants, and to Galois theory through their role in permuting roots of polynomials.

3.4.1 Cycle structure and decomposition

Definition 3.4.1: Symmetric group

The **symmetric group** S_n is the group of all bijections $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ under composition. It has order $n!$.

Definition 3.4.2: Cycle notation

A **k-cycle** (or cycle of length k) is a permutation $(a_1 a_2 \dots a_k)$ that acts by $a_1 \mapsto a_2 \mapsto \dots \mapsto a_k \mapsto a_1$ and fixes all other elements. A 2-cycle is called a **transposition**. Two cycles are **disjoint** if they have no elements in common; disjoint cycles commute.

Theorem 3.4.1 Cycle Decomposition Theorem

Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles, uniquely up to the order of the factors. The decomposition is obtained by writing out the orbits of σ acting on $\{1, \dots, n\}$.

Proof: Existence and uniqueness of cycle decomposition

Existence: Consider the action of $\langle \sigma \rangle$ on $\{1, \dots, n\}$. The orbits partition $\{1, \dots, n\}$. For each orbit $\{a_1, a_2, \dots, a_k\}$ where $\sigma(a_i) = a_{i+1}$ (indices mod k), define the cycle $c = (a_1 a_2 \dots a_k)$. Fixed points give 1-cycles (usually omitted). Since orbits are disjoint, these cycles are disjoint.

To verify σ equals the product of these cycles: for any i , if i is in the orbit giving cycle c , then $c(i) = \sigma(i)$ and all other cycles fix i , so the product sends $i \mapsto \sigma(i)$.

Uniqueness: Suppose $\sigma = c_1 c_2 \dots c_r = d_1 d_2 \dots d_s$ are two decompositions into disjoint cycles. For any i , exactly one c_j moves i (the others fix it), and that c_j must generate the orbit of i under σ . Similarly for the d 's. Thus c_j and the d_k containing i describe the same orbit, hence are the same cycle (up to cyclic rotation of the notation). $\square$

Definition 3.4.3: Cycle type

The **cycle type** of $\sigma \in S_n$ is the partition of n given by the lengths of the cycles in its disjoint cycle decomposition (including 1-cycles for fixed points). We write it as $(1^{m_1} 2^{m_2} \dots n^{m_n})$ where m_k is the number of k -cycles.

Example 3.4.1

In S_6 , the permutation $\sigma = (135)(24)$ has cycle type $(1^1 2^1 3^1)$ since there's one fixed point (6), one 2-cycle, and one 3-cycle. The identity has cycle type (1^6) . A single 6-cycle has type (6^1) .

3.4.2 Conjugacy in S_n

Conjugacy classes are fundamental to representation theory. In S_n , they have an elegant characterization.

Theorem 3.4.2 Conjugacy in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type.

Proof: Proof of conjugacy classification

Key computation: Let $\tau \in S_n$ and $(a_1 a_2 \dots a_k)$ be a cycle. Then

$$\tau(a_1 a_2 \dots a_k) \tau^{-1} = (\tau(a_1) \tau(a_2) \dots \tau(a_k)).$$

Proof: For any i , if $i = \tau(a_j)$, then $\tau(a_1 \dots a_k) \tau^{-1}(i) = \tau(a_{j+1}) = \tau(a_{j+1})$. If $i \notin \{\tau(a_1), \dots, \tau(a_k)\}$, then $\tau^{-1}(i) \notin \{a_1, \dots, a_k\}$, so the cycle fixes $\tau^{-1}(i)$, and conjugation gives $\tau(\tau^{-1}(i)) = i$.

Same type $\Rightarrow$ conjugate: Let $\sigma = (a_1 \dots a_{k_1})(b_1 \dots b_{k_2}) \dots$ and $\rho = (c_1 \dots c_{k_1})(d_1 \dots d_{k_2}) \dots$ have the same cycle type. Define τ by $\tau(a_i) = c_i, \tau(b_j) = d_j$, etc. Then by the key computation, $\tau \sigma \tau^{-1} = \rho$.

Conjugate $\Rightarrow$ same type: If $\rho = \tau \sigma \tau^{-1}$, write σ as a product of disjoint cycles. By the key computation, ρ is the product of the conjugated cycles, which have the same lengths. Disjointness is preserved since τ is a bijection. $\square$

Proposition 3.4.1 Number of conjugacy classes in S_n

The number of conjugacy classes in S_n equals $p(n)$, the number of partitions of n . For example, $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7, p(6) = 11$.

Note:-

This beautiful fact—conjugacy classes correspond to partitions—is one reason why the representation theory of S_n is so rich. The irreducible representations of S_n are also indexed by partitions of n , and the character table can be computed combinatorially. This is a whole subject in itself.

3.4.3 The sign homomorphism and alternating group

Every permutation can be written as a product of transpositions. The number of transpositions is not unique, but its parity is!

Proposition 3.4.2 Transposition generation

S_n is generated by transpositions. More precisely, every k -cycle can be written as a product of $k - 1$ transpositions:

$$(a_1 a_2 \cdots a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2).$$

Proof: Verification of transposition decomposition

Apply the right side to a_1 : $(a_1 a_2)$ sends $a_1 \mapsto a_2$, then subsequent transpositions fix a_2 . So $a_1 \mapsto a_2$. ✓

Apply to a_j for $2 \leq j \leq k - 1$: transpositions $(a_1 a_2), \dots, (a_1 a_{j-1})$ fix a_j . Then $(a_1 a_j)$ sends $a_j \mapsto a_1$. Then $(a_1 a_{j+1})$ sends $a_1 \mapsto a_{j+1}$. Subsequent transpositions fix a_{j+1} . So $a_j \mapsto a_{j+1}$. ✓

Apply to a_k : all transpositions except $(a_1 a_k)$ fix a_k , which sends $a_k \mapsto a_1$. ✓ □

Theorem 3.4.3 Well-definedness of parity

If $\sigma = \tau_1 \cdots \tau_r = \tau'_1 \cdots \tau'_s$ are two expressions of σ as products of transpositions, then $r \equiv s \pmod{2}$.

Proof: Proof of parity well-definedness

Consider the polynomial $P(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$. For $\sigma \in S_n$, define $\sigma \cdot P = \prod_{i < j} (x_{\sigma(i)} - x_{\sigma(j)})$. Each factor $(x_{\sigma(i)} - x_{\sigma(j)})$ equals $\pm(x_k - x_l)$ for some $k < l$, so $\sigma \cdot P = \pm P$.

For a transposition $\tau = (ab)$ with $a < b$: the factor $(x_a - x_b)$ becomes $(x_b - x_a) = -(x_a - x_b)$. For $i < a < b < j$, the factor $(x_i - x_j)$ is unchanged. For $i < a$, the factors $(x_i - x_a)$ and $(x_i - x_b)$ are swapped, contributing no sign change. Similarly for other cases. Careful counting shows $\tau \cdot P = -P$ (the key is that one factor picks up a sign, and swaps come in pairs).

Thus if $\sigma = \tau_1 \cdots \tau_r$, then $\sigma \cdot P = (-1)^r P$. If also $\sigma = \tau'_1 \cdots \tau'_s$, then $(-1)^r = (-1)^s$, so $r \equiv s \pmod{2}$. □

Definition 3.4.4: Sign of a permutation

The **sign** (or **signature**) of $\sigma \in S_n$ is $\text{sgn}(\sigma) = (-1)^r$ where $\sigma = \tau_1 \cdots \tau_r$ is any expression as a product of transpositions. Equivalently, $\text{sgn}(\sigma) = (-1)^{n-c(\sigma)}$ where $c(\sigma)$ is the number of cycles (including 1-cycles). A permutation is **even** if $\text{sgn}(\sigma) = 1$ and **odd** if $\text{sgn}(\sigma) = -1$.

Theorem 3.4.4 Sign is a homomorphism

The map $\text{sgn} : S_n \rightarrow \{1, -1\}$ is a surjective group homomorphism.

Proof: Proof that sign is a homomorphism

If $\sigma = \tau_1 \cdots \tau_r$ and $\rho = \tau'_1 \cdots \tau'_s$, then $\sigma\rho = \tau_1 \cdots \tau_r \tau'_1 \cdots \tau'_s$, a product of $r + s$ transpositions. Thus $\text{sgn}(\sigma\rho) = (-1)^{r+s} = (-1)^r (-1)^s = \text{sgn}(\sigma) \text{sgn}(\rho)$. Surjectivity: any transposition has sign -1 . □

Definition 3.4.5: Alternating group

The **alternating group** $A_n = \ker(\text{sgn}) = \{\sigma \in S_n : \text{sgn}(\sigma) = 1\}$ is the group of even permutations.

Proposition 3.4.3 Properties of A_n

We have:

1. $A_n \trianglelefteq S_n$ with index $[S_n : A_n] = 2$, so $|A_n| = n!/2$.
2. $S_n/A_n \cong \mathbb{Z}/2$.
3. A_n is generated by 3-cycles. In fact, $A_n = \langle (1\ 2\ 3), (1\ 2\ 4), \dots, (1\ 2\ n) \rangle$.
4. For $n \geq 5$, A_n is **simple** (has no proper nontrivial normal subgroups).

Proof: Proof of A_n properties (1)-(3)

(1)-(2) follow from the First Isomorphism Theorem: $S_n/\ker(\text{sgn}) \cong \text{Im}(\text{sgn}) = \{1, -1\} \cong \mathbb{Z}/2$.

(3) Every even permutation is a product of an even number of transpositions. We can pair them: $(a\ b)(c\ d)$. If $\{a, b\} \cap \{c, d\} = \emptyset$, then $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$ (verify!). If $\{a, b\} \cap \{c, d\} = \{b\} = \{c\}$, then $(a\ b)(b\ d) = (a\ b\ d)$. Thus A_n is generated by 3-cycles. For the specific generating set:

$$(a\ b\ c) = (1\ 2\ a)(1\ 2\ b)(1\ 2\ c)(1\ 2\ a)^{-1}(1\ 2\ b)^{-1}$$

(if a, b, c all differ from 1, 2; similar manipulations handle other cases). □

Note:-

The simplicity of A_n for $n \geq 5$ is a deep result with profound consequences. It implies that A_n has no quotients except $\{e\}$ and itself, which means A_n is “irreducible” in a sense. Via Galois theory, this simplicity is precisely why the general quintic polynomial has no solution in radicals: the Galois group of the “generic” degree- n polynomial is S_n , and solvability by radicals requires a composition series with abelian quotients. For $n \geq 5$, A_n blocks this because it’s simple and non-abelian. We will prove simplicity later, after developing more machinery.

Note:-

Connection to linear algebra: the sign homomorphism is intimately related to the determinant. For a matrix $A \in \text{GL}_n(\mathbb{R})$, we have $\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}$. The sign ensures the determinant is alternating in the rows (swapping rows negates the determinant). This is no coincidence: the determinant is the unique alternating multilinear function on n vectors with $\det(I) = 1$, and “alternating” is encoded by the sign.

Question 4

1. Compute the cycle type and sign of $\sigma = (1\ 2\ 3)(4\ 5)(1\ 4)(2\ 5\ 3)$ in S_5 . (First simplify!)
2. How many elements of S_7 have cycle type $(1^1 2^1 4^1)$?
3. Prove that the only normal subgroups of S_3 are $\{e\}, A_3, S_3$.
4. Let $H = \{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \subset S_4$. Prove $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ and $H \trianglelefteq S_4$. What is S_4/H ?

Solution

(1) First, multiply the cycles right-to-left. Trace each element:

$$1 \mapsto 4 \mapsto 5 \mapsto 5 \mapsto 4 \mapsto 1 : \text{wait, let's be systematic.}$$

Write $\sigma = (1\ 2\ 3)(4\ 5)(1\ 4)(2\ 5\ 3)$. Reading right-to-left: start with $(2\ 5\ 3)$, then $(1\ 4)$, then $(4\ 5)$, then $(1\ 2\ 3)$.

Trace 1: $1 \xrightarrow{(2\ 5\ 3)} 1 \xrightarrow{(1\ 4)} 4 \xrightarrow{(4\ 5)} 5 \xrightarrow{(1\ 2\ 3)} 5$. So $1 \mapsto 5$.

Trace 2: $2 \xrightarrow{(2\ 5\ 3)} 5 \xrightarrow{(1\ 4)} 5 \xrightarrow{(4\ 5)} 4 \xrightarrow{(1\ 2\ 3)} 4$. So $2 \mapsto 4$.

Trace 3: $3 \xrightarrow{(2\ 5\ 3)} 2 \xrightarrow{(1\ 4)} 2 \xrightarrow{(4\ 5)} 2 \xrightarrow{(1\ 2\ 3)} 3$. So $3 \mapsto 3$ (fixed).

Trace 4: $4 \xrightarrow{(2\ 5\ 3)} 4 \xrightarrow{(1\ 4)} 1 \xrightarrow{(4\ 5)} 1 \xrightarrow{(1\ 2\ 3)} 2$. So $4 \mapsto 2$.

Trace 5: $5 \xrightarrow{(2\ 5\ 3)} 3 \xrightarrow{(1\ 4)} 3 \xrightarrow{(4\ 5)} 3 \xrightarrow{(1\ 2\ 3)} 1$. So $5 \mapsto 1$.

Thus $\sigma = (1\ 5)(2\ 4)$, cycle type $(1^1 2^2)$. Sign: two 2-cycles = two transpositions, so $\text{sgn}(\sigma) = (-1)^2 = 1$ (even).

(2) Cycle type $(1^1 2^1 4^1)$ means one fixed point, one 2-cycle, one 4-cycle. Choose the fixed point: 7 ways. Choose 4 elements for the 4-cycle from the remaining 6: $\binom{6}{4} = 15$ ways. Arrange them in a 4-cycle: $(4-1)! = 6$ ways. The remaining 2 elements form the 2-cycle: 1 way. Total: $7 \cdot 15 \cdot 6 \cdot 1 = 630$.

(3) S_3 has order 6. By Lagrange, subgroups have order 1, 2, 3, or 6. Normal subgroups must be unions of conjugacy classes. Conjugacy classes in S_3 : $\{e\}$ (size 1), $\{(12), (13), (23)\}$ (size 3), $\{(123), (132)\}$ (size 2).

A normal subgroup of order 2 would need size-1 and size-1 classes, but $\{e\}$ has size 1 and we'd need another element with a size-1 class—none exist. A normal subgroup of order 3 must be $\{e\} \cup \{(123), (132)\} = A_3$. ✓

(4) Check H is a subgroup: $(1\ 2)(3\ 4) \cdot (1\ 3)(2\ 4) = (1\ 4)(2\ 3)$, etc.—closure holds (verify all products give elements of H). Each non-identity element has order 2. So $H \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).

Normality: H consists of e plus all products of two disjoint transpositions. These form a conjugacy class in S_4 (cycle type (2^2)), so H is a union of conjugacy classes, hence normal.

S_4/H has order $24/4 = 6$. The quotient is non-abelian (since S_4 is non-abelian and H is not all of S_4), so $S_4/H \cong S_3$. Explicitly: S_4 acts on the three partitions $\{\{1, 2\}, \{3, 4\}\}, \{\{1, 3\}, \{2, 4\}\}, \{\{1, 4\}, \{2, 3\}\}$, and H is the kernel.

3.5 Exact sequences

The isomorphism theorems reveal a pattern: kernels and images of homomorphisms are intimately related. Exact sequences provide elegant language to express these relationships, and form the foundation of homological algebra—a subject that permeates modern mathematics from algebraic topology to representation theory to algebraic geometry. We introduce the basic concepts here.

3.5.1 Exact sequences and their meaning

Definition 3.5.1: Exact sequence

A sequence of groups and homomorphisms

$$\cdots \rightarrow G_{i-1} \xrightarrow{\varphi_{i-1}} G_i \xrightarrow{\varphi_i} G_{i+1} \rightarrow \cdots$$

is **exact at G_i** if $\text{Im}(\varphi_{i-1}) = \ker(\varphi_i)$. The sequence is **exact** if it is exact at every group (except possibly the endpoints).

The condition $\text{Im}(\varphi_{i-1}) = \ker(\varphi_i)$ says: “everything that comes from the left dies when we go right.” This is stronger than $\text{Im}(\varphi_{i-1}) \subseteq \ker(\varphi_i)$ (which just says $\varphi_i \circ \varphi_{i-1} = 0$, i.e., the composition is the zero map).

Example 3.5.1 (Exactness encodes injectivity and surjectivity)

Consider short sequences with trivial groups $\{e\}$:

1. $\{e\} \rightarrow A \xrightarrow{\varphi} B$ is exact at A iff $\ker(\varphi) = \text{Im}(\{e\} \rightarrow A) = \{e_A\}$, i.e., φ is **injective**.
2. $A \xrightarrow{\varphi} B \rightarrow \{e\}$ is exact at B iff $\text{Im}(\varphi) = \ker(B \rightarrow \{e\}) = B$, i.e., φ is **surjective**.
3. $\{e\} \rightarrow A \xrightarrow{\varphi} B \rightarrow \{e\}$ is exact iff φ is an **isomorphism**.

3.5.2 Short exact sequences

The most important type of exact sequence is the short exact sequence, which captures the structure of quotient groups.

Definition 3.5.2: Short exact sequence

A short exact sequence (SES) is an exact sequence of the form

$$\{e\} \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow \{e\}$$

where exactness means: ι is injective, π is surjective, and $\text{Im}(\iota) = \ker(\pi)$.

Note:-

We often write a short exact sequence as $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ (using 1 for the trivial group in multiplicative notation) or $0 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 0$ (in additive notation for abelian groups).

Proposition 3.5.1 Short exact sequences and quotients

A sequence $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$ is exact if and only if:

1. ι identifies N with a normal subgroup of G (namely $\iota(N) = \ker(\pi) \trianglelefteq G$);
2. π induces an isomorphism $G/\iota(N) \cong Q$.

In other words, short exact sequences encode the statement “ $G/N \cong Q$ ” or “ G is an extension of Q by N .”

Proof: Proof of SES characterization

($\Rightarrow$) Suppose the sequence is exact. Then ι is injective, so $N \cong \iota(N) \subseteq G$. Exactness at G gives $\iota(N) = \ker(\pi)$, which is normal in G . By the First Isomorphism Theorem, $G/\ker(\pi) \cong \text{Im}(\pi) = Q$ (using surjectivity of π).

($\Leftarrow$) If $\iota(N) \trianglelefteq G$ and $G/\iota(N) \cong Q$, take $\pi : G \rightarrow G/\iota(N) \cong Q$ to be the quotient map (composed with the isomorphism). Then $\ker(\pi) = \iota(N) = \text{Im}(\iota)$, giving exactness. $\square$

Example 3.5.2 (Fundamental examples of short exact sequences)

1. **Integers mod n :** $0 \rightarrow n\mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$, where ι is inclusion and π is reduction mod n .
2. **Determinant:** $1 \rightarrow \text{SL}_n(\mathbb{R}) \xrightarrow{\iota} \text{GL}_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R}^* \rightarrow 1$. The kernel of $\det$ is exactly SL_n .
3. **Sign:** $1 \rightarrow A_n \xrightarrow{\iota} S_n \xrightarrow{\text{sgn}} \{\pm 1\} \rightarrow 1$. The alternating group is the kernel of the sign homomorphism.
4. **Center:** For any group G , we have $1 \rightarrow Z(G) \xrightarrow{\iota} G \xrightarrow{\pi} G/Z(G) \rightarrow 1$. The quotient $G/Z(G)$ is called the inner automorphism group (it acts on G by conjugation).
5. **Product decomposition:** $1 \rightarrow N \xrightarrow{\iota} N \times Q \xrightarrow{\pi} Q \rightarrow 1$, where $\iota(n) = (n, e)$ and $\pi(n, q) = q$. This is a “trivial” extension.

3.5.3 Splitting and semidirect products

When does a short exact sequence “come from” a direct product? This leads to the important notion of splitting.

Definition 3.5.3: Split exact sequence

A short exact sequence $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$ is **split** (or **splits**) if any of the following equivalent conditions holds:

1. There exists a homomorphism $s : Q \rightarrow G$ with $\pi \circ s = \text{id}_Q$ (a **section** or **right inverse**);
2. There exists a homomorphism $r : G \rightarrow N$ with $r \circ \iota = \text{id}_N$ (a **retraction** or **left inverse**);
3. $G \cong N \rtimes Q$ for some semidirect product structure.

Note:-

For abelian groups, split exact sequences correspond exactly to direct products: if $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ splits with A, B, C abelian, then $B \cong A \times C$. For non-abelian groups, the situation is more subtle—we get **semidirect** products, which we’ll study in detail later.

Example 3.5.3 (Split vs non-split)

1. **Splits:** $1 \rightarrow A_3 \rightarrow S_3 \rightarrow \mathbb{Z}/2 \rightarrow 1$. A section $s : \mathbb{Z}/2 \rightarrow S_3$ sends the generator to any transposition, e.g., $s(1) = (12)$. Thus $S_3 \cong A_3 \rtimes \mathbb{Z}/2$.
2. **Does not split:** $0 \rightarrow \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \xrightarrow{\text{mod } 2} \mathbb{Z}/2 \rightarrow 0$. If it split, we’d have $\mathbb{Z}/4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, but $\mathbb{Z}/4$ has an element of order 4 while $\mathbb{Z}/2 \times \mathbb{Z}/2$ doesn’t. The sequence is exact but not split.

Proposition 3.5.2 Splitting lemma (partial)

If $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$ has a section $s : Q \rightarrow G$, then every element $g \in G$ can be written uniquely as $g = \iota(n) \cdot s(q)$ for some $n \in N$ and $q \in Q$.

Proof: Proof of splitting decomposition

Existence: Given $g \in G$, let $q = \pi(g) \in Q$. Then $\pi(g \cdot s(q)^{-1}) = \pi(g) \cdot \pi(s(q))^{-1} = q \cdot q^{-1} = e_Q$, so $g \cdot s(q)^{-1} \in \ker(\pi) = \iota(N)$. Thus $g \cdot s(q)^{-1} = \iota(n)$ for some $n \in N$, giving $g = \iota(n) \cdot s(q)$.

Uniqueness: Suppose $\iota(n_1) \cdot s(q_1) = \iota(n_2) \cdot s(q_2)$. Apply π : $e \cdot q_1 = e \cdot q_2$, so $q_1 = q_2$. Then $\iota(n_1) = \iota(n_2)$, and since ι is injective, $n_1 = n_2$. $\square$

3.5.4 The extension problem

The short exact sequence perspective leads to a fundamental question in group theory.

Definition 3.5.4: Group extension

Given groups N and Q , an **extension of Q by N** is a group G fitting into a short exact sequence $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$.

The **extension problem** asks: given N and Q , classify all groups G that are extensions of Q by N . This is a deep problem that leads to:

1. **Semidirect products** (when the sequence splits);
2. **Group cohomology** $H^2(Q, N)$ (classifying non-split extensions);
3. The structure theory of finite groups (building groups from simple groups).

Example 3.5.4 (Extensions of $\mathbb{Z}/2$ by $\mathbb{Z}/2$)

What groups G fit into $1 \rightarrow \mathbb{Z}/2 \rightarrow G \rightarrow \mathbb{Z}/2 \rightarrow 1$? Such G has order 4. We know there are exactly two groups of order 4: $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$. Both work:

- $\mathbb{Z}/4$: the sequence $0 \rightarrow \mathbb{Z}/2 \xrightarrow{\times 2} \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 0$ (non-split).
- $\mathbb{Z}/2 \times \mathbb{Z}/2$: the sequence $0 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/2 \times \mathbb{Z}/2 \rightarrow \mathbb{Z}/2 \rightarrow 0$ (split).

So there are exactly two extensions, corresponding to the two groups of order 4.

Question 5

1. Verify that $1 \rightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\text{mod } n} \mathbb{Z}/n \rightarrow 1$ is exact.
2. Show that the sequence $1 \rightarrow \langle r \rangle \rightarrow D_n \rightarrow \mathbb{Z}/2 \rightarrow 1$ is exact, where $\langle r \rangle \cong \mathbb{Z}/n$ is the rotation subgroup. Does it split?
3. Let G be a group with normal subgroup N . Prove that the sequence $1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$ is exact (with the obvious maps).
4. Suppose $1 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 1$ and $1 \rightarrow C \xrightarrow{h} D \xrightarrow{k} E \rightarrow 1$ are exact. Prove that $1 \rightarrow A \xrightarrow{f} B \xrightarrow{h \circ g} D \xrightarrow{k} E \rightarrow 1$ is exact.

Solution

(1) Let $\iota: \mathbb{Z} \rightarrow \mathbb{Z}$ be $\iota(k) = nk$ and $\pi: \mathbb{Z} \rightarrow \mathbb{Z}/n$ be reduction mod n .

Exactness at first $\mathbb{Z}$: $\ker(\iota) = \{k : nk = 0\} = \{0\}$, and $\text{Im}(1 \rightarrow \mathbb{Z}) = \{0\}$. ✓

Exactness at second $\mathbb{Z}$: $\text{Im}(\iota) = n\mathbb{Z}$ and $\ker(\pi) = n\mathbb{Z}$. ✓

Exactness at $\mathbb{Z}/n$: $\text{Im}(\pi) = \mathbb{Z}/n$ and $\ker(\mathbb{Z}/n \rightarrow 1) = \mathbb{Z}/n$. ✓

(2) The rotation subgroup $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\}$ is normal in D_n (we showed this earlier). The quotient $D_n/\langle r \rangle$ has order $2n/n = 2$, so $D_n/\langle r \rangle \cong \mathbb{Z}/2$. The sequence $1 \rightarrow \langle r \rangle \xrightarrow{\iota} D_n \xrightarrow{\pi} \mathbb{Z}/2 \rightarrow 1$ is exact.

Does it split? Yes! Define $s: \mathbb{Z}/2 \rightarrow D_n$ by $s(0) = e$ and $s(1) = s$ (a reflection). Then $\pi(s(1)) = \pi(s) = 1$ (since $s \notin \langle r \rangle$), so $\pi \circ s = \text{id}$. Thus $D_n \cong \mathbb{Z}/n \rtimes \mathbb{Z}/2$.

(3) Let $\iota: N \hookrightarrow G$ be inclusion and $\pi: G \rightarrow G/N$ be the quotient map.

Exactness at N : ι is injective, so $\ker(\iota) = \{e\} = \text{Im}(1 \rightarrow N)$. ✓

Exactness at G : $\text{Im}(\iota) = N$ and $\ker(\pi) = N$. ✓

Exactness at G/N : π is surjective, so $\text{Im}(\pi) = G/N = \ker(G/N \rightarrow 1)$. ✓

(4) We check exactness at each position.

At A : Same as the original sequence— f is injective.

At B : We need $\text{Im}(f) = \ker(h \circ g)$. We have $\text{Im}(f) = \ker(g)$ (exactness of first sequence). If $b \in \ker(g)$, then $g(b) = e_C$, so $(h \circ g)(b) = h(e_C) = e_D$, giving $b \in \ker(h \circ g)$. Conversely, if $(h \circ g)(b) = e_D$, then $h(g(b)) = e_D$, so $g(b) \in \ker(h) = \{e_C\}$ (since h is injective). Thus $g(b) = e_C$, so $b \in \ker(g) = \text{Im}(f)$.

At D : We need $\text{Im}(h \circ g) = \ker(k)$. From exactness of second sequence, $\text{Im}(h) = \ker(k)$. Since g is surjective, $\text{Im}(h \circ g) = h(g(B)) = h(C) = \text{Im}(h) = \ker(k)$. ✓

At E : Same as the original sequence— k is surjective.

Note:-

Exact sequences are the beginning of homological algebra. The ideas extend to longer exact sequences, to chain complexes (sequences where consecutive compositions are zero, not necessarily exact), and to derived functors that measure the failure of exactness. In representation theory, exact sequences of representations encode how representations decompose. In topology, the long exact sequence of a fibration is a powerful tool. The language you've learned here will reappear throughout mathematics.

3.6 Free groups and presentations

Every group we’ve seen so far has been defined concretely: permutations, matrices, integers mod n . But there’s a fundamentally different approach: describe a group by generators and relations. This leads to group presentations, which are both computationally useful and theoretically profound. The key player is the free group—the “most general” group on a given set of generators.

3.6.1 Words and reduced words

Definition 3.6.1: Alphabet and words

Let S be a set (the **alphabet** or **generating set**). Define $S^{-1} = \{s^{-1} : s \in S\}$, a formal set of “inverse symbols” disjoint from S . A **word** in S is a finite sequence

$$w = a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_n^{\epsilon_n}$$

where each $a_i \in S$ and $\epsilon_i \in \{1, -1\}$. The **empty word**, denoted e or ϵ , is the word with no letters.

Example 3.6.1

If $S = \{a, b\}$, then $ab^{-1}a^{-1}ba$, $aaab$, $b^{-1}b^{-1}a$, and e are all words. We write a^3 for aaa and a^{-2} for $a^{-1}a^{-1}$.

Definition 3.6.2: Concatenation

The **concatenation** of words $w_1 = a_1 \cdots a_m$ and $w_2 = b_1 \cdots b_n$ is $w_1 w_2 = a_1 \cdots a_m b_1 \cdots b_n$. This operation is associative with identity e .

Now, we want words to form a group. The issue: aa^{-1} should equal e , but as sequences they’re different. We need to identify words that “should be equal.”

Definition 3.6.3: Elementary reduction

An **elementary reduction** of a word w is the operation of removing a subword of the form ss^{-1} or $s^{-1}s$ (for $s \in S$). A word is **reduced** if it contains no such subwords—i.e., no letter is immediately followed by its inverse.

Example 3.6.2

The word $ab^{-1}ba^{-1}a$ reduces as follows:

$$ab^{-1}ba^{-1}a \rightarrow ab^{-1}b \cdot e \cdot a^{-1}a \rightarrow ab^{-1}b \rightarrow a \cdot e = a.$$

Wait, let me be more careful: $ab^{-1}ba^{-1}a$. We have $b^{-1}b$, so reduce to $a \cdot e \cdot a^{-1}a = aa^{-1}a$. Then aa^{-1} reduces to e , leaving a . So the reduced form is a .

Theorem 3.6.1 Unique reduced form

Every word can be reduced to a unique reduced word by a sequence of elementary reductions. The order of reductions doesn’t matter.

Proof: Sketch of unique reduction

Termination: Each reduction decreases the length by 2, so the process terminates.

Uniqueness (sketch): This is the key point. Suppose w reduces to both u and v . We use the diamond lemma: if we can perform two different reductions $w \rightarrow w_1$ and $w \rightarrow w_2$, then there exists w' reachable

from both w_1 and w_2 .

The only interesting case is when the two reductions overlap, e.g., $w = \cdots ss^{-1}s \cdots$. Reducing ss^{-1} gives $\cdots s \cdots$; reducing $s^{-1}s$ also gives $\cdots s \cdots$. They agree! So all reduction paths lead to the same result. $\square$

3.6.2 The free group

Definition 3.6.4: Free group

The **free group on S** , denoted $F(S)$ or F_n (where $n = |S|$), is the set of reduced words in S with the group operation:

$$w_1 \cdot w_2 = \text{reduce}(w_1 w_2).$$

The identity is the empty word e , and the inverse of $a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ is $a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$.

Proposition 3.6.1 Free group is a group

The free group $F(S)$ satisfies the group axioms.

Proof: Verification of group axioms for $F(S)$

Closure: The product of two reduced words, after reduction, is a reduced word.

Associativity: This is subtle! We need $(w_1 \cdot w_2) \cdot w_3 = w_1 \cdot (w_2 \cdot w_3)$. Both equal the reduction of $w_1 w_2 w_3$ —but the reductions might happen in different orders. By uniqueness of reduced form, they give the same result.

Identity: $e \cdot w = \text{reduce}(ew) = \text{reduce}(w) = w$ (since w is already reduced). Similarly $w \cdot e = w$.

Inverses: Let $w = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}$ and $w^{-1} = a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Then $ww^{-1} = a_1^{\epsilon_1} \cdots a_n^{\epsilon_n} a_n^{-\epsilon_n} \cdots a_1^{-\epsilon_1}$. Reducing: $a_n^{\epsilon_n} a_n^{-\epsilon_n}$ cancels, then $a_{n-1}^{\epsilon_{n-1}} a_{n-1}^{-\epsilon_{n-1}}$ cancels, etc., until we get e . $\square$

Example 3.6.3 (Free group on one generator)

$F(\{a\}) = F_1$ consists of reduced words in a, a^{-1} : these are $\dots, a^{-2}, a^{-1}, e, a, a^2, \dots$. This is just $\{a^n : n \in \mathbb{Z}\}$ with $a^m \cdot a^n = a^{m+n}$. So $F_1 \cong \mathbb{Z}$.

Example 3.6.4 (Free group on two generators)

$F_2 = F(\{a, b\})$ is much more interesting. Elements include $e, a, b, a^{-1}, b^{-1}, ab, ba, a^2, ab^{-1}, aba, bab^{-1}a^{-1}, \dots$ —infinitely many distinct reduced words. Notably, $ab \neq ba$ (both are reduced but different), so F_2 is non-abelian. In fact, F_2 contains copies of every countable free group!

3.6.3 The universal property

The free group is “free” because it has no relations other than the group axioms. This is captured by a universal property.

Theorem 3.6.2 Universal property of free groups

Let $F(S)$ be the free group on S , and let $\iota : S \rightarrow F(S)$ be the inclusion $s \mapsto s$ (the one-letter word). For any group G and any function $f : S \rightarrow G$, there exists a unique homomorphism $\hat{f} : F(S) \rightarrow G$ such that $\hat{f} \circ \iota = f$.

$$\begin{array}{ccc}
S & \xrightarrow{\iota} & F(S) \\
& \searrow f & \downarrow \exists! \tilde{f} \\
& & G
\end{array}$$

Proof: Proof of universal property

Existence: Define $\tilde{f}(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n}$. This is well-defined on reduced words and is a homomorphism: $\tilde{f}(w_1 w_2) = \tilde{f}(w_1) \tilde{f}(w_2)$ because the reductions in $F(S)$ correspond to the group relations $f(s)f(s)^{-1} = e_G$ in G .

Uniqueness: Any homomorphism $\varphi : F(S) \rightarrow G$ with $\varphi(s) = f(s)$ for all $s \in S$ must satisfy $\varphi(a_1^{\epsilon_1} \cdots a_n^{\epsilon_n}) = \varphi(a_1)^{\epsilon_1} \cdots \varphi(a_n)^{\epsilon_n} = f(a_1)^{\epsilon_1} \cdots f(a_n)^{\epsilon_n} = \tilde{f}(w)$. $\square$

Note:-

The universal property says: to define a homomorphism out of $F(S)$, you just need to say where the generators go—there are no constraints! Any choice of images for S in any group G extends uniquely to a homomorphism. This is why $F(S)$ is “free”—you have complete freedom in defining maps from it. Categorically, the free group functor is left adjoint to the forgetful functor from groups to sets.

3.6.4 Group presentations

The universal property immediately gives us a way to describe groups by generators and relations.

Definition 3.6.5: Group presentation

A presentation of a group G is an expression

$$G = \langle S \mid R \rangle$$

where S is a set of generators and $R \subseteq F(S)$ is a set of relators. This means $G \cong F(S)/N$, where $N = \langle\langle R \rangle\rangle$ is the normal closure of R —the smallest normal subgroup of $F(S)$ containing R . Equivalently, if $R = \{r_1, r_2, \dots\}$, we write $G = \langle S \mid r_1 = e, r_2 = e, \dots \rangle$ or $G = \langle S \mid r_1, r_2, \dots \rangle$.

Note:-

The normal closure $\langle\langle R \rangle\rangle$ consists of all products of conjugates of elements of R and their inverses: $\langle\langle R \rangle\rangle = \{g_1 r_1^{\pm 1} g_1^{-1} \cdots g_k r_k^{\pm 1} g_k^{-1} : g_i \in F(S), r_i \in R\}$. We need normal closure (not just the subgroup generated by R) because we’re quotienting, and quotients require normal subgroups.

Example 3.6.5 (Presentations of familiar groups)

1. **Cyclic group:** $\mathbb{Z}/n = \langle a \mid a^n \rangle$. One generator, one relation.
2. **Infinite cyclic:** $\mathbb{Z} = \langle a \mid \rangle$ (no relations). The free group on one generator.
3. **Klein four-group:** $\mathbb{Z}/2 \times \mathbb{Z}/2 = \langle a, b \mid a^2, b^2, aba^{-1}b^{-1} \rangle$. The last relation says $ab = ba$ (commutativity).
4. **Dihedral group:** $D_n = \langle r, s \mid r^n, s^2, srs^{-1}r \rangle$. The last relation says $srs^{-1} = r^{-1}$, i.e., $sr = r^{-1}s$.
5. **Symmetric group:** $S_n = \langle s_1, \dots, s_{n-1} \mid s_i^2, (s_i s_{i+1})^3, (s_i s_j)^2 \text{ for } |i - j| \geq 2 \rangle$, where $s_i = (i \ i + 1)$. These are the Coxeter relations.
6. **Free abelian group:** $\mathbb{Z}^n = \langle a_1, \dots, a_n \mid [a_i, a_j] \text{ for all } i, j \rangle$, where $[a, b] = aba^{-1}b^{-1}$ is the commutator.

Proposition 3.6.2 Every group has a presentation

For any group G , there exist S and R such that $G \cong \langle S \mid R \rangle$.

Proof: Existence of presentation

Take $S = G$ (yes, all of G !) and define $f : S \rightarrow G$ by $f(g) = g$. By the universal property, this extends to a surjective homomorphism $\tilde{f} : F(G) \rightarrow G$. Let $R = \ker(\tilde{f})$. Then $G \cong F(G)/R = \langle G \mid R \rangle$.

This is a “stupid” presentation (huge!), but it exists. Finding nice presentations is an art. $\square$

3.6.5 The word problem

Presentations raise a fundamental computational question.

Definition 3.6.6: Word problem

Given a presentation $G = \langle S \mid R \rangle$ and two words w_1, w_2 in S , the **word problem** asks: do w_1 and w_2 represent the same element of G ? Equivalently: is $w_1 w_2^{-1}$ in the normal closure of R ?

Theorem 3.6.3 Undecidability of the word problem (Novikov-Boone)

There exists a finitely presented group $G = \langle S \mid R \rangle$ (with S, R finite) for which the word problem is **undecidable**—no algorithm can determine, for all pairs of words, whether they’re equal in G .

Note:-

This is a remarkable and sobering result. Group presentations look like a nice computational tool, but they’re actually extremely powerful—so powerful that they can encode arbitrary computation. The word problem for G can simulate a Turing machine! This doesn’t mean presentations are useless; for specific groups (like S_n , D_n , free groups), the word problem is decidable. But there’s no universal algorithm.

Question 6

1. Prove that F_2 contains a subgroup isomorphic to F_3 .
2. Show that $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$.
3. Find a presentation for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.
4. Prove that $\langle a, b \mid a^2, b^3, (ab)^2 \rangle \cong S_3$.

Solution

(1) In $F_2 = F(\{a, b\})$, consider the subgroup $H = \langle a^2, ab, b^2 \rangle$. We claim $H \cong F_3$.

To see this, define $f : \{x, y, z\} \rightarrow F_2$ by $f(x) = a^2, f(y) = ab, f(z) = b^2$. By the universal property, this extends to a homomorphism $\tilde{f} : F_3 \rightarrow F_2$ with image H . To show $\tilde{f}$ is injective, we must show no nontrivial reduced word in x, y, z maps to e .

Suppose $w(x, y, z) \mapsto e$ in F_2 . Substituting: $w(a^2, ab, b^2) = e$ as a reduced word in a, b . The key observation: a^2, ab, b^2 have the property that no product of them (with exponents ± 1) reduces to e unless the product was empty. (This can be shown by analyzing how cancellation works—the words have distinct “shapes.”) This is the Schreier basis approach. Thus $\tilde{f}$ is injective, and $H \cong F_3$.

(2) The presentation $G = \langle a, b \mid aba^{-1}b^{-1} \rangle$ has one relation: $aba^{-1}b^{-1} = e$, i.e., $ab = ba$. So G is the free group on a, b modulo the relation that a and b commute. This is exactly the free abelian group on two generators, which is $\mathbb{Z}^2$.

Explicitly: define $\varphi : G \rightarrow \mathbb{Z}^2$ by $\varphi(a) = (1, 0)$ and $\varphi(b) = (0, 1)$. This is well-defined (the relation $ab = ba$ holds in $\mathbb{Z}^2$). It’s surjective since $(1, 0), (0, 1)$ generate $\mathbb{Z}^2$. For injectivity: any element of G can be written as $a^m b^n$ (using commutativity to collect), and $\varphi(a^m b^n) = (m, n)$; if this equals

$(0, 0)$, then $m = n = 0$.

(3) The quaternion group Q_8 has elements $\{1, -1, i, -i, j, -j, k, -k\}$ with relations $i^2 = j^2 = k^2 = ijk = -1$.

A presentation: $Q_8 = \langle i, j \mid i^4, i^2j^{-2}, iji^{-1}j \rangle$.

Let's verify: $i^4 = 1$ says i has order dividing 4 (actually 4). $i^2 = j^2$ says both square to the same element (which will be -1). $iji^{-1}j = e$ means $ij = j^{-1}i^{-1} = j^{-1}i^3 = j^{-1}(i^{-1}) = (ji)^{-1}$, which combined with $i^2 = j^2$ gives $ij = k$ where $k = i^{-1}j^{-1}$, and one can derive all the quaternion relations.

Alternative (cleaner): $Q_8 = \langle x, y \mid x^4, x^2y^{-2}, yxy^{-1}x \rangle$.

(4) Let $G = \langle a, b \mid a^2, b^3, (ab)^2 \rangle$. We have $|a| \mid 2$, $|b| \mid 3$, and $(ab)^2 = e$ means $abab = e$, so $ab = b^{-1}a^{-1} = b^{-1}a = b^2a$ (using $a^2 = e$).

So $ab = b^2a$, i.e., $aba^{-1} = b^2 = b^{-1}$. This is exactly the dihedral relation! Compare with $D_3 = \langle r, s \mid r^3, s^2, srs^{-1} = r^{-1} \rangle$. Setting $r = b, s = a$, we get the same relations. Thus $G \cong D_3 \cong S_3$.

Vector Spaces and Linear Algebra

This lecture marks our transition from the purely algebraic study of groups to the more geometric realm of linear algebra. Vector spaces are arguably the most tractable algebraic structures—unlike groups, where classification is hopelessly complex, finite-dimensional vector spaces over a fixed field are completely classified by a single integer: the dimension. We begin by establishing the algebraic prerequisites (ring and field homomorphisms, polynomial rings) before developing the theory of vector spaces systematically.

4.1 Ring and Field Homomorphisms

Just as group homomorphisms preserve the group structure, ring and field homomorphisms preserve both operations simultaneously. These maps are fundamental for understanding how algebraic structures relate to one another.

Definition 4.1.1: Ring homomorphism

A **ring homomorphism** is a map $\varphi : R \rightarrow S$ between rings that respects both operations:

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$;
2. $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$;
3. $\varphi(1_R) = 1_S$.

A **field homomorphism** is a ring homomorphism between fields.

Note:-

Condition (1) implies that φ is a group homomorphism $(R, +) \rightarrow (S, +)$, so we automatically get $\varphi(0) = 0$ and $\varphi(-a) = -\varphi(a)$. However, condition (3) does not follow from condition (2), even for fields! Consider $\varphi = 0$: this satisfies $\varphi(ab) = 0 = 0 \cdot 0 = \varphi(a)\varphi(b)$, but $\varphi(1) = 0 \neq 1$. We must include (3) as a separate axiom.

Proposition 4.1.1 Field homomorphisms are injective

If $\varphi : k \rightarrow K$ is a field homomorphism, then φ is injective.

Proof: Let $a \in k$ with $a \neq 0$. Since k is a field, there exists $b = a^{-1}$ such that $ab = 1_k$. Applying φ :

$$\varphi(a)\varphi(b) = \varphi(ab) = \varphi(1_k) = 1_K \neq 0_K.$$

This implies $\varphi(a) \neq 0_K$ (if $\varphi(a) = 0$, then $\varphi(a)\varphi(b) = 0 \neq 1_K$).

Thus $\ker(\varphi) = \{a \in k : \varphi(a) = 0_K\} = \{0_k\}$, so φ is injective. $\square$

Note:-

This means every field homomorphism is an embedding. There are no “collapsing” maps between fields. This stands in sharp contrast to group and ring homomorphisms, which can have large kernels. The reason is that fields have no nontrivial ideals (a consequence of having multiplicative inverses), so the kernel of any field homomorphism must be either $\{0\}$ or the entire field—but the latter is ruled out by $\varphi(1) = 1 \neq 0$.

4.2 Polynomial Rings and Field Extensions

Polynomials are central objects connecting algebra to analysis, number theory, and geometry. Given a field k , we construct the ring of polynomials $k[x]$ and explore how to enlarge k to contain roots of polynomials.

4.2.1 The polynomial ring $k[x]$

Definition 4.2.1: Polynomial ring

Given a field k , the **polynomial ring in one variable** is

$$k[x] := \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in k, n \in \mathbb{N}\}.$$

Here x is a **formal variable**—it is not an element of k or any particular set. A polynomial is simply a finite sequence of coefficients, and we add and multiply term by term using the familiar rules.

Note:-

The adjective “formal” is important. When we write $x^2 + 1 \in \mathbb{R}^x$, we are not thinking of x as a real number—we are thinking of $x^2 + 1$ as a symbolic expression. However, we can evaluate a polynomial at any element of k (or any field extension of k): given $f(x) = a_0 + a_1x + \cdots + a_nx^n$ and $r \in k$, we define $f(r) = a_0 + a_1r + \cdots + a_nr^n \in k$.

The polynomial ring $k[x]$ is indeed a ring: it is closed under addition and multiplication, has identity 1, and satisfies all the ring axioms. However, it is not a field.

Proposition 4.2.1 $k[x]$ is not a field

The polynomial ring $k[x]$ has no multiplicative inverses for non-constant polynomials. For instance, x has no inverse in $k[x]$.

Proof: Suppose $f(x) \in k[x]$ satisfies $x \cdot f(x) = 1$. Then $\deg(x \cdot f(x)) = 1 + \deg(f(x)) \geq 1$, but $\deg(1) = 0$. Contradiction. $\square$

4.2.2 The field of rational functions

To make $k[x]$ into a field, we must allow “fractions” of polynomials, just as we pass from $\mathbb{Z}$ to $\mathbb{Q}$.

Definition 4.2.2: Field of rational functions

The **field of rational functions** over k is

$$k(x) := \left\{ \frac{p}{q} \mid p, q \in k[x], q \neq 0 \right\} / \sim$$

where $\frac{p}{q} \sim \frac{p'}{q'}$ iff $pq' = p'q$. Addition and multiplication are defined as for ordinary fractions.

This construction generalizes naturally to polynomials in any number of variables, giving $k(x_1, \dots, x_n)$, the field of rational functions in n variables.

4.2.3 Formal power series

Another way to extend $k[x]$ is to allow infinite sums—formal power series.

Definition 4.2.3: Formal power series

The ring of formal power series over k is

$$k[[x]] := \left\{ \sum_{i=0}^{\infty} a_i x^i \mid a_i \in k \right\}.$$

Addition and multiplication are defined term by term: given $f = \sum a_i x^i$ and $g = \sum b_i x^i$, we have $f + g = \sum (a_i + b_i) x^i$ and $fg = \sum c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$.

Note:-

The product formula makes sense because each coefficient c_k is a finite sum—only finitely many pairs (i, j) satisfy $i + j = k$ with $i, j \geq 0$. This is why we call these “formal” power series: we never worry about convergence. The ring $k[[x]]$ contains $k[x]$ as a subring (polynomials are power series with only finitely many nonzero coefficients).

Unlike $k[x]$, the ring $k[[x]]$ has many more invertible elements.

Lemma 4.2.1 Invertibility in $k[[x]]$

A power series $f = \sum_{i=0}^{\infty} a_i x^i \in k[[x]]$ has a multiplicative inverse if and only if $a_0 \neq 0$.

Proof: $(\Rightarrow)$ If $fg = 1$ for some $g = \sum b_i x^i$, then comparing constant terms gives $a_0 b_0 = 1$, so $a_0 \neq 0$.

$(\Leftarrow)$ Assume $a_0 \neq 0$. We construct $g = \sum b_i x^i$ satisfying $fg = 1$ inductively.

The constant term equation is $a_0 b_0 = 1$, so $b_0 = a_0^{-1}$ (which exists since k is a field).

The coefficient of x^1 equation is $a_0 b_1 + a_1 b_0 = 0$, so $b_1 = -a_0^{-1}(a_1 b_0) = -a_0^{-1} a_1 a_0^{-1}$.

In general, the coefficient of x^n gives $\sum_{i=0}^n a_i b_{n-i} = 0$ for $n \geq 1$, i.e.,

$$a_0 b_n + \sum_{i=1}^n a_i b_{n-i} = 0 \implies b_n = -a_0^{-1} \sum_{i=1}^n a_i b_{n-i}.$$

Each b_n depends only on $b_0, \dots, b_{n-1}$, so we can solve inductively. □

Example 4.2.1

The geometric series: in $k[[x]]$, we have $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots = \sum_{i=0}^{\infty} x^i$. This is verified directly: $(1 - x)(1 + x + x^2 + \dots) = 1$ since all higher terms cancel.

4.2.4 Laurent series and field of fractions

Every nonzero element of $k[[x]]$ can be written as $x^m(a_m + a_{m+1}x + \dots)$ where $a_m \neq 0$. To get a field, we need to invert x^m —i.e., allow negative powers.

Definition 4.2.4: Field of Laurent series

The field of Laurent series over k is

$$k((x)) := \left\{ \sum_{i=m}^{\infty} a_i x^i \mid m \in \mathbb{Z}, a_i \in k \right\}.$$

Elements are “power series starting at some (possibly negative) power of x .”

Proposition 4.2.2 $k((x))$ is a field

Every nonzero Laurent series has a multiplicative inverse.

Proof: Let $f = \sum_{i=m}^{\infty} a_i x^i$ with $a_m \neq 0$. Then $f = x^m g$ where $g = a_m + a_{m+1}x + \cdots \in k[[x]]$ has nonzero constant term. By the lemma, g^{-1} exists in $k[[x]]$. Thus $f^{-1} = x^{-m} g^{-1} \in k((x))$. $\square$

4.2.5 Roots of polynomials and field extensions

Given a polynomial $f \in k[x]$ with $\deg(f) \geq 1$, we can ask: does f have a root in k ? If not, can we extend k to a larger field K containing a root?

Definition 4.2.5: Root and field extension

Let $f \in k[x]$ and K be a field containing k . An element $r \in K$ is a root of f if $f(r) = 0$. We call K a field extension of k , written K/k or $k \subset K$.

Example 4.2.2

The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ has no roots in $\mathbb{Q}$, but we can form the extension

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\},$$

which is a field containing $\mathbb{Q}$ and the root $\sqrt{2}$. To verify this is a field, we need to check multiplicative inverses: for $a + b\sqrt{2} \neq 0$,

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2} \sqrt{2} \in \mathbb{Q}(\sqrt{2}).$$

Note that $a^2 - 2b^2 \neq 0$ when $(a, b) \neq (0, 0)$, since $\sqrt{2} \notin \mathbb{Q}$.

Example 4.2.3

Similarly, $x^2 + 1 \in \mathbb{R}[x]$ has no real roots. The extension

$$\mathbb{R}(i) = \mathbb{R}(\sqrt{-1}) = \{a + bi : a, b \in \mathbb{R}\} = \mathbb{C}$$

is the field of complex numbers, which contains roots of $x^2 + 1$ (namely $\pm i$).

Definition 4.2.6: Algebraically closed field

A field k is algebraically closed if every non-constant polynomial $f \in k[x]$ has a root in k . Equivalently, every polynomial factors completely into linear factors over k .

Note:-

The Fundamental Theorem of Algebra states that $\mathbb{C}$ is algebraically closed. Over an algebraically closed field, every non-constant polynomial has a root, and there are no further algebraic extensions. This is why $\mathbb{C}$ is often called the “algebraic closure” of $\mathbb{R}$. In contrast, $\mathbb{Q}$ is very far from algebraically closed—there are infinitely many distinct algebraic extensions of $\mathbb{Q}$.

4.3 Characteristic of a Field

Every field k has a distinguished ring homomorphism from the integers, whose kernel determines an important invariant.

Proposition 4.3.1 Universal map from $\mathbb{Z}$

For any field k , there is a unique ring homomorphism $\varphi : \mathbb{Z} \rightarrow k$ defined by $\varphi(1) = 1_k$. Explicitly, $\varphi(n) = n \cdot 1_k$ (add 1_k to itself n times for $n > 0$; negate for $n < 0$).

Proof: Uniqueness: Any ring homomorphism must satisfy $\varphi(1) = 1_k$, and since $\mathbb{Z} = \langle 1 \rangle$ as an additive group, φ is determined by its value on 1.

Existence: Define $\varphi(n) = n \cdot 1_k$. Then $\varphi(m+n) = (m+n) \cdot 1_k = m \cdot 1_k + n \cdot 1_k = \varphi(m) + \varphi(n)$ and $\varphi(mn) = (mn) \cdot 1_k = (m \cdot 1_k)(n \cdot 1_k) = \varphi(m)\varphi(n)$ (verify by induction). Also $\varphi(1) = 1_k$. $\square$

The kernel of φ is a subgroup of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for some $n \geq 0$.

Proposition 4.3.2 Kernel is zero or prime

The kernel $\ker(\varphi)$ equals $n\mathbb{Z}$ where either $n = 0$ or $n = p$ is prime.

Proof: Suppose $n > 0$ and n is not prime, so $n = ab$ with $1 < a, b < n$. Then

$$\varphi(n) = \varphi(ab) = \varphi(a)\varphi(b) = 0_k.$$

In a field, $xy = 0$ implies $x = 0$ or $y = 0$. So $\varphi(a) = 0$ or $\varphi(b) = 0$, meaning $a \in \ker(\varphi)$ or $b \in \ker(\varphi)$. But $\ker(\varphi) = n\mathbb{Z}$ and $a, b < n$, so a or b is in $n\mathbb{Z}$ only if $a = 0$ or $b = 0$ —contradicting $a, b > 1$.

Thus if $n > 0$, then n must be prime. $\square$

Definition 4.3.1: Characteristic

The **characteristic** of a field k , denoted $\text{char}(k)$, is:

- 0 if $\ker(\varphi) = \{0\}$, i.e., $\varphi : \mathbb{Z} \hookrightarrow k$ is injective;
- p if $\ker(\varphi) = p\mathbb{Z}$ for a prime p .

Equivalently, $\text{char}(k) = p$ means $\underbrace{1_k + 1_k + \cdots + 1_k}_{p \text{ times}} = 0_k$, and p is the smallest such positive integer.

Example 4.3.1

- $\text{char}(\mathbb{Q}) = \text{char}(\mathbb{R}) = \text{char}(\mathbb{C}) = 0$. The map $\varphi : \mathbb{Z} \rightarrow \mathbb{Q}$ is injective.
- $\text{char}(\mathbb{Z}/p) = p$ for prime p . Indeed, $p \cdot 1 = 0$ in $\mathbb{Z}/p$.
- $\text{char}(k(x)) = \text{char}(k[[x]]) = \text{char}(k((x))) = \text{char}(k)$.

Note:-

The field $\mathbb{Z}/p$ is often denoted $\mathbb{F}_p$ and called the **prime field** of characteristic p . Every field of characteristic p contains a copy of $\mathbb{F}_p$; every field of characteristic 0 contains a copy of $\mathbb{Q}$. Thus $\mathbb{F}_p$ and $\mathbb{Q}$ are the “minimal” fields in their respective characteristics.

Theorem 4.3.1 Existence and uniqueness of finite fields

For every prime p and $n \geq 1$, there exists a unique (up to isomorphism) field with exactly p^n elements, denoted $\mathbb{F}_{p^n}$ or $\text{GF}(p^n)$. These are all the finite fields.

Note:-

We will not prove this theorem here. The finite field $\mathbb{F}_{p^n}$ can be constructed as a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$. There are also infinite fields of characteristic p —for example, $\mathbb{F}_p(x)$, the field of rational functions

over $\mathbb{F}_p$.

4.4 Vector Spaces

We now arrive at the central structure of linear algebra. A vector space is a set with two operations—addition and scalar multiplication—satisfying axioms that generalize familiar properties of $\mathbb{R}^n$.

Definition 4.4.1: Vector space

Fix a field k . A **vector space over k** (or **k -vector space**) is a set V equipped with two operations:

1. **Vector addition:** $+: V \times V \rightarrow V$;
2. **Scalar multiplication:** $\cdot: k \times V \rightarrow V$;

satisfying the following axioms:

1. $(V, +)$ is an abelian group with identity 0 (the **zero vector**);
2. $1 \cdot v = v$ for all $v \in V$ (identity for scalar multiplication);
3. $(ab) \cdot v = a \cdot (b \cdot v)$ for all $a, b \in k$ and $v \in V$ (associativity);
4. $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in k$ and $v \in V$ (distributivity over scalar addition);
5. $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in k$ and $v, w \in V$ (distributivity over vector addition).

Elements of V are called **vectors**; elements of k are called **scalars**.

Note:-

The distributive laws (4) and (5) connect the two operations. Note that axiom (1) implies $0_V + v = v$ and $v + (-v) = 0_V$, while axiom (2)-(5) give us things like $0_k \cdot v = 0_V$ (proof: $0_k \cdot v = (0_k + 0_k) \cdot v = 0_k \cdot v + 0_k \cdot v$, so $0_k \cdot v = 0_V$ by cancellation). We also get $(-1) \cdot v = -v$.

4.4.1 Fundamental examples

Example 4.4.1

The standard vector space k^n : For any $n \geq 1$, the set

$$k^n = \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

is a vector space with componentwise operations:

$$\begin{aligned} (a_1, \dots, a_n) + (b_1, \dots, b_n) &= (a_1 + b_1, \dots, a_n + b_n), \\ c \cdot (a_1, \dots, a_n) &= (ca_1, \dots, ca_n). \end{aligned}$$

This is the prototypical n -dimensional vector space.

Example 4.4.2

Infinite sequences k^∞ : The set of all sequences in k ,

$$k^\infty = \{(a_i)_{i \in \mathbb{N}} : a_i \in k\} = \{(a_1, a_2, a_3, \dots) : a_i \in k\},$$

is a vector space with componentwise operations. It contains the subspace of **eventually zero sequences**—those with $a_i = 0$ for all sufficiently large i .

Example 4.4.3

Function spaces: For any set S , the set

$$k^S = \{f : S \rightarrow k\}$$

of all functions from S to k is a vector space with pointwise operations: $(f + g)(s) = f(s) + g(s)$ and $(c \cdot f)(s) = c \cdot f(s)$.

When $S = \mathbb{N}$, we recover k^∞ . When $S = \{1, \dots, n\}$, we recover k^n .

Example 4.4.4

Polynomial and power series spaces:

- $k[x]$ is a k -vector space (addition is polynomial addition; scalar multiplication is multiplying each coefficient by the scalar).
- $k[[x]]$ is a k -vector space.
- For each $d \geq 0$, the set $k[x]_{\leq d} = \{f \in k[x] : \deg(f) \leq d\}$ of polynomials of degree at most d is a k -vector space.

Example 4.4.5

Spaces of maps $\mathbb{R} \rightarrow \mathbb{R}$: Consider $V = \{f : \mathbb{R} \rightarrow \mathbb{R}\}$, all functions from $\mathbb{R}$ to $\mathbb{R}$. This contains many interesting subspaces:

- Continuous functions $C(\mathbb{R})$;
- Differentiable functions $C^1(\mathbb{R})$;
- Smooth (infinitely differentiable) functions $C^\infty(\mathbb{R})$;
- Polynomials $\mathbb{R}^x$.

Each is closed under addition and scalar multiplication, forming a vector space.

4.4.2 Subspaces

Definition 4.4.2: Subspace

A **subspace** of a vector space V is a nonempty subset $W \subseteq V$ that is closed under addition and scalar multiplication:

1. $w_1, w_2 \in W \implies w_1 + w_2 \in W$;
2. $w \in W, a \in k \implies a \cdot w \in W$.

Equivalently, W is a subspace if it is a vector space under the inherited operations.

Proposition 4.4.1 Subspace criterion

A nonempty subset $W \subseteq V$ is a subspace if and only if $aw_1 + bw_2 \in W$ for all $a, b \in k$ and $w_1, w_2 \in W$.

Proof: ($\implies$) If W is a subspace, then $aw_1 \in W$ and $bw_2 \in W$ by scalar multiplication closure, and $aw_1 + bw_2 \in W$ by addition closure.

($\impliedby$) Taking $a = b = 1$ gives closure under addition. Taking $b = 0$ gives closure under scalar multiplication. $\square$

Note:-

The condition “nonempty” is required. Alternatively, we can replace it with “ $0 \in W$,” which follows from taking $a = 0$ in the scalar multiplication closure. So a subspace is precisely a nonempty subset closed under linear combinations.

Example 4.4.6

- $\{0\}$ and V itself are always subspaces (the **trivial subspaces**).
- In $\mathbb{R}^3$, subspaces are: $\{0\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.
- In $k[x]$, the polynomials of degree $\leq n$ form a subspace. The polynomials of degree exactly n do not form a subspace (not closed under addition: $x^n + (-x^n) = 0$).

4.4.3 Direct Sums of Subspaces

When can we decompose a vector space into “independent” pieces? The notion of direct sum captures exactly this.

Definition 4.4.3: Sum of subspaces

Given subspaces $U, W \subseteq V$, their **sum** is

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is the smallest subspace of V containing both U and W .

Proposition 4.4.2 Sum is a subspace

For subspaces $U, W \subseteq V$, the sum $U + W$ is a subspace of V .

Proof: Let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ be elements of $U + W$, and let $\lambda \in k$. Then

$$v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2) \in U + W$$

since $u_1 + u_2 \in U$ and $w_1 + w_2 \in W$. Similarly, $\lambda v_1 = \lambda u_1 + \lambda w_1 \in U + W$. The zero vector $0 = 0 + 0 \in U + W$. $\square$

Definition 4.4.4: Direct sum of subspaces

The sum $U + W$ is called a **direct sum**, written $U \oplus W$, if $U \cap W = \{0\}$. Equivalently, every element of $U + W$ can be written uniquely as $u + w$ with $u \in U$ and $w \in W$.

Proposition 4.4.3 Characterizations of direct sum

For subspaces $U, W \subseteq V$, the following are equivalent:

1. $U + W$ is a direct sum (i.e., $U \cap W = \{0\}$).
2. Every $v \in U + W$ has a unique representation $v = u + w$ with $u \in U, w \in W$.
3. $u + w = 0$ with $u \in U, w \in W$ implies $u = w = 0$.

Proof: (1) $\Rightarrow$ (2): Suppose $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. The left side is in U , the right side is in W , so both are in $U \cap W = \{0\}$. Thus $u_1 = u_2$ and $w_1 = w_2$.

(2) $\Rightarrow$ (3): If $u + w = 0$, then $u + w = 0 + 0$. By uniqueness, $u = 0$ and $w = 0$.

(3) $\Rightarrow$ (1): If $v \in U \cap W$, then $v + (-v) = 0$ with $v \in U$ and $-v \in W$. By (3), $v = 0$. $\square$

Theorem 4.4.1 Dimension formula for sums

For finite-dimensional subspaces $U, W \subseteq V$:

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

In particular, if $U + W = U \oplus W$ is a direct sum, then $\dim(U \oplus W) = \dim(U) + \dim(W)$.

Proof: Let $\{v_1, \dots, v_k\}$ be a basis of $U \cap W$. Extend to a basis $\{v_1, \dots, v_k, u_1, \dots, u_m\}$ of U and to a basis $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ of W .

Claim: $\{v_1, \dots, v_k, u_1, \dots, u_m, w_1, \dots, w_n\}$ is a basis of $U + W$.

Spanning: Any $x \in U + W$ is $x = u + w$ with $u \in U, w \in W$. Express u and w in terms of the bases above; the result is a linear combination of all the listed vectors.

Independence: Suppose $\sum a_i v_i + \sum b_j u_j + \sum c_\ell w_\ell = 0$. Then

$$\sum c_\ell w_\ell = -\sum a_i v_i - \sum b_j u_j \in U.$$

But $\sum c_\ell w_\ell \in W$, so $\sum c_\ell w_\ell \in U \cap W = \text{Span}\{v_1, \dots, v_k\}$. Since $\{v_1, \dots, v_k, w_1, \dots, w_n\}$ is independent, all $c_\ell = 0$. Similarly, all $b_j = 0$, and then all $a_i = 0$.

Thus $\dim(U + W) = k + m + n = (k + m) + (k + n) - k = \dim(U) + \dim(W) - \dim(U \cap W)$. $\square$

Definition 4.4.5: Internal direct sum decomposition

We say V is the **(internal) direct sum** of subspaces U and W , written $V = U \oplus W$, if:

1. $V = U + W$ (every vector is a sum of elements from U and W), and
2. $U \cap W = \{0\}$ (the decomposition is unique).

More generally, $V = U_1 \oplus U_2 \oplus \dots \oplus U_n$ if every $v \in V$ can be written uniquely as $v = u_1 + u_2 + \dots + u_n$ with $u_i \in U_i$.

Example 4.4.7

- In $\mathbb{R}^3$, let $U = \{(x, 0, 0) : x \in \mathbb{R}\}$ (the x -axis) and $W = \{(0, y, z) : y, z \in \mathbb{R}\}$ (the yz -plane). Then $\mathbb{R}^3 = U \oplus W$.
- Let $V = k^n$ and $U = \text{Span}\{e_1, \dots, e_m\}$, $W = \text{Span}\{e_{m+1}, \dots, e_n\}$. Then $k^n = U \oplus W$.
- If V is the space of $n \times n$ matrices, let $S =$ symmetric matrices and $A =$ antisymmetric matrices. Then $V = S \oplus A$ (any matrix $M = \frac{1}{2}(M + M^T) + \frac{1}{2}(M - M^T)$).

Note:-

When we diagonalize a linear operator, we decompose the vector space as a direct sum of eigenspaces. When we study quotient spaces, we often use the fact that $V = U \oplus W$ implies $V/U \cong W$; if that doesn't make sense right now, don't worry, we'll get to it. The direct sum structure makes computations tractable by breaking a problem into independent pieces.

4.4.4 External Direct Sums and Products

We can also construct new vector spaces from old ones by taking products or direct sums “externally.”

Definition 4.4.6: Direct product (external direct sum)

Given vector spaces $V_1, \dots, V_n$ over the same field k , their **direct product** (or **external direct sum**) is

$$V_1 \times V_2 \times \cdots \times V_n = \{(v_1, v_2, \dots, v_n) : v_i \in V_i\}$$

with componentwise operations:

$$\begin{aligned}(v_1, \dots, v_n) + (w_1, \dots, w_n) &= (v_1 + w_1, \dots, v_n + w_n), \\ \lambda(v_1, \dots, v_n) &= (\lambda v_1, \dots, \lambda v_n).\end{aligned}$$

This is also denoted $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ or $\bigoplus_{i=1}^n V_i$.

Proposition 4.4.4 Dimension of direct product

If each V_i is finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_n) = \dim(V_1) + \cdots + \dim(V_n).$$

Proof: If $\mathcal{B}_i = \{v_{i,1}, \dots, v_{i,d_i}\}$ is a basis of V_i , then

$$\{(v_{1,1}, 0, \dots, 0), \dots, (v_{1,d_1}, 0, \dots, 0), (0, v_{2,1}, 0, \dots, 0), \dots, (0, \dots, 0, v_{n,d_n})\}$$

is a basis of $V_1 \times \cdots \times V_n$. This has $d_1 + \cdots + d_n$ elements. □

Note:-

The external direct sum $V_1 \times V_2$ and the internal direct sum $U \oplus W \subseteq V$ are related as follows: if $V = U \oplus W$, then the map

$$U \times W \rightarrow V, \quad (u, w) \mapsto u + w$$

is an isomorphism. Conversely, $V_1 \times V_2$ contains copies of V_1 and V_2 as subspaces (embedded as $V_1 \times \{0\}$ and $\{0\} \times V_2$), and $V_1 \times V_2$ is their internal direct sum.

Example 4.4.8

The space $\mathbb{R}^{m+n}$ is isomorphic to $\mathbb{R}^m \times \mathbb{R}^n$ via $(x_1, \dots, x_m, y_1, \dots, y_n) \mapsto ((x_1, \dots, x_m), (y_1, \dots, y_n))$. This is why we freely identify $\mathbb{R}^m \oplus \mathbb{R}^n$ with $\mathbb{R}^{m+n}$.

4.5 Span, Linear Independence, and Basis

These three concepts—span, linear independence, and basis—are the building blocks of linear algebra. They allow us to describe vector spaces in terms of generating sets and coordinates.

4.5.1 Linear combinations and span

Definition 4.5.1: Linear combination

A **linear combination** of vectors $v_1, \dots, v_n \in V$ is any vector of the form

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

where $a_1, \dots, a_n \in k$ are scalars.

Definition 4.5.2: Span

Given vectors $v_1, \dots, v_n \in V$, the span of $\{v_1, \dots, v_n\}$ is the set of all linear combinations:

$$\text{Span}(v_1, \dots, v_n) = \{a_1 v_1 + \dots + a_n v_n : a_i \in k\}.$$

This is the **smallest subspace** of V containing $v_1, \dots, v_n$.

We say $v_1, \dots, v_n$ span V (or are a spanning set for V) if $\text{Span}(v_1, \dots, v_n) = V$.

Proposition 4.5.1 Span is a subspace

For any $v_1, \dots, v_n \in V$, the span $\text{Span}(v_1, \dots, v_n)$ is a subspace of V .

Proof: Let $W = \text{Span}(v_1, \dots, v_n)$. Clearly $0 = 0 \cdot v_1 + \dots + 0 \cdot v_n \in W$, so $W \neq \emptyset$. For closure: if $w = \sum a_i v_i$ and $w' = \sum a'_i v_i$ are in W , and $c, c' \in k$, then

$$cw + c'w' = \sum (ca_i + c'a'_i)v_i \in W.$$

□

The span can also be defined for infinite (or even arbitrary) sets of vectors.

Definition 4.5.3: Span of an arbitrary set

For $S \subseteq V$, the span of S is the set of all finite linear combinations of elements of S :

$$\text{Span}(S) = \left\{ \sum_{i=1}^n a_i v_i : n \geq 0, a_i \in k, v_i \in S \right\}.$$

When $n = 0$, the empty sum is 0, so $0 \in \text{Span}(S)$ always.

4.5.2 Linear independence

Definition 4.5.4: Linear independence

Vectors $v_1, \dots, v_n \in V$ are linearly independent if the only way to write 0 as a linear combination is the trivial way:

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0 \implies a_1 = a_2 = \dots = a_n = 0.$$

Otherwise, they are linearly dependent—there exist scalars, not all zero, such that $\sum a_i v_i = 0$.

Note:-

Linear dependence means some vector can be expressed as a linear combination of the others. If $a_1 v_1 + \dots + a_n v_n = 0$ with (say) $a_1 \neq 0$, then

$$v_1 = -\frac{a_2}{a_1} v_2 - \dots - \frac{a_n}{a_1} v_n.$$

So v_1 is “redundant”—it’s already in the span of the others.

Proposition 4.5.2 Equivalent characterizations of linear independence

The following are equivalent for $v_1, \dots, v_n \in V$:

1. $v_1, \dots, v_n$ are linearly independent.

2. No v_i is a linear combination of the others.
3. The linear map $\phi : k^n \rightarrow V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is injective.

Proof: (1) $\Leftrightarrow$ (3): $\ker(\phi) = \{(a_1, \dots, a_n) : \sum a_i v_i = 0\}$. Linear independence says $\ker(\phi) = \{0\}$, which is equivalent to ϕ being injective.

(1) $\Rightarrow$ (2): Suppose $v_j = \sum_{i \neq j} b_i v_i$. Then $\sum_{i \neq j} b_i v_i + (-1)v_j = 0$ is a nontrivial linear combination (coefficient of v_j is $-1 \neq 0$), contradicting independence.

(2) $\Rightarrow$ (1): Suppose $\sum a_i v_i = 0$ with some $a_j \neq 0$. Then $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, contradicting (2). $\square$

Similarly, we can characterize spanning:

Proposition 4.5.3 Characterization of spanning

The following are equivalent:

1. $v_1, \dots, v_n$ span V .
2. Every $v \in V$ can be written as $\sum a_i v_i$ for some $a_i \in k$.
3. The linear map $\phi : k^n \rightarrow V$ defined by $\phi(a_1, \dots, a_n) = \sum a_i v_i$ is surjective.

4.5.3 Basis

Definition 4.5.5: Basis

A **basis** for V is a set $\{v_1, \dots, v_n\}$ of vectors that are both linearly independent and span V . Equivalently, the linear map $\phi : k^n \rightarrow V$ is bijjective.

Proposition 4.5.4 Unique representation

If $\{v_1, \dots, v_n\}$ is a basis for V , then every vector $v \in V$ can be written uniquely as

$$v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

for some $a_i \in k$. The scalars $(a_1, \dots, a_n)$ are called the coordinates of v with respect to this basis.

Proof: Existence: Since $\{v_i\}$ spans V , every v is some linear combination.

Uniqueness: If $v = \sum a_i v_i = \sum a'_i v_i$, then $\sum (a_i - a'_i) v_i = 0$. By linear independence, $a_i - a'_i = 0$ for all i , so $a_i = a'_i$. $\square$

Example 4.5.1

Standard basis of k^n : The vectors

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

form the standard basis of k^n . Any $(a_1, \dots, a_n) = a_1 e_1 + \dots + a_n e_n$.

Example 4.5.2

Non-standard basis of k^2 : In k^2 , the vectors $(1, 0)$ and $(0, 1)$ form the standard basis. But so do $(1, 1)$ and $(1, -1)$ (for most k —we need $\text{char}(k) \neq 2$ for these to be independent).

Example 4.5.3

Basis of $k[x]_{\leq n}$: The polynomials $\{1, x, x^2, \dots, x^n\}$ form a basis for $k[x]_{\leq n}$, the space of polynomials of degree $\leq n$.

Question 1

1. Does $k[x]$ have a finite basis? What is a basis for $k[x]$?
2. Does $k[[x]]$ have a basis?

Solution

(1) The polynomial ring $k[x]$ does not have a finite basis, because any finite set $\{p_1, \dots, p_n\}$ of polynomials has bounded degree, so cannot span polynomials of higher degree. However, $k[x]$ has the infinite basis $\{1, x, x^2, x^3, \dots\}$. Every polynomial is a finite linear combination of these monomials, and they are linearly independent (a nontrivial linear combination would give a nonzero polynomial equaling 0).

(2) This is subtle. By the axiom of choice, $k[[x]]$ has a basis (every vector space does). But this basis cannot be explicitly described—it is uncountably infinite and highly non-constructive. Unlike $k[x]$, the monomials $\{1, x, x^2, \dots\}$ do not form a basis of $k[[x]]$: while they are linearly independent, they do not span (a power series like $\sum_{i=0}^{\infty} x^i$ is not a finite linear combination of monomials).

4.6 Finite-Dimensional Vector Spaces and Dimension

The notion of dimension will appear constantly throughout linear algebra and beyond. Unlike groups, where classification is intractable, finite-dimensional vector spaces are completely determined by a single integer—their dimension. We establish this remarkable fact through a sequence of lemmas about the interplay between spanning sets and linearly independent sets.

Definition 4.6.1: Finite-dimensional

A vector space V is **finite-dimensional** if there exists a finite spanning set for V , i.e., vectors $v_1, \dots, v_n$ such that $\text{Span}(v_1, \dots, v_n) = V$. Otherwise, V is **infinite-dimensional**.

4.6.1 Extracting bases from spanning sets

Our first key result shows that every spanning set contains a basis.

Lemma 4.6.1 Spanning sets contain bases

If $\{v_1, \dots, v_n\}$ spans V , then some subset of $\{v_1, \dots, v_n\}$ is a basis for V .

Proof: We construct a basis by iteratively removing redundant vectors.

If $\{v_1, \dots, v_n\}$ is linearly independent, we are done—it is already a basis.

Otherwise, there exists a nontrivial relation $\sum_{i=1}^n a_i v_i = 0$ with some $a_j \neq 0$. Then

$$v_j = -\frac{1}{a_j} \sum_{i \neq j} a_i v_i \in \text{Span}(\{v_i : i \neq j\}).$$

So $\{v_i : i \neq j\}$ still spans V (any linear combination involving v_j can be rewritten using the other vectors).

Remove v_j and repeat. Since we start with finitely many vectors, this process terminates. When it does, the remaining vectors are linearly independent (else we could remove another) and still span V , hence form a basis. $\square$

Corollary 4.6.1 Finite-dimensional spaces have bases

Every finite-dimensional vector space has a basis.

4.6.2 Extending linearly independent sets to bases

The dual result shows that any linearly independent set can be enlarged to a basis.

Lemma 4.6.2 Linearly independent sets extend to bases

If V is finite-dimensional and $\{v_1, \dots, v_m\}$ is linearly independent, then there exists a basis of V containing $\{v_1, \dots, v_m\}$.

Proof: Let $\{w_1, \dots, w_r\}$ be a finite spanning set for V (exists by finite-dimensionality). We enlarge $\{v_1, \dots, v_m\}$ to a basis using vectors from $\{w_j\}$.

Define $W_0 = \text{Span}(v_1, \dots, v_m)$, with basis $\{v_1, \dots, v_m\}$ (our independent set).

Induction step: Suppose we have constructed a basis $\mathcal{B}_{j-1}$ for $W_{j-1} = \text{Span}(v_1, \dots, v_m, w_{i_1}, \dots, w_{i_{j-1}})$.

Consider w_j :

- If $w_j \in W_{j-1}$, set $W_j = W_{j-1}$ and $\mathcal{B}_j = \mathcal{B}_{j-1}$.
- If $w_j \notin W_{j-1}$, then $\mathcal{B}_{j-1} \cup \{w_j\}$ is linearly independent (otherwise w_j would be a linear combination of $\mathcal{B}_{j-1}$, contradicting $w_j \notin W_{j-1}$). Set $W_j = \text{Span}(\mathcal{B}_{j-1} \cup \{w_j\})$ and $\mathcal{B}_j = \mathcal{B}_{j-1} \cup \{w_j\}$.

After processing all w_j , we have $W_r = \text{Span}(v_1, \dots, v_m, w_1, \dots, w_r) \supseteq \text{Span}(w_1, \dots, w_r) = V$. So $\mathcal{B}_r$ is a linearly independent set spanning V , i.e., a basis containing $\{v_1, \dots, v_m\}$. $\square$

4.6.3 Invariance of basis size

The following theorem shows that the size of a basis is intrinsic to the vector space, not dependent on the choice of basis.

Theorem 4.6.1 All bases have the same size

If V has a basis with n elements and another basis with m elements, then $m = n$.

Proof: Let $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_m\}$ be bases for V . We show $m \leq n$ and $n \leq m$, hence $m = n$.

Claim: $\{v_1, \dots, v_{m-1}, w_j\}$ is a basis for some choice of reordering, for each j we process.

We proceed by a “replacement” argument. Start with the basis $\{v_1, \dots, v_n\}$.

Since $w_1 \in V = \text{Span}(v_1, \dots, v_n)$, we can write $w_1 = \sum_{i=1}^n a_i v_i$. Not all $a_i = 0$ (else $w_1 = 0$, contradicting independence of $\{w_j\}$). By reordering, assume $a_n \neq 0$.

Then

$$v_n = a_n^{-1}(w_1 - \sum_{i=1}^{n-1} a_i v_i) \in \text{Span}(v_1, \dots, v_{n-1}, w_1).$$

So

$$\text{Span}(v_1, \dots, v_{n-1}, w_1) \supseteq \text{Span}(v_1, \dots, v_n) = V,$$

and thus $\{v_1, \dots, v_{n-1}, w_1\}$ spans V . Since it has n elements and spans, it contains a basis—but removing any element would give $n - 1$ vectors. We verify it’s actually linearly independent: if

$$\sum_{i=1}^{n-1} b_i v_i + c w_1 = 0$$

with $c \neq 0$, then $w_1 \in \text{Span}(v_1, \dots, v_{n-1})$, so $v_n \in \text{Span}(v_1, \dots, v_{n-1})$, contradicting independence of $\{v_i\}$. Thus $c = 0$, and then all $b_i = 0$. So $\{v_1, \dots, v_{n-1}, w_1\}$ is a basis.

Repeat: $w_2 = \sum_{i=1}^{n-1} a_i v_i + c w_1$. Some $a_i \neq 0$ (else $w_2 \in \text{Span}(w_1)$, contradicting independence). Replace that v_i with w_2 .

Continue this process, replacing one v with one w each time. After m steps, we have a basis

$$\{v_{i_1}, \dots, v_{i_{n-m}}, w_1, \dots, w_m\}$$

(assuming $m \leq n$).

If $m > n$, the process would terminate before incorporating all w_j , but then some w_j would be in the span of earlier w 's, contradicting independence. Hence $m \leq n$.

By symmetry, $n \leq m$. Thus $m = n$. □

Definition 4.6.2: Dimension

The **dimension** of a finite-dimensional vector space V , denoted $\dim(V)$ or $\dim_k(V)$, is the number of elements in any basis. The theorem above shows this is well-defined. By convention, $\dim(\{0\}) = 0$ (the empty set is a basis for the zero space).

Corollary 4.6.2 Isomorphism with k^n

Every finite-dimensional vector space V with $\dim(V) = n$ is isomorphic to k^n .

Proof: Let $\{v_1, \dots, v_n\}$ be a basis for V . The map

$$\phi : k^n \rightarrow V, \quad (a_1, \dots, a_n) \mapsto \sum_{i=1}^n a_i v_i$$

is linear (easily verified), injective (by linear independence), and surjective (by spanning). Hence ϕ is an isomorphism. □

Note:-

The isomorphism $\phi : k^n \xrightarrow{\sim} V$ depends on the choice of basis. Different bases give different isomorphisms. This is the key insight of linear algebra: choosing a basis amounts to choosing coordinates, and we represent elements of V by column vectors $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in k^n$.

Example 4.6.1

Dimensions:

- $\dim(k^n) = n$. The standard basis has n elements.
- $\dim(k[x]_{\leq d}) = d + 1$. The monomials $\{1, x, \dots, x^d\}$ form a basis.
- $k[x]$ is infinite-dimensional: any finite set of polynomials has bounded degree and cannot span.
- $k[[x]]$ is infinite-dimensional (and in a “larger” sense—its bases are uncountable).

4.7 Linear Maps

Having established the structure of vector spaces, we now study the morphisms between them. Just as group homomorphisms preserve the group operation, linear maps preserve both vector space operations.

Definition 4.7.1: Linear map

Let V and W be vector spaces over k . A **linear map** (or **k -linear map**, or **homomorphism of vector spaces**) is a function $\varphi : V \rightarrow W$ satisfying:

1. $\varphi(u + v) = \varphi(u) + \varphi(v)$ for all $u, v \in V$ (additivity);
2. $\varphi(\lambda v) = \lambda \varphi(v)$ for all $\lambda \in k$ and $v \in V$ (homogeneity).

Equivalently, φ preserves all linear combinations:

$$\varphi\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i \varphi(v_i).$$

Note:-

A linear map is completely determined by its values on a basis. If $\{v_1, \dots, v_n\}$ is a basis for V , and we specify $\varphi(v_i) = w_i \in W$ for each i , then for any $v = \sum a_i v_i$:

$$\varphi(v) = \varphi\left(\sum a_i v_i\right) = \sum a_i \varphi(v_i) = \sum a_i w_i.$$

This is the **universal property** of bases: specifying a linear map from V is equivalent to choosing where to send the basis vectors.

Definition 4.7.2: Space of linear maps

The set of all linear maps from V to W is denoted

$$\text{Hom}_k(V, W) \quad \text{or simply} \quad \text{Hom}(V, W).$$

When $V = W$, we write $\text{End}(V) = \text{Hom}(V, V)$ and call elements **endomorphisms**.

Proposition 4.7.1 $\text{Hom}(V, W)$ is a vector space

The set $\text{Hom}(V, W)$ is itself a vector space over k , with operations:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ for all $v \in V$;
- **Scalar multiplication:** $(\lambda \varphi)(v) = \lambda \cdot \varphi(v)$ for all $v \in V$.

Proof: We must verify that $\varphi + \psi$ and $\lambda \varphi$ are linear maps when φ, ψ are.

For $\varphi + \psi$: Let $u, v \in V$ and $\mu \in k$.

$$\begin{aligned} (\varphi + \psi)(u + v) &= \varphi(u + v) + \psi(u + v) = \varphi(u) + \varphi(v) + \psi(u) + \psi(v) \\ &= (\varphi(u) + \psi(u)) + (\varphi(v) + \psi(v)) = (\varphi + \psi)(u) + (\varphi + \psi)(v). \\ (\varphi + \psi)(\mu v) &= \varphi(\mu v) + \psi(\mu v) = \mu \varphi(v) + \mu \psi(v) = \mu(\varphi(v) + \psi(v)) = \mu(\varphi + \psi)(v). \end{aligned}$$

For $\lambda \varphi$: Similarly,

$$\begin{aligned} (\lambda \varphi)(u + v) &= \lambda(\varphi(u + v)) = \lambda(\varphi(u) + \varphi(v)) = \lambda \varphi(u) + \lambda \varphi(v) = (\lambda \varphi)(u) + (\lambda \varphi)(v). \\ (\lambda \varphi)(\mu v) &= \lambda(\varphi(\mu v)) = \lambda(\mu \varphi(v)) = \mu(\lambda \varphi(v)) = \mu(\lambda \varphi)(v). \end{aligned}$$

The vector space axioms for $\text{Hom}(V, W)$ follow from those of W (operations are defined pointwise). The zero vector is the zero map $v \mapsto 0_W$, and additive inverses are $(-\varphi)(v) = -\varphi(v)$. $\square$

Note:-

The verification that $\varphi + \psi$ and $\lambda\varphi$ are linear is “rather boring, but worth checking if you’re not sure.” The operations on $\text{Hom}(V, W)$ are inherited pointwise from W , so the axioms follow routinely.

Theorem 4.7.1 Dimension of $\text{Hom}(V, W)$

If $\dim(V) = n$ and $\dim(W) = m$, then

$$\dim(\text{Hom}(V, W)) = mn.$$

The proof of this theorem will emerge naturally when we establish the correspondence between linear maps and matrices in the next section.

Note:-

We will soon see: if $\dim(V) = n$ and $\dim(W) = m$, then $\dim(\text{Hom}(V, W)) = mn$. In terms of bases for V and W , linear maps become $m \times n$ matrices!

4.7.1 Dependence on the scalar field

The notions of basis, dimension, and linearity depend crucially on the choice of scalar field k . The same set can be a vector space over different fields, with different dimensions.

Example 4.7.1

$\mathbb{C}$ as a vector space over different fields: Consider $\mathbb{C}$ as a vector space:

- Over $\mathbb{C}$: $\dim_{\mathbb{C}}(\mathbb{C}) = 1$, with basis $\{1\}$. Every complex number z is $z \cdot 1$.
- Over $\mathbb{R}$: $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, with basis $\{1, i\}$. Every complex number $a + bi$ is the $\mathbb{R}$ -linear combination $a \cdot 1 + b \cdot i$.

Note:-

More generally, if $k' \subset k$ is a subfield (e.g., $\mathbb{R} \subset \mathbb{C}$ or $\mathbb{Q} \subset \mathbb{R}$), then any k -vector space V is automatically a k' -vector space by “restriction of scalars”—we simply forget that we can multiply by elements of $k \setminus k'$. The k' -dimension is always at least as large as the k -dimension.

Proposition 4.7.2 Linearity over subfields

If V and W are k -vector spaces and $k' \subset k$ is a subfield, then:

1. V is a k' -vector space (by restriction of scalars).
2. Any k -linear map $\varphi : V \rightarrow W$ is also k' -linear.
3. The converse fails: not every k' -linear map is k -linear.

Thus $\text{Hom}_k(V, W) \subsetneq \text{Hom}_{k'}(V, W)$ in general.

Example 4.7.2

Complex conjugation is $\mathbb{R}$ -linear but not $\mathbb{C}$ -linear: Consider the map

$$\bar{} : \mathbb{C} \rightarrow \mathbb{C}, \quad z = a + bi \mapsto \bar{z} = a - bi.$$

$\mathbb{R}$ -linearity: For $z_1, z_2 \in \mathbb{C}$ and $r \in \mathbb{R}$:

- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ ✓
- $\overline{r \cdot z} = r \cdot \bar{z}$ (since $\bar{r} = r$ for $r \in \mathbb{R}$) ✓

Not $\mathbb{C}$ -linear: For $\lambda = i$ and $z = 1$:

$$\overline{i \cdot 1} = \bar{i} = -i, \quad \text{but} \quad i \cdot \bar{1} = i \cdot 1 = i.$$

Since $-i \neq i$, complex conjugation is not $\mathbb{C}$ -linear.

This example illustrates a profound point: the “natural” maps between spaces can depend on what scalars we consider. Complex conjugation is an $\mathbb{R}$ -linear automorphism of $\mathbb{C}$ that is not $\mathbb{C}$ -linear—it does not commute with multiplication by i .

4.8 Matrix Representation of Linear Maps

The correspondence between linear maps and matrices is one of the most powerful ideas in linear algebra. It reduces abstract questions about linear maps to concrete computations with arrays of numbers.

4.8.1 The matrix of a linear map

Fix bases $\mathcal{B} = (v_1, \dots, v_n)$ for V and $\mathcal{C} = (w_1, \dots, w_m)$ for W . Given a linear map $\varphi : V \rightarrow W$, we can express each image $\varphi(v_j)$ as a linear combination of the basis vectors of W :

$$\varphi(v_j) = \sum_{i=1}^m a_{ij} w_i = a_{1j} w_1 + a_{2j} w_2 + \cdots + a_{mj} w_m.$$

Definition 4.8.1: Matrix of a linear map

The **matrix of φ** with respect to bases $\mathcal{B}$ and $\mathcal{C}$ is the $m \times n$ matrix

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathcal{M}_{m \times n}(k).$$

The j -th **column** of A is the coordinate vector of $\varphi(v_j)$ in the basis $\mathcal{C}$.

Note:-

The key insight: **columns of A are the images of basis vectors, expressed in coordinates.** This is worth memorizing: the j -th column of the matrix tells you where the j -th basis vector goes.

4.8.2 Matrix-vector multiplication

The power of the matrix representation is that applying φ corresponds to matrix multiplication.

Represent a vector $x \in V$ by its coordinate column vector: if $x = \sum_{j=1}^n x_j v_j$, we write

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in k^n.$$

Similarly, $y = \varphi(x) \in W$ has coordinates $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$ where $y = \sum_{i=1}^m y_i w_i$.

Proposition 4.8.1 Linear maps as matrix multiplication

With the notation above, $Y = AX$, i.e.,

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

Proof: Compute:

$$\begin{aligned} \varphi(x) &= \varphi\left(\sum_{j=1}^n x_j v_j\right) = \sum_{j=1}^n x_j \varphi(v_j) = \sum_{j=1}^n x_j \left(\sum_{i=1}^m a_{ij} w_i\right) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j\right) w_i. \end{aligned}$$

Reading off the coefficient of w_i , we get $y_i = \sum_{j=1}^n a_{ij} x_j$, which is exactly the i -th component of AX . $\square$

Note:-

As a memory aid: the isomorphism $k^n \xrightarrow{\sim} V$ given by a basis can be written symbolically as

$$(v_1 \ v_2 \ \cdots \ v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{j=1}^n x_j v_j.$$

Similarly, φ acts as

$$\varphi((v_1 \ \cdots \ v_n)X) = (w_1 \ \cdots \ w_m)AX.$$

This notation makes the matrix formula $Y = AX$ visually clear.

4.8.3 The isomorphism $\text{Hom}(V, W) \cong \mathcal{M}_{m \times n}(k)$ **Theorem 4.8.1** Linear maps correspond to matrices

Fix bases $\mathcal{B}$ for V and $\mathcal{C}$ for W . The map

$$\Phi : \text{Hom}(V, W) \rightarrow \mathcal{M}_{m \times n}(k), \quad \varphi \mapsto [\varphi]_{\mathcal{C}}^{\mathcal{B}}$$

sending a linear map to its matrix is an isomorphism of vector spaces.

Proof: We verify Φ is linear, injective, and surjective.

Linearity: Let $\varphi, \psi \in \text{Hom}(V, W)$ and $\lambda \in k$. The j -th column of $[\varphi + \psi]$ is the coordinate vector of $(\varphi + \psi)(v_j) = \varphi(v_j) + \psi(v_j)$, which is the sum of the j -th columns of $[\varphi]$ and $[\psi]$. Hence $[\varphi + \psi] = [\varphi] + [\psi]$. Similarly $[\lambda\varphi] = \lambda[\varphi]$.

Injectivity: If $[\varphi] = 0$ (the zero matrix), then every column is zero, meaning $\varphi(v_j) = 0$ for all basis vectors v_j . Since φ is linear and determined by its values on a basis, $\varphi = 0$.

Surjectivity: Given any matrix $A \in \mathcal{M}_{m \times n}(k)$, define $\varphi : V \rightarrow W$ by specifying $\varphi(v_j) = \sum_i a_{ij} w_i$ (the j -th column of A gives the coordinates of $\varphi(v_j)$). Extend linearly. Then $[\varphi] = A$. $\square$

Corollary 4.8.1 Dimension formula

$$\dim(\text{Hom}(V, W)) = \dim(\mathcal{M}_{m \times n}(k)) = mn.$$

Proof: The space $\mathcal{M}_{m \times n}(k)$ has the standard basis $\{E_{ij}\}$, where E_{ij} is the matrix with 1 in position (i, j) and 0 elsewhere. There are mn such matrices. $\square$

Note:-

This result says: **linear maps are matrices, once we choose coordinates.** The abstract theory of linear maps reduces to the concrete theory of matrix computations. However, the matrix depends on the choice of bases—changing bases changes the matrix (we will study this “change of basis” in the near future).

The interplay between the abstract (linear maps) and the concrete (matrices) is what makes linear algebra, well linear algebra (duh). Our goal is to understand how much of a linear map’s structure is intrinsic (independent of basis choice) versus coordinate-dependent.

4.9 Change of basis

What if we want to choose a different basis for V and/or W ? That is to say, what if we want to take vectors expressible in one basis to the same vectors expressed in another basis? Well, say that we are given two bases: (v_i) and (v'_i) in V . If given an expression of (v_i) we want to plug in and get an expression in terms of v'_i , we write this as

$$v'_j = \sum_n p_{nj} v_n.$$

So we get an $n \times m$ matrix P with entries (p_{ij}) , whose j^{th} column is the expression of v'_j in the basis (v_i) as a column vector. So symbolically, we can think of this as $(v'_i) = (v_i)P$. So $(v'_i)X' = (v_i)PX'$, i.e. the element of V represented by a column vector X' in new basis (v'_i) is represented by $X = PX'$ in the old basis (v_i) . We can think of this more conceptually as P being a linear map—i.e. a matrix transformation—by $P = \mathcal{M}(\text{id}_V, (v'_i), (v_i))$. Breaking this down, we have the identity mapping taking the vector to itself (from one basis to another, but the same vector(s)); thus we are taking things from (v'_i) , here, to the targets of (v_i) . We can additionally do the same for W in reverse! Let $Q = \mathcal{M}(\text{id}_W, (w_i), (w'_i))$, i.e. $(w_i) = (w'_i)Q$. Hence:

$$\varphi((v'_i)X') = \varphi((v_i)PX') = (w_i)APX' = (w'_i)QAPX'.$$

That is to say, $\mathcal{M}(\varphi, (v'), (w')) = QAP$.

Note:-

Matrix of identity from (v_i) to (v'_i) would be P^{-1} which is the matrix such that PP^{-1} is the identity matrix.

If $V = W$ and we change basis using a single new basis (v'_i) for both domain and codomain, then a remarkable simplification occurs. Let $\varphi \in \text{End}(V) = \text{Hom}(V, V)$ be an **endomorphism** (a linear map from a space to itself). With $P = \mathcal{M}(\text{id}_V, (v'_i), (v_i))$ as above, and using the same basis change for both domain and codomain, we have $Q = P$ (since we’re using (v_i) and (v'_i) for W as well). Thus the formula $\mathcal{M}(\varphi, (v'), (w')) = QAP$ becomes:

Theorem 4.9.1 Change of basis for endomorphisms

Let $\varphi \in \text{End}(V)$, let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$ be two bases of V , and let $P = \mathcal{M}(\text{id}_V, \mathcal{B}', \mathcal{B})$ be the change-of-basis matrix. If $A = \mathcal{M}(\varphi, \mathcal{B})$ and $A' = \mathcal{M}(\varphi, \mathcal{B}')$, then

$$A' = P^{-1}AP.$$

Proof: We have established that $\mathcal{M}(\varphi, \mathcal{B}', \mathcal{B}') = QAP$ where $Q = \mathcal{M}(\text{id}_V, (v_i), (v'_i))$. But $Q = P^{-1}$ since applying P then Q (or vice versa) gives the identity transformation on coordinate vectors. Explicitly: if $X' = P^{-1}X$ converts old coordinates to new, and $X = PX'$ converts new to old, then composing identity maps gives $\mathcal{M}(\text{id}_V, (v_i), (v'_i)) = P^{-1}$.

Thus $A' = P^{-1}AP$. $\square$

This formula tells us that the matrices representing the same linear operator in different bases are related by conjugation by an invertible matrix.

4.9.1 Similar matrices

Definition 4.9.1: Similar matrices

Two $n \times n$ matrices A and B over a field k are similar if there exists an invertible matrix $P \in GL_n(k)$ such that

$$B = P^{-1}AP.$$

We write $A \sim B$.

Proposition 4.9.1 Similarity is an equivalence relation

The similarity relation on $\mathcal{M}_{n \times n}(k)$ is an equivalence relation:

1. **Reflexive:** $A \sim A$ (take $P = I$).
2. **Symmetric:** If $A \sim B$, then $B \sim A$ (if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$).
3. **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (if $B = P^{-1}AP$ and $C = Q^{-1}BQ$, then $C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ)$).

As we can see, similarity classes of matrices correspond exactly to linear operators on k^n : two matrices are similar if and only if they represent the same linear operator in different bases.

Note:-

From the categorical perspective, similar matrices represent the same morphism in $\mathbf{Vect}_k$ —they are simply different coordinate descriptions of one intrinsic object. This is why similarity-invariant properties (trace, determinant, eigenvalues, rank) are so important: they are properties of the operator itself, not of any particular matrix representation.

4.9.2 Invariants under similarity

What properties of a matrix are preserved under similarity? These are called similarity invariants—they depend only on the underlying linear operator, not on the choice of basis.

Definition 4.9.2: Trace of a matrix

The trace of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries:

$$\text{Tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Lemma 4.9.1 Cyclic property of trace

For any matrices A and B such that both AB and BA are defined and square, we have $\text{Tr}(AB) = \text{Tr}(BA)$.

Proof: Let A be $m \times n$ and B be $n \times m$. Then

$$\text{Tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji}.$$

Similarly,

$$\text{Tr}(BA) = \sum_{j=1}^n (BA)_{jj} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij}.$$

These double sums are identical (just reorder the summation), so $\text{Tr}(AB) = \text{Tr}(BA)$. $\square$

Proposition 4.9.2 Trace is a similarity invariant

If $A \sim B$, then $\text{Tr}(A) = \text{Tr}(B)$. In particular, the trace of a linear operator $\varphi \in \text{End}(V)$ is well-defined: $\text{Tr}(\varphi) := \text{Tr}(\mathcal{M}(\varphi, \mathcal{B}))$ for any basis $\mathcal{B}$.

Proof: If $B = P^{-1}AP$, then using the cyclic property:

$$\text{Tr}(B) = \text{Tr}(P^{-1}AP) = \text{Tr}((P^{-1}A)P) = \text{Tr}(P(P^{-1}A)) = \text{Tr}((PP^{-1})A) = \text{Tr}(A).$$

$\square$

Proposition 4.9.3 Determinant is a similarity invariant

If $A \sim B$, then $\det(A) = \det(B)$. Hence the determinant of a linear operator is well-defined.

Proof: If $B = P^{-1}AP$, then by the multiplicativity of determinants:

$$\det(B) = \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

$\square$

Proposition 4.9.4 Eigenvalues are similarity invariants

If $A \sim B$, then A and B have the same eigenvalues (with the same algebraic multiplicities). In particular, the eigenvalues of a linear operator are intrinsic to the operator, not dependent on the choice of basis.

Proof: A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$. If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}IP = P^{-1}(A - \lambda I)P.$$

Thus $B - \lambda I \sim A - \lambda I$, so $\det(B - \lambda I) = \det(A - \lambda I)$ by the previous result. Hence A and B have identical characteristic polynomials, and therefore the same eigenvalues with the same multiplicities. $\square$

Note:-

Rank is also a similarity invariant: $\text{rank}(P^{-1}AP) = \text{rank}(A)$ since multiplying by invertible matrices doesn't change rank. More sophisticated invariants include the **minimal polynomial**, the **Jordan normal form**, and the **rational canonical form**—these completely classify operators up to similarity over algebraically closed fields. We will encounter these when we study eigenvalues in depth.

Example 4.9.1

Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ over $\mathbb{R}$. Both have $\text{Tr} = 4$, $\det = 4$, and eigenvalue $\lambda = 2$ with multiplicity 2. Yet $A \not\sim B$! The reason: $B = 2I$, so $P^{-1}BP = P^{-1}(2I)P = 2I = B$ for any P . Since $A \neq B$, they cannot be similar.

This shows that trace, determinant, and eigenvalues are necessary but not sufficient conditions for similarity. The difference lies in the **geometric multiplicity** of the eigenvalue 2: for B , the eigenspace

$E_2 = \ker(B - 2I) = \mathbb{R}^2$ has dimension 2; for A , $E_2 = \ker(A - 2I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span}\{(1, 0)\}$ has

dimension 1. Geometric multiplicity is also a similarity invariant!

Question 2

1. Let $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map with matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ with respect to the standard basis. Find the matrix of φ with respect to the basis $\mathcal{B}' = \{(1, 1), (1, -1)\}$.
2. Prove that an $n \times n$ matrix A is similar to a diagonal matrix if and only if there exists a basis of k^n consisting of eigenvectors of A .
3. Show that $\text{Tr}(AB) = \text{Tr}(BA)$ but in general $\det(A + B) \neq \det(A) + \det(B)$. Give a counterexample.

Solution

(1) First, find the change-of-basis matrix P from $\mathcal{B}'$ to the standard basis $\mathcal{E}$. The columns of P are the vectors of $\mathcal{B}'$ expressed in $\mathcal{E}$:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$\text{Then } P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}.$$

The matrix of φ with respect to $\mathcal{B}'$ is:

$$A' = P^{-1}AP = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$\text{Computing } AP \text{ first: } AP = \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix}.$$

$$\text{Then } P^{-1}(AP) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 7 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}.$$

So $A' = \begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$. We can verify: $\text{Tr}(A') = 5 = \text{Tr}(A)$ and $\det(A') = 0 - 2 = -2 = \det(A)$. ✓

(2) ($\Rightarrow$) Suppose A is similar to a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$ for some invertible P . Let v_i be the i -th column of P . Then $AP = PD$, which gives $Av_i = \lambda_i v_i$. So each column of P is an eigenvector of A . Since P is invertible, its columns form a basis of k^n .

($\Leftarrow$) Suppose $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $Av_i = \lambda_i v_i$. Let P be the matrix with columns v_i . Then $AP = PD$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, so $D = P^{-1}AP$.

(3) We proved $\text{Tr}(AB) = \text{Tr}(BA)$ above. For the determinant counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $\det(A) = 0$, $\det(B) = 0$, but $A + B = I$, so $\det(A + B) = 1 \neq 0 = \det(A) + \det(B)$.

4.10 Kernel and Image

Every linear map carries two intrinsic subspaces that reveal its fundamental structure: the kernel (what gets annihilated) and the image (what gets hit). These are the linear-algebraic analogues of concepts we've seen before: the kernel of a group homomorphism and the image of any function. The intermingling betwixt (what a fun word) kernel and image is governed by the **Rank-Nullity Theorem**.

Note:-

From the categorical viewpoint, kernel and image are two ways to “measure” a morphism. The kernel measures the failure of injectivity—how much information the map destroys. The image measures the failure of surjectivity—how much of the codomain the map misses. The Rank-Nullity Theorem says these two failures are complementary: the more a linear map kills, the less it can produce, and vice versa. A linear map $\varphi : V \rightarrow W$ is completely determined by: (1) which subspace it kills (the kernel), and (2) which subspace it hits (the image). Once you know these, there’s essentially only one way to fill in the rest—the map restricted to any complement of the kernel is an isomorphism onto the image. This is different from nonlinear maps, where knowing the zero set tells you almost nothing about the map’s behavior elsewhere.

4.10.1 The Kernel (Null Space)

Definition 4.10.1: Kernel of a linear map

Let $\varphi : V \rightarrow W$ be a linear map. The kernel (or null space) of φ is

$$\ker(\varphi) = \{v \in V : \varphi(v) = 0_W\}.$$

This is the set of all vectors that φ maps to zero.

Note:-

The terminology “kernel” comes from algebra (group homomorphisms), while “null space” is more common in applied linear algebra and matrix theory. We use both interchangeably. In categorical language, the kernel is the equalizer of φ and the zero map.

Example 4.10.1 (Examples of kernels)

1. The zero map $0 : V \rightarrow W$ has $\ker(0) = V$.
2. The identity map $\text{id}_V : V \rightarrow V$ has $\ker(\text{id}_V) = \{0\}$.
3. Let $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by $\varphi(x, y, z) = x + 2y + 3z$. Then $\ker(\varphi) = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}$, a plane through the origin.
4. Let $D : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ be differentiation, $Dp = p'$. Then $\ker(D)$ is the set of constant polynomials.
5. Let $T : \mathcal{M}_{n \times n}(k) \rightarrow k$ be the trace map, $T(A) = \text{Tr}(A)$. Then $\ker(T)$ is the space of traceless matrices.

Proposition 4.10.1 The kernel is a subspace

Let $\varphi : V \rightarrow W$ be a linear map. Then $\ker(\varphi)$ is a subspace of V .

Proof: We verify the subspace criteria:

1. **Contains zero:** Since $\varphi(0_V) = 0_W$ (every linear map preserves zero), we have $0_V \in \ker(\varphi)$.
2. **Closed under addition:** If $u, v \in \ker(\varphi)$, then $\varphi(u+v) = \varphi(u) + \varphi(v) = 0 + 0 = 0$, so $u+v \in \ker(\varphi)$.
3. **Closed under scalar multiplication:** If $v \in \ker(\varphi)$ and $\lambda \in k$, then $\varphi(\lambda v) = \lambda \varphi(v) = \lambda \cdot 0 = 0$, so $\lambda v \in \ker(\varphi)$.

□

Note:-

Compare this with group theory: the kernel of a group homomorphism $f : G \rightarrow H$ is a normal subgroup of G . For linear maps, every subspace is “normal” in the appropriate sense—vector spaces are abelian, so there’s no distinction between left and right cosets.

The kernel provides a clean characterization of injectivity:

Theorem 4.10.1 Injectivity via the kernel

Let $\varphi : V \rightarrow W$ be a linear map. Then

$$\varphi \text{ is injective} \iff \ker(\varphi) = \{0\}.$$

Proof: $(\Rightarrow)$ Suppose φ is injective. Let $v \in \ker(\varphi)$. Then $\varphi(v) = 0 = \varphi(0)$. Since φ is injective, $v = 0$. Thus $\ker(\varphi) \subseteq \{0\}$, and since $0 \in \ker(\varphi)$ always, we have $\ker(\varphi) = \{0\}$.

$(\Leftarrow)$ Suppose $\ker(\varphi) = \{0\}$. Let $u, v \in V$ with $\varphi(u) = \varphi(v)$. Then

$$\varphi(u - v) = \varphi(u) - \varphi(v) = 0,$$

so $u - v \in \ker(\varphi) = \{0\}$. Thus $u - v = 0$, i.e., $u = v$. Hence φ is injective. $\square$

Corollary 4.10.1

An injective linear map from V to W embeds V as a subspace of W . More precisely, if $\varphi : V \rightarrow W$ is injective, then $V \cong \text{Im}(\varphi) \subseteq W$.

Note:-

Here the kernel has a role more than merely verifying injectivity—it measures the failure of injectivity in a precise sense. Two vectors $u, v \in V$ have the same image under φ if and only if $u - v \in \ker(\varphi)$. Thus the fibers of φ (preimages of points) are exactly the cosets $v + \ker(\varphi)$. The map φ is injective precisely when these cosets are singletons, i.e., when $\ker(\varphi) = \{0\}$. This is why we can “mod out by the kernel” to obtain an injective map—the quotient $V/\ker(\varphi)$ collapses each fiber to a point.

4.10.2 The Image (Range)**Definition 4.10.2: Image of a linear map**

Let $\varphi : V \rightarrow W$ be a linear map. The image (or range) of φ is

$$\text{Im}(\varphi) = \{\varphi(v) : v \in V\} = \{w \in W : w = \varphi(v) \text{ for some } v \in V\}.$$

This is the set of all vectors that φ “hits.”

Example 4.10.2 (Examples of images)

1. The zero map $0 : V \rightarrow W$ has $\text{Im}(0) = \{0_W\}$.
2. The identity map $\text{id}_V : V \rightarrow V$ has $\text{Im}(\text{id}_V) = V$.
3. Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be projection $\pi(x, y, z) = (x, y)$. Then $\text{Im}(\pi) = \mathbb{R}^2$.
4. Let $D : \mathcal{P}_n(\mathbb{R}) \rightarrow \mathcal{P}_{n-1}(\mathbb{R})$ be differentiation. Then $\text{Im}(D) = \mathcal{P}_{n-1}(\mathbb{R})$ (every polynomial of degree $\leq n-1$ is the derivative of some polynomial of degree $\leq n$).
5. Let $A \in \mathcal{M}_{m \times n}(k)$ define $L_A : k^n \rightarrow k^m$. Then $\text{Im}(L_A) = \text{ColSpace}(A)$, the column space of A .

Proposition 4.10.2 The image is a subspace

Let $\varphi : V \rightarrow W$ be a linear map. Then $\text{Im}(\varphi)$ is a subspace of W .

Proof: We verify the subspace criteria:

1. **Contains zero:** $\varphi(0_V) = 0_W$, so $0_W \in \text{Im}(\varphi)$.
2. **Closed under addition:** If $w_1, w_2 \in \text{Im}(\varphi)$, then $w_1 = \varphi(v_1)$ and $w_2 = \varphi(v_2)$ for some $v_1, v_2 \in V$. Thus $w_1 + w_2 = \varphi(v_1) + \varphi(v_2) = \varphi(v_1 + v_2) \in \text{Im}(\varphi)$.
3. **Closed under scalar multiplication:** If $w \in \text{Im}(\varphi)$ and $\lambda \in k$, then $w = \varphi(v)$ for some $v \in V$, so $\lambda w = \lambda \varphi(v) = \varphi(\lambda v) \in \text{Im}(\varphi)$.

□

Definition 4.10.3: Surjectivity

A linear map $\varphi : V \rightarrow W$ is **surjective** (or **onto**) if $\text{Im}(\varphi) = W$, i.e., every element of W is hit by some element of V .

Proposition 4.10.3 Characterization of surjectivity

Let $\varphi : V \rightarrow W$ be a linear map. The following are equivalent:

1. φ is surjective.
2. $\text{Im}(\varphi) = W$.
3. If $\mathcal{B}$ spans V , then $\varphi(\mathcal{B}) = \{\varphi(v) : v \in \mathcal{B}\}$ spans W .

4.10.3 The Rank-Nullity Theorem

We now come to the central result connecting kernel and image. This theorem quantifies the fundamental trade-off in linear algebra: the more a map “kills,” the less it can “produce.”

Definition 4.10.4: Rank and nullity

Let $\varphi : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. The **rank** of φ is

$$\text{rank}(\varphi) = \dim(\text{Im}(\varphi)),$$

and the **nullity** of φ is

$$\text{null}(\varphi) = \dim(\ker(\varphi)).$$

For a matrix A , we define $\text{rank}(A)$ and $\text{null}(A)$ to be the rank and nullity of the corresponding linear map.

Theorem 4.10.2 The Rank-Nullity Theorem (Fundamental Theorem of Linear Maps)

Let $\varphi : V \rightarrow W$ be a linear map with V finite-dimensional. Then $\text{Im}(\varphi)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(\varphi)) + \dim(\text{Im}(\varphi)).$$

Equivalently: $\dim(V) = \text{null}(\varphi) + \text{rank}(\varphi)$.

Proof: Let $\{u_1, \dots, u_m\}$ be a basis for $\ker(\varphi)$. By the basis extension theorem, we can extend this to a basis

$$\{u_1, \dots, u_m, v_1, \dots, v_r\}$$

of V , where $\dim(V) = m + r$.

Claim: $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\text{Im}(\varphi)$.

Spanning: Let $w \in \text{Im}(\varphi)$, say $w = \varphi(x)$ for some $x \in V$. Write

$$x = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_r v_r.$$

Applying φ :

$$w = \varphi(x) = a_1 \varphi(u_1) + \dots + a_m \varphi(u_m) + b_1 \varphi(v_1) + \dots + b_r \varphi(v_r).$$

Since each $u_i \in \ker(\varphi)$, we have $\varphi(u_i) = 0$, so

$$w = b_1 \varphi(v_1) + \dots + b_r \varphi(v_r).$$

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ spans $\text{Im}(\varphi)$.

Linear independence: Suppose

$$c_1 \varphi(v_1) + \dots + c_r \varphi(v_r) = 0.$$

Then $\varphi(c_1 v_1 + \dots + c_r v_r) = 0$, so $c_1 v_1 + \dots + c_r v_r \in \ker(\varphi)$. Since $\{u_1, \dots, u_m\}$ is a basis for $\ker(\varphi)$, we can write

$$c_1 v_1 + \dots + c_r v_r = d_1 u_1 + \dots + d_m u_m$$

for some scalars d_j . Rearranging:

$$d_1 u_1 + \dots + d_m u_m - c_1 v_1 - \dots - c_r v_r = 0.$$

Since $\{u_1, \dots, u_m, v_1, \dots, v_r\}$ is linearly independent (it's a basis of V), all coefficients are zero. In particular, $c_1 = \dots = c_r = 0$.

Thus $\{\varphi(v_1), \dots, \varphi(v_r)\}$ is a basis for $\text{Im}(\varphi)$, so $\dim(\text{Im}(\varphi)) = r$. We conclude:

$$\dim(V) = m + r = \dim(\ker(\varphi)) + \dim(\text{Im}(\varphi)).$$

□

Note:-

The Rank-Nullity Theorem is the linear algebra version of the **First Isomorphism Theorem** from group theory. Recall that for a group homomorphism $f : G \rightarrow H$, we have $G/\ker(f) \cong \text{Im}(f)$. The linear version states $V/\ker(\varphi) \cong \text{Im}(\varphi)$, and “counting dimensions” gives $\dim(V) - \dim(\ker(\varphi)) = \dim(\text{Im}(\varphi))$.

4.10.4 Consequences of the Rank-Nullity Theorem

The Rank-Nullity Theorem has powerful implications for understanding linear maps.

Corollary 4.10.2 Dimension constraints on injectivity

Let $\varphi : V \rightarrow W$ be a linear map with $\dim(V) = n$ and $\dim(W) = m$.

1. If $n > m$, then φ cannot be injective.
2. If $n < m$, then φ cannot be surjective.

Proof: (1) If φ were injective, then $\dim(\ker(\varphi)) = 0$, so by Rank-Nullity:

$$\dim(\text{Im}(\varphi)) = \dim(V) = n > m = \dim(W).$$

But $\text{Im}(\varphi) \subseteq W$, so $\dim(\text{Im}(\varphi)) \leq \dim(W) = m$. Contradiction.

(2) If φ were surjective, then $\dim(\text{Im}(\varphi)) = \dim(W) = m$, so by Rank-Nullity:

$$\dim(\ker(\varphi)) = \dim(V) - \dim(\text{Im}(\varphi)) = n - m < 0.$$

But dimensions are non-negative. Contradiction.

□

Corollary 4.10.3 Equal dimensions: bijectivity criteria

Let $\varphi : V \rightarrow W$ be a linear map with $\dim(V) = \dim(W) = n$. The following are equivalent:

1. φ is injective.
2. φ is surjective.
3. φ is bijective (an isomorphism).
4. $\ker(\varphi) = \{0\}$.
5. $\text{Im}(\varphi) = W$.
6. $\text{rank}(\varphi) = n$.

Proof: By Rank-Nullity, $\text{null}(\varphi) + \text{rank}(\varphi) = n$. Since $\text{rank}(\varphi) \leq \dim(W) = n$:

- φ injective $\iff \text{null}(\varphi) = 0 \iff \text{rank}(\varphi) = n$.
- φ surjective $\iff \text{rank}(\varphi) = n \iff \text{null}(\varphi) = 0$.

Thus injectivity, surjectivity, and bijectivity are equivalent when dimensions match. $\square$

Note:-

For linear maps between equal-dimensional spaces, “one-to-one” and “onto” are the same condition. This fails for general functions and even for linear maps between infinite-dimensional spaces. Consider the **right shift** $R : k^\infty \rightarrow k^\infty$ defined by $R(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots)$. This is injective (kernel is trivial) but not surjective (the sequence $(1, 0, 0, \dots)$ is not in the image). Conversely, the **left shift** $L(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$ is surjective but not injective. In infinite dimensions, there’s “room” for one without the other. The finite-dimensional equivalence is a manifestation of the pigeonhole principle applied to basis vectors.

Corollary 4.10.4 Existence of right/left inverses

Let $\varphi : V \rightarrow W$ be a linear map between finite-dimensional spaces.

1. φ has a left inverse if and only if φ is injective.
2. φ has a right inverse if and only if φ is surjective.

4.10.5 Matrix Interpretation

For a matrix $A \in \mathcal{M}_{m \times n}(k)$, the associated linear map is $L_A : k^n \rightarrow k^m$ given by $L_A(x) = Ax$. The Rank-Nullity Theorem becomes:

Proposition 4.10.4 Rank-Nullity for matrices

For $A \in \mathcal{M}_{m \times n}(k)$:

1. $\ker(L_A) = \{x \in k^n : Ax = 0\}$ is the null space of A .
2. $\text{Im}(L_A) = \{Ax : x \in k^n\}$ is the column space of A (the span of the columns).
3. $\text{null}(A) + \text{rank}(A) = n$ (the number of columns).

Note:-

The rank of a matrix equals the number of linearly independent columns, which equals the number of linearly independent rows. The column space has dimension equal to the rank, and the null space has dimension equal to $n - \text{rank}(A)$.

Looking ahead: when we study eigenvalues, the kernel will reappear in a crucial role. For a linear operator $T : V \rightarrow V$ and scalar λ , the **eigenspace** $E_\lambda = \ker(T - \lambda I)$ is precisely the kernel of the shifted operator. The dimension of this eigenspace (the geometric multiplicity of λ) measures “how degenerate” the eigenvalue is. The interplay between algebraic and geometric multiplicities—and the failure of these to match—is what makes Jordan normal form necessary.

Example 4.10.3 (Computing kernel and image)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix}$ over $\mathbb{R}$.

Finding $\ker(A)$: Solve $Ax = 0$. Row reduce:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Back-substituting: $x_2 = -2x_3$, $x_1 = -2x_2 - 3x_3 = 4x_3 - 3x_3 = x_3$. So

$$\ker(A) = \text{Span}\{(1, -2, 1)^T\}, \quad \text{null}(A) = 1.$$

Finding $\text{Im}(A)$: The non-zero rows in RREF correspond to pivot columns. Columns 1 and 2 are pivot columns, so

$$\text{Im}(A) = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad \text{rank}(A) = 2.$$

Verification: $\text{null}(A) + \text{rank}(A) = 1 + 2 = 3 = n$. ✓

Question 3

1. Let $D : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ be the differentiation operator. Find $\ker(D)$, $\text{Im}(D)$, $\text{null}(D)$, and $\text{rank}(D)$. Verify the Rank-Nullity Theorem.
2. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 - x_2, x_2 - x_3, x_3 - x_4)$. Determine whether T is injective, surjective, or neither.
3. Prove that if $\varphi : V \rightarrow V$ is a linear operator on a finite-dimensional space with $\varphi^2 = \varphi$, then $V = \ker(\varphi) \oplus \text{Im}(\varphi)$.
4. Let V and W be finite-dimensional vector spaces with $\dim(V) = \dim(W)$. Prove that a linear map $\varphi : V \rightarrow W$ is injective if and only if for every linearly independent list in V , its image under φ is linearly independent in W .

Solution

(1) The space $\mathcal{P}_3(\mathbb{R})$ has basis $\{1, x, x^2, x^3\}$, so $\dim(\mathcal{P}_3(\mathbb{R})) = 4$.

Kernel: $Dp = 0$ if and only if p is a constant polynomial. So $\ker(D) = \text{Span}\{1\}$, which are the constant functions. Thus $\text{null}(D) = 1$.

Image: Differentiating: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. The non-zero outputs span $\text{Span}\{1, x, x^2\} = \mathcal{P}_2(\mathbb{R})$. Thus $\text{Im}(D) = \mathcal{P}_2(\mathbb{R})$ and $\text{rank}(D) = 3$.

Verification: $\text{null}(D) + \text{rank}(D) = 1 + 3 = 4 = \dim(\mathcal{P}_3(\mathbb{R}))$. ✓

(2) **Injectivity:** Find $\ker(T)$. Solve $T(x_1, x_2, x_3, x_4) = (0, 0, 0)$:

$$x_1 - x_2 = 0, \quad x_2 - x_3 = 0, \quad x_3 - x_4 = 0.$$

This gives $x_1 = x_2 = x_3 = x_4$. So $\ker(T) = \{(t, t, t, t) : t \in \mathbb{R}\} = \text{Span}\{(1, 1, 1, 1)\}$. Since $\ker(T) \neq \{0\}$, T is **not injective**.

Surjectivity: We have $\text{null}(T) = 1$, so $\text{rank}(T) = 4 - 1 = 3 = \dim(\mathbb{R}^3)$. Thus $\text{Im}(T) = \mathbb{R}^3$, and T is **surjective**.

(3) We show $V = \ker(\varphi) + \text{Im}(\varphi)$ and $\ker(\varphi) \cap \text{Im}(\varphi) = \{0\}$.

Sum equals V: Let $v \in V$. Write $v = (v - \varphi(v)) + \varphi(v)$. Note that $\varphi(v) \in \text{Im}(\varphi)$. Also:

$$\varphi(v - \varphi(v)) = \varphi(v) - \varphi^2(v) = \varphi(v) - \varphi(v) = 0,$$

so $v - \varphi(v) \in \ker(\varphi)$. Thus $v \in \ker(\varphi) + \text{Im}(\varphi)$.

Intersection is trivial: Let $w \in \ker(\varphi) \cap \text{Im}(\varphi)$. Then $\varphi(w) = 0$ and $w = \varphi(u)$ for some $u \in V$. So:

$$w = \varphi(u) = \varphi^2(u) = \varphi(\varphi(u)) = \varphi(w) = 0.$$

Thus $V = \ker(\varphi) \oplus \text{Im}(\varphi)$.

(4) ($\Rightarrow$) Suppose φ is injective and $\{v_1, \dots, v_m\}$ is linearly independent. Suppose $\sum c_i \varphi(v_i) = 0$. Then $\varphi(\sum c_i v_i) = 0$, so $\sum c_i v_i \in \ker(\varphi) = \{0\}$. Thus $\sum c_i v_i = 0$, and by linear independence of the v_i , all $c_i = 0$. Hence $\{\varphi(v_1), \dots, \varphi(v_m)\}$ is linearly independent.

($\Leftarrow$) Suppose φ maps every linearly independent set to a linearly independent set. If $v \neq 0$, then $\{v\}$ is linearly independent, so $\{\varphi(v)\}$ is linearly independent, meaning $\varphi(v) \neq 0$. Thus $\ker(\varphi) = \{0\}$, so φ is injective.

4.11 Duality

The dual space of a vector space V consists of all linear maps from V to the base field k —these are called linear functionals. While this might seem like a minor variation on Hom -spaces, duality reveals useful structure: every basis has a “dual basis,” every linear map has a “dual map,” and the matrix of the dual map is the transpose of the original. In more advanced contexts (differential geometry, functional analysis, representation theory), duality becomes indispensable.

Note:-

The dual space construction is the first example of a **contravariant functor** you’ll encounter. When we pass from V to V' , linear maps $T : V \rightarrow W$ don’t induce maps $V' \rightarrow W'$ —instead, they induce maps $W' \rightarrow V'$, going the “wrong way.” This reversal of arrows is the hallmark of contravariance. In category theory, the dual space functor $(-)' : \mathbf{Vect}_k^{\text{op}} \rightarrow \mathbf{Vect}_k$ is representable: $V' = \text{Hom}(V, k)$ is “the functor represented by k .”

4.11.1 Linear Functionals and the Dual Space

Definition 4.11.1: Linear functional

A **linear functional** on a vector space V over a field k is a linear map $\ell : V \rightarrow k$. In other words, a linear functional is an element of $\text{Hom}(V, k)$.

Example 4.11.1 (Examples of linear functionals)

1. **Coordinate projections:** On k^n , define $\pi_j(x_1, \dots, x_n) = x_j$. Each π_j is a linear functional.
2. **Weighted sums:** Fix $(c_1, \dots, c_n) \in k^n$. Define $\ell(x_1, \dots, x_n) = c_1 x_1 + \dots + c_n x_n$. This is a linear functional on k^n , and every linear functional on k^n has this form.
3. **Evaluation:** On $\mathcal{P}(\mathbb{R})$, define $\text{ev}_a(p) = p(a)$ for fixed $a \in \mathbb{R}$. This is a linear functional.
4. **Integration:** On $\mathcal{P}(\mathbb{R})$, define $\ell(p) = \int_0^1 p(x) dx$. Linearity of the integral makes this a linear functional.

5. **Trace:** On $\mathcal{M}_{n \times n}(k)$, the trace $\text{Tr}(A) = \sum_{i=1}^n a_{ii}$ is a linear functional.

Definition 4.11.2: Dual space

The **dual space** of V , denoted V^* or V' , is the vector space of all linear functionals on V :

$$V^* = \text{Hom}(V, k).$$

This is a vector space under pointwise addition and scalar multiplication: $(\ell_1 + \ell_2)(v) = \ell_1(v) + \ell_2(v)$ and $(\lambda\ell)(v) = \lambda \cdot \ell(v)$.

Note:-

We use V^* notation (common in differential geometry and physics) interchangeably with V' (common in algebra). Both mean the same thing. Some texts reserve V^* for the continuous dual in functional analysis, but for finite-dimensional spaces this distinction is irrelevant.

Proposition 4.11.1 Dimension of the dual space

If V is finite-dimensional with $\dim(V) = n$, then $\dim(V^*) = n$. Thus $V^* \cong V$ as vector spaces (non-canonically).

Proof: By our earlier result on Hom-spaces:

$$\dim(V^*) = \dim(\text{Hom}(V, k)) = \dim(V) \cdot \dim(k) = n \cdot 1 = n.$$

□

Note:-

The isomorphism $V \cong V^*$ is non-canonical—it depends on a choice of basis. There’s no “natural” way to identify vectors with linear functionals without extra structure. However, the double dual $V^{**} = (V^*)^*$ does have a canonical isomorphism with V , given by evaluation: $v \mapsto (\ell \mapsto \ell(v))$.

4.11.2 The Dual Basis

Given a basis of V , there’s a natural associated basis of V^* .

Definition 4.11.3: Dual basis

Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of V . The **dual basis** is the list $\mathcal{B}^* = \{\varphi_1, \dots, \varphi_n\}$ of linear functionals in V^* defined by

$$\varphi_j(v_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

In other words, φ_j “picks out” the j -th coordinate: if $v = c_1v_1 + \dots + c_nv_n$, then $\varphi_j(v) = c_j$.

Proposition 4.11.2 Dual basis is a basis

The dual basis $\{\varphi_1, \dots, \varphi_n\}$ is indeed a basis of V^* .

Proof: Linear independence: Suppose $a_1\varphi_1 + \dots + a_n\varphi_n = 0$ (the zero functional). Evaluating at v_k :

$$0 = (a_1\varphi_1 + \dots + a_n\varphi_n)(v_k) = a_1\varphi_1(v_k) + \dots + a_n\varphi_n(v_k) = a_k.$$

So all $a_k = 0$.

Spanning: Since we have n linearly independent vectors in V^* and $\dim(V^*) = n$, they form a basis. □

Proposition 4.11.3 Dual basis extracts coordinates

Let $\{v_1, \dots, v_n\}$ be a basis of V with dual basis $\{\varphi_1, \dots, \varphi_n\}$. For any $v \in V$:

$$v = \varphi_1(v)v_1 + \varphi_2(v)v_2 + \dots + \varphi_n(v)v_n.$$

That is, $\varphi_j(v)$ is the j -th coordinate of v with respect to the basis $\{v_i\}$.

Proof: Write $v = c_1v_1 + \dots + c_nv_n$. Applying φ_j :

$$\varphi_j(v) = c_1\varphi_j(v_1) + \dots + c_n\varphi_j(v_n) = c_j.$$

Substituting back: $v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n$. □

Example 4.11.2 (Dual basis of the standard basis)

Let $\{e_1, \dots, e_n\}$ be the standard basis of k^n . The dual basis consists of the coordinate projections:

$$\varphi_j(x_1, \dots, x_n) = x_j.$$

These are the “row vectors” $\varphi_j = (0, \dots, 0, 1, 0, \dots, 0)$ (with 1 in position j).

4.11.3 The Dual Map

Just as a linear map $T : V \rightarrow W$ induces a map on Hom-spaces, it induces a map on dual spaces—but in the opposite direction.

Definition 4.11.4: Dual map (transpose)

Let $T : V \rightarrow W$ be a linear map. The **dual map** (or **transpose**) is the linear map $T^* : W^* \rightarrow V^*$ defined by

$$T^*(\psi) = \psi \circ T$$

for each $\psi \in W^*$. In other words, $T^*(\psi)$ is the linear functional on V given by $v \mapsto \psi(T(v))$.

Note:-

The dual map goes “backwards”: $T : V \rightarrow W$ yields $T^* : W^* \rightarrow V^*$. This is because we’re precomposing with T . If you think of linear functionals as “measurements,” then $T^*(\psi)$ is the measurement you get by first applying T , then measuring with ψ . The reversal of direction is the essence of contravariance.

Proposition 4.11.4 The dual map is linear

For $T : V \rightarrow W$, the dual map $T^* : W^* \rightarrow V^*$ is indeed a linear map.

Proof: For $\psi_1, \psi_2 \in W^*$ and $\lambda \in k$:

$$\begin{aligned} T^*(\psi_1 + \psi_2) &= (\psi_1 + \psi_2) \circ T = \psi_1 \circ T + \psi_2 \circ T = T^*(\psi_1) + T^*(\psi_2), \\ T^*(\lambda\psi) &= (\lambda\psi) \circ T = \lambda(\psi \circ T) = \lambda T^*(\psi). \end{aligned}$$

□

Proposition 4.11.5 Algebraic properties of the dual map

Let $S : V \rightarrow W$ and $T : W \rightarrow U$ be linear maps.

1. $(S + T)^* = S^* + T^*$ (when both are defined).
2. $(\lambda T)^* = \lambda T^*$ for $\lambda \in k$.

3. $(T \circ S)^* = S^* \circ T^*$ (order reverses!).
4. $(\text{id}_V)^* = \text{id}_{V^*}$.

Proof: (3) is the key property. For $\psi \in U^*$:

$$(T \circ S)^*(\psi) = \psi \circ (T \circ S) = (\psi \circ T) \circ S = S^*(\psi \circ T) = S^*(T^*(\psi)) = (S^* \circ T^*)(\psi).$$

The other properties are similar. □

Note:-

Property (3) says the dual is a contravariant functor: it reverses the order of composition. Compare with the transpose of matrices: $(AB)^T = B^T A^T$. This is not a coincidence—the matrix of the dual map is the transpose!

4.11.4 Matrix of the Dual Map

The connection between dual maps and matrix transposes clarifies why the transpose operation is natural.

Theorem 4.11.1 Matrix of the dual map is the transpose

Let $T : V \rightarrow W$ be a linear map between finite-dimensional spaces. Fix bases $\mathcal{B} = \{v_1, \dots, v_n\}$ of V and $\mathcal{C} = \{w_1, \dots, w_m\}$ of W , with dual bases $\mathcal{B}^* = \{\varphi_1, \dots, \varphi_n\}$ and $\mathcal{C}^* = \{\psi_1, \dots, \psi_m\}$. If

$$A = \mathcal{M}(T; \mathcal{B}, \mathcal{C}) \in \mathcal{M}_{m \times n}(k)$$

is the matrix of T , then the matrix of T^* with respect to the dual bases is the transpose:

$$\mathcal{M}(T^*; \mathcal{C}^*, \mathcal{B}^*) = A^T \in \mathcal{M}_{n \times m}(k).$$

Proof: We need to show that $T^*(\psi_j) = \sum_{i=1}^n A_{ji} \varphi_i$ for each j .

By definition of the matrix A , we have $T(v_k) = \sum_{i=1}^m A_{ik} w_i$. Now compute:

$$T^*(\psi_j)(v_k) = \psi_j(T(v_k)) = \psi_j\left(\sum_{i=1}^m A_{ik} w_i\right) = \sum_{i=1}^m A_{ik} \psi_j(w_i) = A_{jk}.$$

On the other hand, if $T^*(\psi_j) = \sum_{i=1}^n c_i \varphi_i$, then

$$T^*(\psi_j)(v_k) = \sum_{i=1}^n c_i \varphi_i(v_k) = c_k.$$

Comparing: $c_k = A_{jk}$. Thus $T^*(\psi_j) = \sum_{k=1}^n A_{jk} \varphi_k$, which means the j -th column of $\mathcal{M}(T^*)$ is the j -th row of A . Hence $\mathcal{M}(T^*) = A^T$. □

Note:-

This theorem explains why the transpose is natural: it's not just a matrix operation, it's the coordinate representation of the dual map. The reversal of dimensions ($m \times n$ becomes $n \times m$) reflects the contravariance of the dual construction.

Corollary 4.11.1 Rank of transpose equals rank

For any matrix A , $\text{rank}(A^T) = \text{rank}(A)$.

Proof: The rank of A is $\dim(\text{Im}(T))$ where T is the associated linear map. By a result we'll see shortly, $\dim(\text{Im}(T^*)) = \dim(\text{Im}(T))$. Since the matrix of T^* is A^T , we get $\text{rank}(A^T) = \text{rank}(A)$. □

4.11.5 Kernel and Image of the Dual Map

The dual map interchanges the roles of kernel and image in a precise way.

Proposition 4.11.6 Relationships between T and T^*

Let $T : V \rightarrow W$ be a linear map between finite-dimensional spaces.

1. $\ker(T^*) = (\text{Im}(T))^\circ$, the annihilator of the image.
2. $\text{Im}(T^*) = (\ker(T))^\circ$, the annihilator of the kernel.
3. $\dim(\ker(T^*)) = \dim(W) - \text{rank}(T)$.
4. $\dim(\text{Im}(T^*)) = \text{rank}(T)$.

Here, for a subspace $U \subseteq V$, the annihilator is $U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}$.

Note:-

The annihilator U° consists of all linear functionals that vanish on U . If $\dim(V) = n$ and $\dim(U) = k$, then $\dim(U^\circ) = n - k$. The annihilator provides a duality between subspaces of V and subspaces of V^* : bigger subspaces have smaller annihilators.

Question 4

1. Let $V = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Write down the dual basis explicitly as functions $\mathbb{R}^3 \rightarrow \mathbb{R}$.
2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y + z)$. Find the matrix of T with respect to the standard bases, then find the matrix of T^* with respect to the dual bases.
3. Prove that if $T : V \rightarrow W$ is surjective, then $T^* : W^* \rightarrow V^*$ is injective.
4. Prove that if V is finite-dimensional, the map $\Phi : V \rightarrow V^{**}$ defined by $\Phi(v)(\ell) = \ell(v)$ is an isomorphism.

Solution

(1) The dual basis consists of the coordinate projections:

$$\varphi_1(x, y, z) = x, \quad \varphi_2(x, y, z) = y, \quad \varphi_3(x, y, z) = z.$$

These are “row vectors” $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$.

(2) The matrix of T is:

$$A = \mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The matrix of T^* is the transpose:

$$A^T = \mathcal{M}(T^*) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Explicitly, if ψ_1, ψ_2 are the dual basis of $\mathbb{R}^2$ and $\varphi_1, \varphi_2, \varphi_3$ are the dual basis of $\mathbb{R}^3$:

$$T^*(\psi_1) = \varphi_1 + \varphi_2, \quad T^*(\psi_2) = \varphi_2 + \varphi_3.$$

(3) Suppose T is surjective and $T^*(\psi) = 0$, i.e., $\psi \circ T = 0$. For any $w \in W$, since T is surjective, $w = T(v)$ for some $v \in V$. Then:

$$\psi(w) = \psi(T(v)) = (\psi \circ T)(v) = 0.$$

Since this holds for all $w \in W$, we have $\psi = 0$. Thus $\ker(T^*) = \{0\}$, so T^* is injective.

(4) Well-defined and linear: For $v \in V$, $\Phi(v) : V^* \rightarrow k$ is defined by $\Phi(v)(\ell) = \ell(v)$. This is linear in ℓ (check!), so $\Phi(v) \in V^{**}$. Moreover, Φ is linear: $\Phi(v_1 + v_2)(\ell) = \ell(v_1 + v_2) = \ell(v_1) + \ell(v_2) = \Phi(v_1)(\ell) + \Phi(v_2)(\ell)$.

Injective: Suppose $\Phi(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis $\{v, v_2, \dots, v_n\}$. The dual basis element φ_1 satisfies $\varphi_1(v) = 1 \neq 0$, contradiction. So $v = 0$.

Surjective: Since $\dim(V) = \dim(V^*) = \dim(V^{**})$, injectivity implies surjectivity.

4.11.6 The Double Dual and Natural Isomorphisms

We've seen that $V \cong V^*$ for finite-dimensional V , but this isomorphism required a choice of basis. The double dual $V^{**} = (V^*)^*$ enjoys a more canonical relationship with V .

Definition 4.11.5: Evaluation map

The **evaluation map** is $\text{ev} : V \rightarrow V^{**}$ defined by

$$\text{ev}(v)(\ell) = \ell(v)$$

for $v \in V$ and $\ell \in V^*$. That is, $\text{ev}(v)$ is the linear functional on V^* that “evaluates at v .”

Proposition 4.11.7 Properties of the evaluation map

The evaluation map $\text{ev} : V \rightarrow V^{**}$ satisfies:

1. ev is linear.
2. ev is injective.
3. If V is finite-dimensional, ev is an isomorphism.

Proof: (1) For $v, w \in V$, $\lambda \in k$, and $\ell \in V^*$:

$$\text{ev}(v + \lambda w)(\ell) = \ell(v + \lambda w) = \ell(v) + \lambda \ell(w) = \text{ev}(v)(\ell) + \lambda \text{ev}(w)(\ell).$$

(2) Suppose $\text{ev}(v) = 0$, meaning $\ell(v) = 0$ for all $\ell \in V^*$. If $v \neq 0$, extend $\{v\}$ to a basis and let ℓ be the dual basis element with $\ell(v) = 1$. Contradiction. So $v = 0$.

(3) For finite-dimensional V : $\dim(V^{**}) = \dim(V^*) = \dim(V)$. Since ev is injective, it's an isomorphism. $\square$

Note:-

The crucial point: ev is natural—it doesn't depend on any choices. Given bases $\{e_1, \dots, e_n\}$ of V , dual basis $\{e_1^*, \dots, e_n^*\}$ of V^* , and double dual basis $\{e_1^{**}, \dots, e_n^{**}\}$ of V^{**} , one can verify that $\text{ev}(e_i) = e_i^{**}$. The evaluation map sends each basis element to its “double dual counterpart.”

Note:-

For infinite-dimensional V , the situation is strikingly different. If V has a basis $\{e_i\}_{i \in I}$ indexed by an infinite set I , then every element of V is a finite linear combination $\sum_{i \in I} x_i e_i$ (with only finitely many $x_i \neq 0$). However, elements of V^* can be arbitrary: given any function $\alpha : I \rightarrow k$, the map $\ell_\alpha(\sum x_i e_i) = \sum x_i \alpha_i$ is a well-defined linear functional. Thus $V^* \cong k^I = \prod_{i \in I} k$, the full product, which is much larger than the direct sum $V \cong \bigoplus_{i \in I} k$. The evaluation map $\text{ev} : V \rightarrow V^{**}$ is still injective, but far from surjective.

4.11.7 Annihilators Revisited

The annihilator provides a precise duality between subspaces of V and subspaces of V^* .

Definition 4.11.6: Annihilator

For a subspace $U \subseteq V$, the **annihilator** of U is

$$\text{Ann}(U) = U^\circ = \{\ell \in V^* : \ell(u) = 0 \text{ for all } u \in U\}.$$

Theorem 4.11.2 Dimension of the annihilator

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(\text{Ann}(U)) = \dim(V) - \dim(U).$$

Proof: The restriction map $\rho : V^* \rightarrow U^*$ defined by $\rho(\ell) = \ell|_U$ is a linear map with $\ker(\rho) = \text{Ann}(U)$. We claim ρ is surjective: given $\psi \in U^*$, extend a basis of U to a basis of V , define ℓ to agree with ψ on U and be zero on the complementary basis vectors. Then $\rho(\ell) = \psi$.

By rank-nullity:

$$\dim(V^*) = \dim(\text{Ann}(U)) + \dim(U^*).$$

Since $\dim(V^*) = \dim(V)$ and $\dim(U^*) = \dim(U)$, we get $\dim(\text{Ann}(U)) = \dim(V) - \dim(U)$. $\square$

Theorem 4.11.3 Annihilator isomorphism

For a subspace $U \subseteq V$, there is a natural isomorphism

$$\text{Ann}(U) \cong (V/U)^*.$$

Proof: Define $\Phi : \text{Ann}(U) \rightarrow (V/U)^*$ by $\Phi(\ell)(v + U) = \ell(v)$.

Well-defined: If $v + U = v' + U$, then $v - v' \in U$, so $\ell(v) - \ell(v') = \ell(v - v') = 0$ since $\ell \in \text{Ann}(U)$.

Injective: If $\Phi(\ell) = 0$, then $\ell(v) = 0$ for all $v \in V$, so $\ell = 0$.

Surjective: Given $\psi \in (V/U)^*$, define $\ell = \psi \circ q$ where $q : V \rightarrow V/U$ is the quotient map. Then $\ell \in \text{Ann}(U)$ and $\Phi(\ell) = \psi$. $\square$

4.12 Quotient Spaces

Quotient spaces provide a systematic way to “collapse” a subspace to zero, creating a smaller vector space that captures the structure “perpendicular” to the subspace. This construction is fundamental in linear algebra and pervades abstract algebra.

4.12.1 Construction of the Quotient Space

Definition 4.12.1: Cosets and quotient space

Let V be a vector space over k and $U \subseteq V$ a subspace. For $v \in V$, the **coset** of v modulo U is

$$v + U = \{v + u : u \in U\}.$$

The **quotient space** is the set of all cosets:

$$V/U = \{v + U : v \in V\}.$$

Note:-

Two cosets $v + U$ and $w + U$ are equal if and only if $v - w \in U$. Equivalently, the cosets partition V into equivalence classes under the relation $v \sim w \Leftrightarrow v - w \in U$.

Proposition 4.12.1 Vector space structure on the quotient

The quotient V/U becomes a vector space over k under the operations:

1. **Addition:** $(v + U) + (w + U) = (v + w) + U$.
2. **Scalar multiplication:** $\lambda(v + U) = (\lambda v) + U$.

The zero element is $0 + U = U$, and $-(v + U) = (-v) + U$.

Proof: The key is well-definedness: if $v + U = v' + U$ and $w + U = w' + U$, then $v - v' \in U$ and $w - w' \in U$, so $(v + w) - (v' + w') = (v - v') + (w - w') \in U$, meaning $(v + w) + U = (v' + w') + U$.

Similarly for scalar multiplication: if $v + U = v' + U$, then $v - v' \in U$, so $\lambda(v - v') = \lambda v - \lambda v' \in U$, giving $\lambda v + U = \lambda v' + U$.

The vector space axioms follow from those in V . □

4.12.2 The Quotient Map and Exact Sequences**Definition 4.12.2: Quotient map**

The **quotient map** (or **canonical projection**) is $q : V \rightarrow V/U$ defined by $q(v) = v + U$.

Proposition 4.12.2 Properties of the quotient map

The quotient map $q : V \rightarrow V/U$ satisfies:

1. q is linear.
2. q is surjective.
3. $\ker(q) = U$.

Proof: (1) $q(v + \lambda w) = (v + \lambda w) + U = (v + U) + \lambda(w + U) = q(v) + \lambda q(w)$.

(2) Every coset $v + U$ is $q(v)$.

(3) $v \in \ker(q) \Leftrightarrow v + U = 0 + U \Leftrightarrow v \in U$. □

Note:-

This gives a **short exact sequence**:

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{q} V/U \longrightarrow 0$$

where $\iota : U \hookrightarrow V$ is the inclusion. “Exact” means: $\text{Im}(\iota) = \ker(q)$, ι is injective, and q is surjective. Short exact sequences are a central organizing principle in algebra.

Theorem 4.12.1 Dimension of the quotient space

If V is finite-dimensional and $U \subseteq V$ is a subspace, then

$$\dim(V/U) = \dim(V) - \dim(U).$$

Proof: Apply rank-nullity to $q : V \rightarrow V/U$:

$$\dim(V) = \dim(\ker(q)) + \dim(\text{Im}(q)) = \dim(U) + \dim(V/U).$$

□

Example 4.12.1 (Concrete example)

Let $V = \mathbb{R}^3$ and $U = \text{Span}\{(1, 0, 0)\}$, the x -axis. Then $V/U \cong \mathbb{R}^2$:

The coset $(a, b, c) + U$ consists of all points $\{(a + t, b, c) : t \in \mathbb{R}\}$ —a horizontal line through (a, b, c) . Two points have the same coset iff they have the same y and z coordinates. The map $(a, b, c) + U \mapsto (b, c)$ is an isomorphism $V/U \xrightarrow{\sim} \mathbb{R}^2$.

4.12.3 Universal Property of the Quotient

The quotient space is characterized by a universal property that explains its importance.

Theorem 4.12.2 Universal property of the quotient

Let $\varphi : V \rightarrow W$ be a linear map with $U \subseteq \ker(\varphi)$. Then there exists a unique linear map $\bar{\varphi} : V/U \rightarrow W$ such that $\varphi = \bar{\varphi} \circ q$:

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ q \downarrow & \nearrow \exists! \bar{\varphi} & \\ V/U & & \end{array}$$

Moreover, $\bar{\varphi}$ is defined by $\bar{\varphi}(v + U) = \varphi(v)$.

Proof: **Well-defined:** If $v + U = v' + U$, then $v - v' \in U \subseteq \ker(\varphi)$, so $\varphi(v) = \varphi(v')$.

Linearity: $\bar{\varphi}((v + U) + \lambda(w + U)) = \bar{\varphi}((v + \lambda w) + U) = \varphi(v + \lambda w) = \varphi(v) + \lambda\varphi(w) = \bar{\varphi}(v + U) + \lambda\bar{\varphi}(w + U)$.

Uniqueness: Any $\psi : V/U \rightarrow W$ with $\varphi = \psi \circ q$ must satisfy $\psi(v + U) = \psi(q(v)) = \varphi(v) = \bar{\varphi}(v + U)$. □

Corollary 4.12.1 First Isomorphism Theorem

For any linear map $\varphi : V \rightarrow W$:

$$V/\ker(\varphi) \cong \text{Im}(\varphi).$$

The isomorphism is $\bar{\varphi} : v + \ker(\varphi) \mapsto \varphi(v)$.

Proof: By the universal property, $\bar{\varphi} : V/\ker(\varphi) \rightarrow W$ is well-defined. Its image is exactly $\text{Im}(\varphi)$. Injectivity: if $\bar{\varphi}(v + \ker(\varphi)) = 0$, then $\varphi(v) = 0$, so $v \in \ker(\varphi)$, meaning $v + \ker(\varphi) = 0 + \ker(\varphi)$. □

Note:-

This theorem is remarkably useful. It says that every surjective linear map $\varphi : V \twoheadrightarrow W$ is “really” a quotient map: $V \rightarrow V/\ker(\varphi) \cong W$. Dually, every injective map $\iota : U \hookrightarrow V$ makes U isomorphic to a subspace of V .

4.12.4 Correspondence Between Subspaces

The quotient construction establishes a bijection between certain subspaces.

Theorem 4.12.3 Lattice correspondence

Let $U \subseteq V$ be a subspace. There is a bijection

$$\{\text{subspaces } W \subseteq V \text{ with } U \subseteq W\} \longleftrightarrow \{\text{subspaces of } V/U\}$$

given by:

- Forward: $W \mapsto W/\mathcal{U} = \{w + \mathcal{U} : w \in W\}$.
- Backward: $\bar{W} \mapsto q^{-1}(\bar{W}) = \{v \in V : v + \mathcal{U} \in \bar{W}\}$.

Moreover, this bijection preserves inclusions: $W_1 \subseteq W_2 \Leftrightarrow W_1/\mathcal{U} \subseteq W_2/\mathcal{U}$.

Proof: We verify the maps are inverses.

Forward then backward: Starting with $W \supseteq \mathcal{U}$, we get $W/\mathcal{U}$, then $q^{-1}(W/\mathcal{U}) = \{v \in V : v + \mathcal{U} \in W/\mathcal{U}\}$. But $v + \mathcal{U} \in W/\mathcal{U}$ iff $v + \mathcal{U} = w + \mathcal{U}$ for some $w \in W$, iff $v - w \in \mathcal{U} \subseteq W$, iff $v \in W$. So $q^{-1}(W/\mathcal{U}) = W$.

Backward then forward: Starting with $\bar{W} \subseteq V/\mathcal{U}$, let $W = q^{-1}(\bar{W})$. Note $\mathcal{U} = q^{-1}(0) \subseteq q^{-1}(\bar{W}) = W$. Then $W/\mathcal{U} = q(W) = q(q^{-1}(\bar{W})) = \bar{W}$ (surjectivity of q ensures $q(q^{-1}(S)) = S$ for any $S \subseteq V/\mathcal{U}$). $\square$

Note:-

Maximal proper subspaces of V containing $\mathcal{U}$ correspond to maximal proper subspaces of $V/\mathcal{U}$ —i.e., hyperplanes (codimension-1 subspaces). This perspective becomes useful in algebraic geometry and representation theory.

4.13 Linear Operators and Endomorphisms

When the domain and codomain of a linear map coincide, we gain additional algebraic structure: the ability to compose maps with themselves.

4.13.1 The Ring of Endomorphisms

Definition 4.13.1: Linear operator / Endomorphism

A **linear operator** or **endomorphism** of a vector space V is a linear map $\varphi : V \rightarrow V$. We write

$$\text{End}(V) = \text{Hom}(V, V)$$

for the set of all endomorphisms of V .

Proposition 4.13.1 Ring structure on $\text{End}(V)$

$\text{End}(V)$ is a **ring** (with identity) under:

- **Addition:** $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$.
- **Multiplication:** $(\varphi \circ \psi)(v) = \varphi(\psi(v))$ (composition).
- **Identity:** The identity map id_V .

Moreover, $\text{End}(V)$ is a k -algebra: multiplication distributes over scalar multiplication.

Proof: Addition makes $\text{End}(V)$ an abelian group (it's a subgroup of $\text{Hom}(V, V)$ as a vector space). Composition is associative with identity id_V . Distributivity: $\varphi \circ (\psi + \rho) = \varphi \circ \psi + \varphi \circ \rho$ and $(\varphi + \psi) \circ \rho = \varphi \circ \rho + \psi \circ \rho$. These follow from linearity. $\square$

Note:-

The ring $\text{End}(V)$ is noncommutative unless $\dim(V) \leq 1$. For $V = k^n$, $\text{End}(V) \cong M_n(k)$, the ring of $n \times n$ matrices. The noncommutativity of matrix multiplication reflects the noncommutativity of $\text{End}(V)$.

4.13.2 Polynomials of Operators

Since endomorphisms can be composed, we can form “powers” and evaluate polynomials.

Definition 4.13.2: Powers and polynomials of operators

For $\varphi \in \text{End}(V)$ and $n \in \mathbb{Z}_{\geq 0}$:

$$\varphi^0 = \text{id}_V, \quad \varphi^n = \underbrace{\varphi \circ \varphi \circ \cdots \circ \varphi}_{n \text{ times}}.$$

For a polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n \in k[x]$, define

$$p(\varphi) = a_0\text{id}_V + a_1\varphi + \cdots + a_n\varphi^n \in \text{End}(V).$$

Proposition 4.13.2 Evaluation homomorphism

The map $\text{ev}_\varphi : k[x] \rightarrow \text{End}(V)$ defined by $p(x) \mapsto p(\varphi)$ is a ring homomorphism. That is:

1. $(p + q)(\varphi) = p(\varphi) + q(\varphi)$.
2. $(pq)(\varphi) = p(\varphi) \circ q(\varphi)$.
3. $1(\varphi) = \text{id}_V$.

Proof: (1) and (3) are immediate. For (2), let $p(x) = \sum a_i x^i$ and $q(x) = \sum b_j x^j$. Then $(pq)(x) = \sum_k c_k x^k$ where $c_k = \sum_{i+j=k} a_i b_j$. So

$$(pq)(\varphi) = \sum_k c_k \varphi^k = \sum_k \left(\sum_{i+j=k} a_i b_j \right) \varphi^k = \left(\sum_i a_i \varphi^i \right) \circ \left(\sum_j b_j \varphi^j \right) = p(\varphi) \circ q(\varphi).$$

□

Note:-

This makes $\text{End}(V)$ a $k[x]$ -module: $p(x) \cdot v = p(\varphi)(v)$. The study of this module structure (particularly the kernel of ev_φ , called the minimal polynomial) is central to the theory of canonical forms.

4.14 Invariant Subspaces

Understanding an operator often begins with finding subspaces on which its behavior is simpler.

Definition 4.14.1: Invariant subspace

A subspace $W \subseteq V$ is **invariant** under $\varphi \in \text{End}(V)$ (or **φ -invariant**) if

$$\varphi(W) \subseteq W,$$

i.e., $\varphi(w) \in W$ for all $w \in W$.

Example 4.14.1 (Basic examples of invariant subspaces)

For any $\varphi \in \text{End}(V)$:

- $\{0\}$ and V are always φ -invariant.
- $\ker(\varphi)$ is φ -invariant: if $\varphi(v) = 0$, then $\varphi(\varphi(v)) = \varphi(0) = 0$.
- $\text{Im}(\varphi)$ is φ -invariant: $\varphi(\text{Im}(\varphi)) = \text{Im}(\varphi^2) \subseteq \text{Im}(\varphi)$.

- More generally, $\ker(\varphi^n)$ and $\text{Im}(\varphi^n)$ are φ -invariant for all $n \geq 0$.

Proposition 4.14.1 Restriction to an invariant subspace

If $W \subseteq V$ is φ -invariant, the **restriction** $\varphi|_W : W \rightarrow W$ is a well-defined linear operator on W .

4.14.1 Block Diagonal Structure

Invariant subspaces lead to simpler matrix representations.

Theorem 4.14.1 Direct sum of invariant subspaces

Suppose $V = V_1 \oplus V_2$ where both V_1 and V_2 are φ -invariant. Choose bases $\mathcal{B}_1 = \{e_1, \dots, e_m\}$ of V_1 and $\mathcal{B}_2 = \{e_{m+1}, \dots, e_n\}$ of V_2 , forming a basis $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ of V . Then the matrix of φ with respect to $\mathcal{B}$ is **block diagonal**:

$$\mathcal{M}(\varphi; \mathcal{B}) = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

where $A_1 = \mathcal{M}(\varphi|_{V_1}; \mathcal{B}_1)$ and $A_2 = \mathcal{M}(\varphi|_{V_2}; \mathcal{B}_2)$.

Proof: For $i \leq m$: $\varphi(e_i) \in V_1$ (by φ -invariance), so $\varphi(e_i)$ is a linear combination of $e_1, \dots, e_m$ only. For $i > m$: $\varphi(e_i) \in V_2$, so $\varphi(e_i)$ is a linear combination of $e_{m+1}, \dots, e_n$ only. This gives the block structure. $\square$

Note:-

Conversely, if a matrix is block diagonal with blocks A_1 and A_2 , then the spans of the corresponding basis vectors are invariant subspaces. The goal of much of spectral theory is to decompose V into a direct sum of invariant subspaces on which φ acts simply.

Corollary 4.14.1 Block triangular from one invariant subspace

If V_1 is φ -invariant (but we don't assume V_1 has an invariant complement), then extending a basis of V_1 to a basis of V gives a **block upper-triangular** matrix:

$$\mathcal{M}(\varphi) = \begin{pmatrix} A_1 & * \\ 0 & A_2 \end{pmatrix}.$$

The upper-right block $*$ may be nonzero; it measures the “interaction” between V_1 and V/V_1 .

4.15 Eigenvalues and Eigenvectors

The simplest invariant subspaces are one-dimensional. These give rise to the fundamental notion of eigenvalues.

4.15.1 Definitions and Basic Properties

Definition 4.15.1: Eigenvector and eigenvalue

Let $\varphi \in \text{End}(V)$. A nonzero vector $v \in V$ is an **eigenvector** of φ if

$$\varphi(v) = \lambda v$$

for some scalar $\lambda \in \mathbb{k}$. The scalar λ is called the **eigenvalue** corresponding to v .

Note:-

The condition $\varphi(v) = \lambda v$ says that φ acts on the line spanned by v as scaling by λ . Equivalently, $\text{Span}\{v\}$ is a one-dimensional φ -invariant subspace. Conversely, any one-dimensional invariant subspace $U = \text{Span}\{v\}$ consists of eigenvectors (with the same eigenvalue) plus zero.

Proposition 4.15.1 Characterization of eigenvalues

For $\varphi \in \text{End}(V)$ and $\lambda \in k$, the following are equivalent:

1. λ is an eigenvalue of φ .
2. $\varphi - \lambda \text{id}_V$ is not injective.
3. $\ker(\varphi - \lambda \text{id}_V) \neq \{0\}$.
4. (If V is finite-dimensional) $\varphi - \lambda \text{id}_V$ is not surjective.
5. (If V is finite-dimensional) $\det(\varphi - \lambda \text{id}_V) = 0$.

Proof: (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3): λ is an eigenvalue iff there exists nonzero v with $\varphi(v) = \lambda v$, iff $(\varphi - \lambda \text{id})(v) = 0$ for some nonzero v , iff $\ker(\varphi - \lambda \text{id}) \neq \{0\}$.

(3) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5): In finite dimensions, a linear operator is injective iff surjective iff invertible iff has nonzero determinant. So $\ker(\varphi - \lambda \text{id}) \neq \{0\}$ iff $\det(\varphi - \lambda \text{id}) = 0$. $\square$

Definition 4.15.2: Eigenspace

For an eigenvalue λ of φ , the **eigenspace** is

$$E(\lambda, \varphi) = \ker(\varphi - \lambda \text{id}_V) = \{v \in V : \varphi(v) = \lambda v\}.$$

This is a subspace of V (it includes 0, though 0 is not an eigenvector).

Theorem 4.15.1 Eigenvectors with distinct eigenvalues are linearly independent

Let $v_1, \dots, v_m$ be eigenvectors of φ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Then $v_1, \dots, v_m$ are linearly independent.

Proof: Induction on m . Base case $m = 1$: a single eigenvector is nonzero, hence linearly independent. Suppose the result holds for $m - 1$ eigenvectors. Assume $\sum_{i=1}^m c_i v_i = 0$. Apply φ :

$$\sum_{i=1}^m c_i \lambda_i v_i = 0.$$

Multiply the original equation by λ_m and subtract:

$$\sum_{i=1}^{m-1} c_i (\lambda_i - \lambda_m) v_i = 0.$$

By the induction hypothesis (since $v_1, \dots, v_{m-1}$ are eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_{m-1}$), we have $c_i (\lambda_i - \lambda_m) = 0$ for $i < m$. Since $\lambda_i \neq \lambda_m$, we get $c_i = 0$. Substituting back: $c_m v_m = 0$, so $c_m = 0$ (since $v_m \neq 0$). $\square$

Corollary 4.15.1 Bound on number of eigenvalues

A linear operator on an n -dimensional space has at most n distinct eigenvalues.

Proof: If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues with eigenvectors $v_1, \dots, v_m$, then $v_1, \dots, v_m$ are linearly independent. In an n -dimensional space, any linearly independent set has at most n elements. $\square$

Corollary 4.15.2 Sum of eigenspaces is direct

If $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of φ , then

$$E(\lambda_1, \varphi) + \dots + E(\lambda_m, \varphi) = E(\lambda_1, \varphi) \oplus \dots \oplus E(\lambda_m, \varphi).$$

is a direct sum. Furthermore, $\sum_i \dim E(\lambda_i, \varphi) \leq \dim V$.

4.15.2 Diagonalizable Operators

The best-case scenario: V has a basis of eigenvectors.

Definition 4.15.3: Diagonalizable operator

An operator $\varphi \in \text{End}(V)$ is **diagonalizable** if there exists a basis of V consisting entirely of eigenvectors of φ . Equivalently, φ has a diagonal matrix with respect to some basis.

Note:-

If $\{v_1, \dots, v_n\}$ is a basis of eigenvectors with $\varphi(v_i) = \lambda_i v_i$, then the matrix of φ is

$$\mathcal{M}(\varphi) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

The eigenvalues appear on the diagonal (possibly with repetition).

Theorem 4.15.2 Characterizations of diagonalizability

For $\varphi \in \text{End}(V)$ with $\dim(V) = n$ and distinct eigenvalues $\lambda_1, \dots, \lambda_m$, the following are equivalent:

1. φ is diagonalizable.
2. V has a basis of eigenvectors of φ .
3. $V = E(\lambda_1, \varphi) \oplus \dots \oplus E(\lambda_m, \varphi)$.
4. $\dim E(\lambda_1, \varphi) + \dots + \dim E(\lambda_m, \varphi) = n$.

Proof: (1) $\Leftrightarrow$ (2): By definition.

(2) $\Rightarrow$ (3): If V has a basis of eigenvectors, partition it by eigenvalue. Each part is a basis of the corresponding eigenspace, and together they span V .

(3) $\Rightarrow$ (4): Take dimensions: $\dim V = \dim(E_1 \oplus \dots \oplus E_m) = \sum \dim E_i$.

(4) $\Rightarrow$ (2): The sum $E_1 + \dots + E_m$ is direct (earlier corollary), and has dimension $\sum \dim E_i = n = \dim V$. So $E_1 \oplus \dots \oplus E_m = V$. Combining bases of each E_i gives a basis of V consisting of eigenvectors. $\square$

Corollary 4.15.3 Enough distinct eigenvalues implies diagonalizability

If $\varphi \in \text{End}(V)$ has $\dim(V)$ distinct eigenvalues, then φ is diagonalizable.

Proof: Each eigenspace has dimension at least 1 (it contains an eigenvector). With $n = \dim V$ distinct eigenvalues, $\sum \dim E(\lambda_i) \geq n$. But this sum is also $\leq n$, so equality holds, and criterion (4) above is satisfied. $\square$

4.15.3 Examples: Diagonalizable and Non-Diagonalizable

Example 4.15.1 (Diagonal matrices)

The identity matrix I_n has eigenvalue 1 with eigenspace all of k^n . More generally, a diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_n)$ has the standard basis vectors as eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$. Such a matrix is (trivially) diagonalizable—it's already diagonal!

Example 4.15.2 (A non-diagonalizable matrix)

Consider $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues satisfy $\det(A - \lambda I) = (1 - \lambda)^2 = 0$, so $\lambda = 1$ is the only eigenvalue.

The eigenspace: $E(1, \varphi) = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span}\{e_1\}$.

This has dimension 1, but $\dim(\mathbb{R}^2) = 2$. So φ is **not diagonalizable**.

Geometrically, φ is a “shear”: it fixes the x -axis pointwise and slides other points parallel to the x -axis. The only eigenvector direction is along the x -axis.

Example 4.15.3 (No eigenvectors over $\mathbb{R}$)

Consider rotation by 90° in $\mathbb{R}^2$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots. So R has no eigenvalues over $\mathbb{R}$, hence no eigenvectors. Over $\mathbb{C}$, the eigenvalues are $\pm i$ with eigenvectors $(1, \mp i)^T$, and R is diagonalizable as a complex matrix. The eigenfairy doesn't visit $\mathbb{R}$ (sad).

This illustrates that diagonalizability depends on the scalar field!

Note:-

The rotation example shows that not every operator is diagonalizable. Even over $\mathbb{C}$, where every polynomial has roots, an operator may fail to be diagonalizable (like the shear above). Understanding non-diagonalizable operators requires the theory of Jordan normal form, which we'll develop later.

Question 5: Exercises on quotient spaces, operators, and eigenvalues

1. Let $V = \mathbb{R}^4$ and $U = \{(x, y, z, w) : x + y = 0, z = w\}$. Find $\dim(U)$, $\dim(V/U)$, and describe V/U explicitly.
2. Prove that if $\varphi, \psi \in \text{End}(V)$ commute ($\varphi \circ \psi = \psi \circ \varphi$), then every eigenspace of φ is ψ -invariant.
3. Let $\varphi \in \text{End}(V)$ be diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $p(\varphi) = 0$ where $p(x) = \prod_{i=1}^m (x - \lambda_i)$.
4. Show that the quotient map $q : V \rightarrow V/U$ induces a bijection between $\{\text{linear maps } V/U \rightarrow W\}$ and $\{\text{linear maps } V \rightarrow W \text{ vanishing on } U\}$.
5. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Find the eigenvalues and eigenspaces of A . Is A diagonalizable? If so, find a matrix P such that $P^{-1}AP$ is diagonal.

(1) The constraints $x + y = 0$ and $z = w$ give 2 independent linear equations on $\mathbb{R}^4$, so $\dim(\mathcal{U}) = 4 - 2 = 2$. A basis: $\{(-1, 1, 0, 0), (0, 0, 1, 1)\}$.

By the dimension formula: $\dim(V/\mathcal{U}) = \dim(V) - \dim(\mathcal{U}) = 4 - 2 = 2$.

The map $\pi : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ defined by $\pi(x, y, z, w) = (x + y, z - w)$ has $\ker(\pi) = \mathcal{U}$. By the first isomorphism theorem, $V/\mathcal{U} \cong \mathbb{R}^2$. Explicitly, the coset $(x, y, z, w) + \mathcal{U}$ is determined by $(x + y, z - w)$.

(2) Let v be an eigenvector of φ with eigenvalue λ : $\varphi(v) = \lambda v$. We need $\psi(v) \in E(\lambda, \varphi)$, i.e., $\varphi(\psi(v)) = \lambda \psi(v)$.

Compute: $\varphi(\psi(v)) = \psi(\varphi(v)) = \psi(\lambda v) = \lambda \psi(v)$. ✓

(We used commutativity in the first equality.)

(3) Let $p(x) = \prod_{i=1}^m (x - \lambda_i)$. Since φ is diagonalizable, $V = \bigoplus_{i=1}^m E(\lambda_i, \varphi)$. For any eigenvector $v \in E(\lambda_j, \varphi)$:

$$(\varphi - \lambda_j \text{id})(v) = 0.$$

So $p(\varphi)(v) = (\varphi - \lambda_1) \cdots (\varphi - \lambda_m)(v) = 0$ because one factor gives zero.

Since this holds for all eigenvectors and they span V , we have $p(\varphi) = 0$.

(4) Define $\Phi : \text{Hom}(V/\mathcal{U}, W) \rightarrow \{T \in \text{Hom}(V, W) : \mathcal{U} \subseteq \ker(T)\}$ by $\Phi(\bar{T}) = \bar{T} \circ q$.

Injective: If $\bar{T} \circ q = 0$, then $\bar{T}(v + \mathcal{U}) = \bar{T}(q(v)) = 0$ for all v , so $\bar{T} = 0$.

Surjective: Given $T : V \rightarrow W$ with $\mathcal{U} \subseteq \ker(T)$, define $\bar{T} : V/\mathcal{U} \rightarrow W$ by $\bar{T}(v + \mathcal{U}) = T(v)$. Well-defined because $v - v' \in \mathcal{U} \Rightarrow T(v) = T(v')$. Then $\Phi(\bar{T}) = \bar{T} \circ q = T$.

(5) Characteristic polynomial: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$.

Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.

Eigenspaces:

• $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = 2y$. Eigenspace: $\text{Span}\{(2, 1)^T\}$.

• $\lambda = 2$: $(A - 2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$, so $x = -y$. Eigenspace: $\text{Span}\{(1, -1)^T\}$.

Since we have 2 linearly independent eigenvectors in $\mathbb{R}^2$, A is diagonalizable.

Let $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ (columns are eigenvectors). Then $P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$.

4.16 Let's get general up in here!